GEK-83798

Instructions

Troubleshooting Your μdrill™ 3000 Equipment

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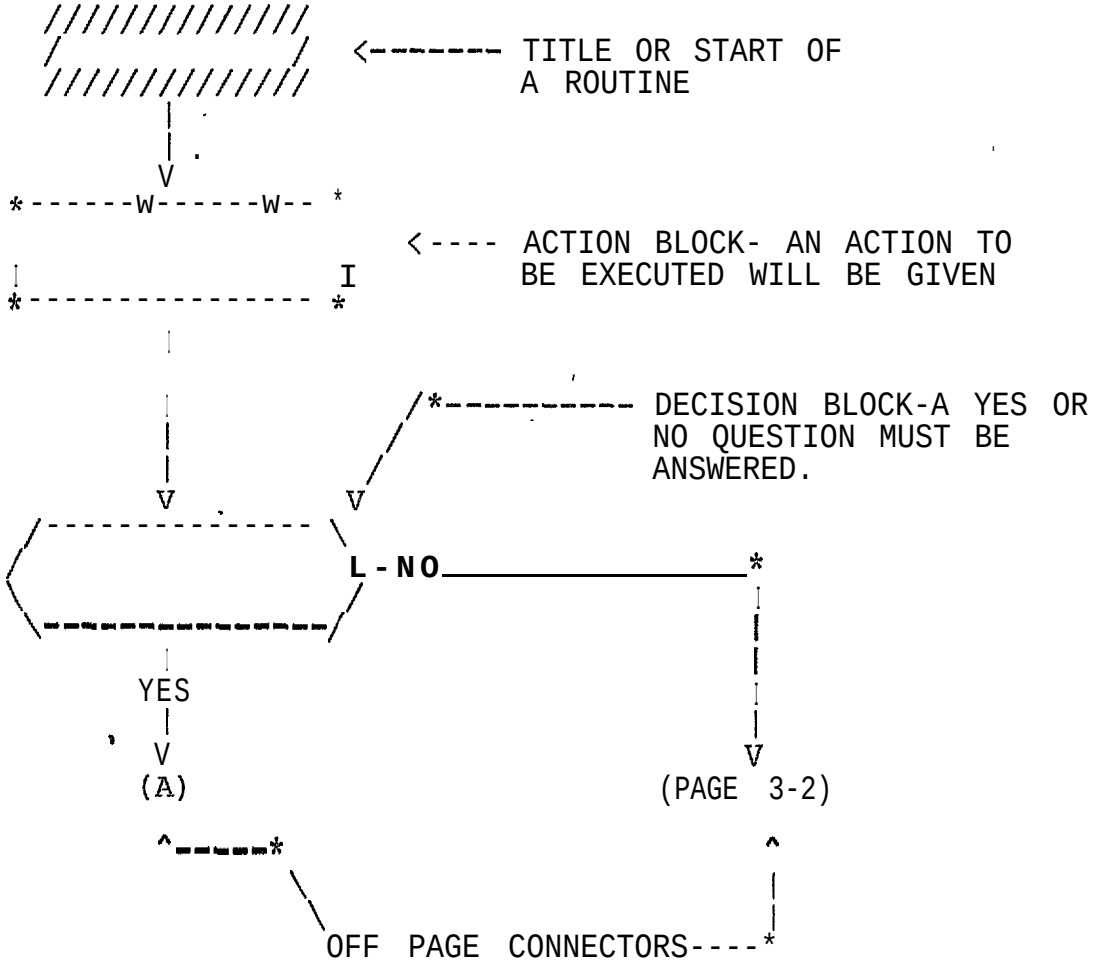
INTRODUCTION

THE PURPOSE OF THIS DOCUMENT IS TO AID THE DRILLING RIG ELECTRICAL MAINTENANCE PERSONNEL IN NARROWING DOWN A FAULT OR PROBLEM TO A PARTICULAR AREA. IT WILL COVER THE MAJORITY OF POSSIBLE PROBLEMS WHICH MAY OCCUR.

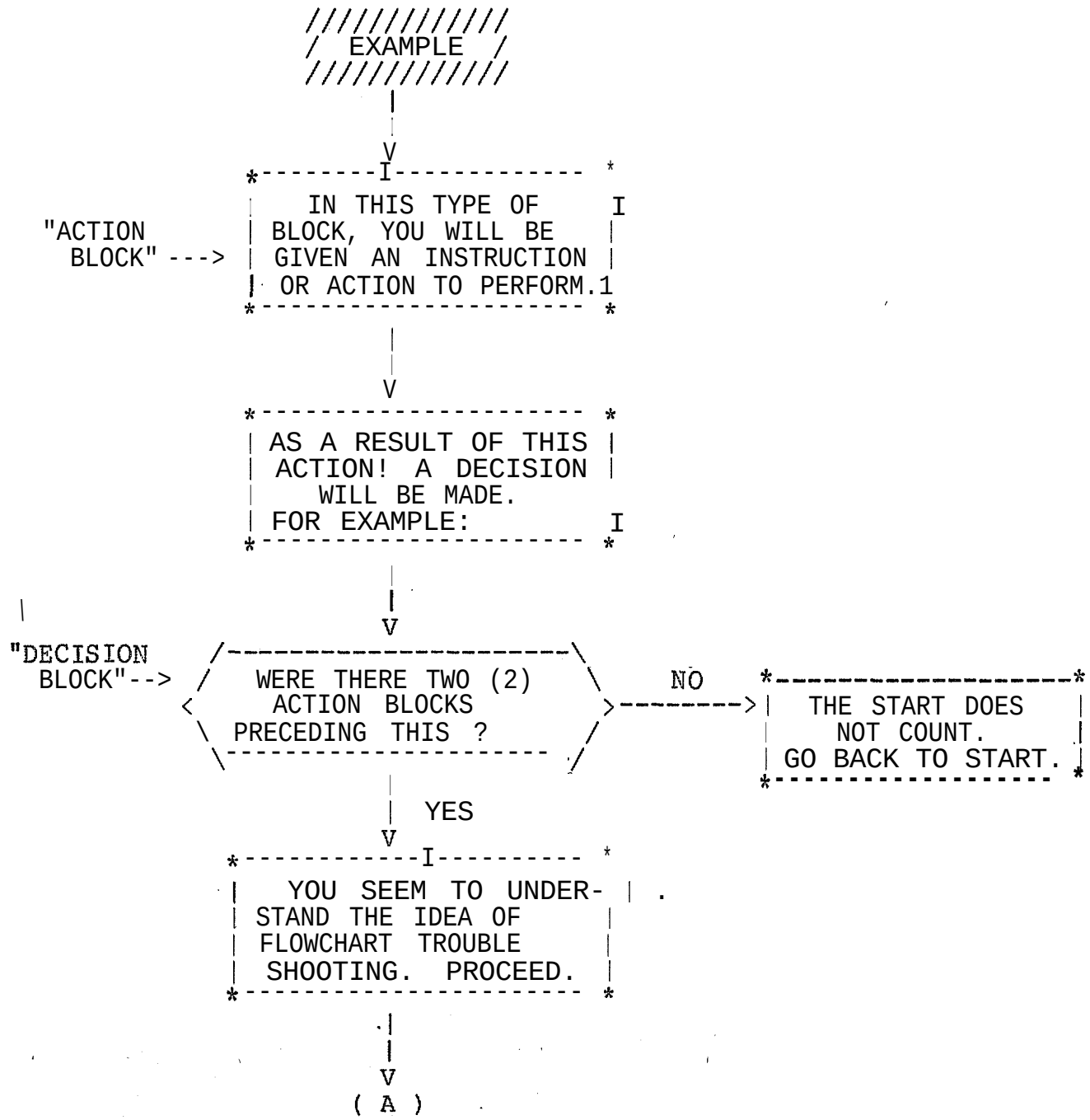
IT CANNOT BE A SUBSTITUTE FOR FAMILIARITY WITH THE EQUIPMENT AND/OR TROUBLESHOOTING TOOLS.

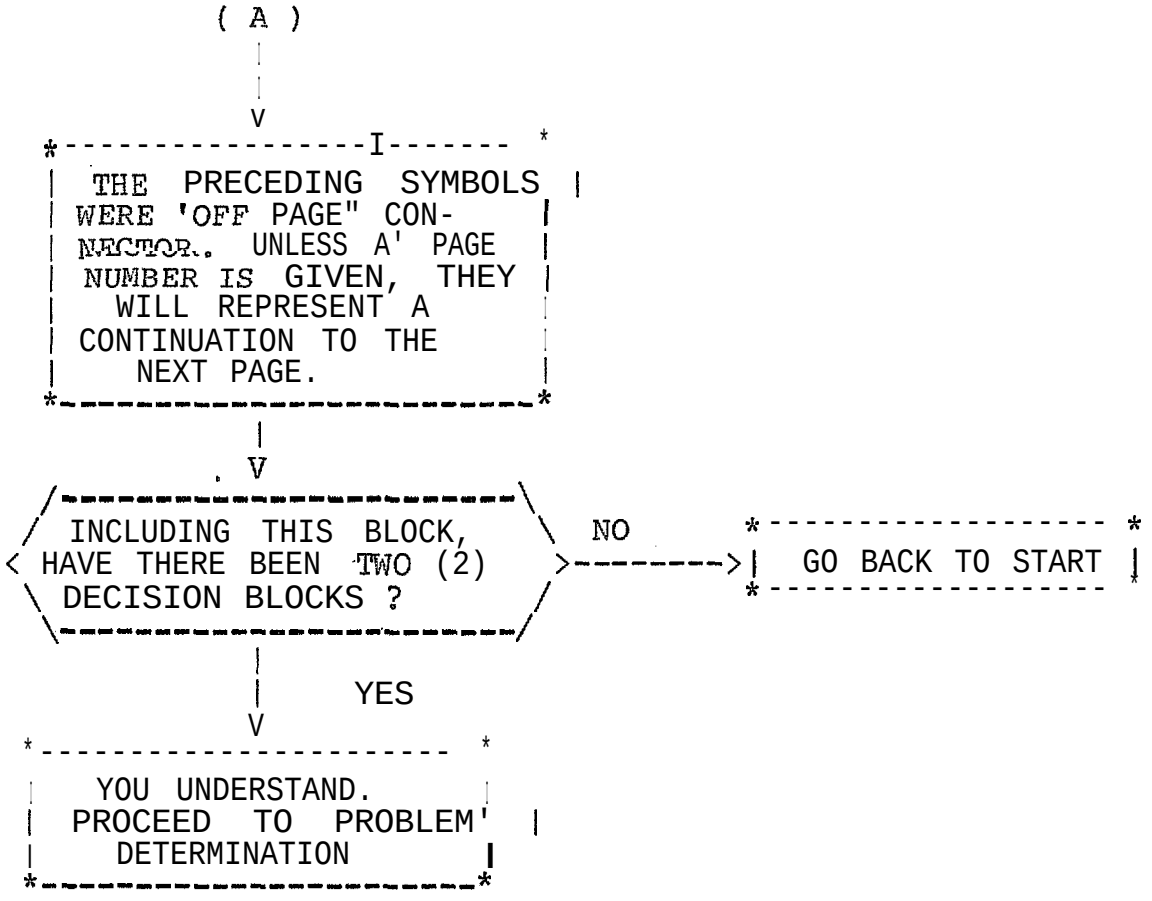
LIKE ANY TOOL, PROPER USE AND UNDERSTANDING OF THAT TOOL WILL MAKE THE USER MORE EFFICIENT AT THE TASK BEING PERFORMED. THEREFORE, FOR A COMPLETE UNDERSTANDING OF THIS DOCUMENT, THE ENTIRE INTRODUCTION SHOULD BE READ.

TROUBLESHOOTING WILL BE PERFORMED USING A FLOWCHART APPROACH. GIVEN BELOW ARE THE SYMBOLS USED THROUGHOUT THIS DOCUMENT, FOLLOWING THAT, AN EXAMPLE IS GIVEN TO AID IN THE UNDERSTANDING OF THE FLOWCHART METHOD FOR THOSE WHO HAVE NEVER USED IT.



THE FOLLOWING FLOWCHART WILL GIVE YOU AN INTRODUCTION INTO FLOWCHART TROUBLE SHOOTING. THIS TYPE OF FLOWCHART WILL BE USED THROUGHOUT THIS MANUAL SO UNDERSTANDING IT WILL BE ESSENTIAL.



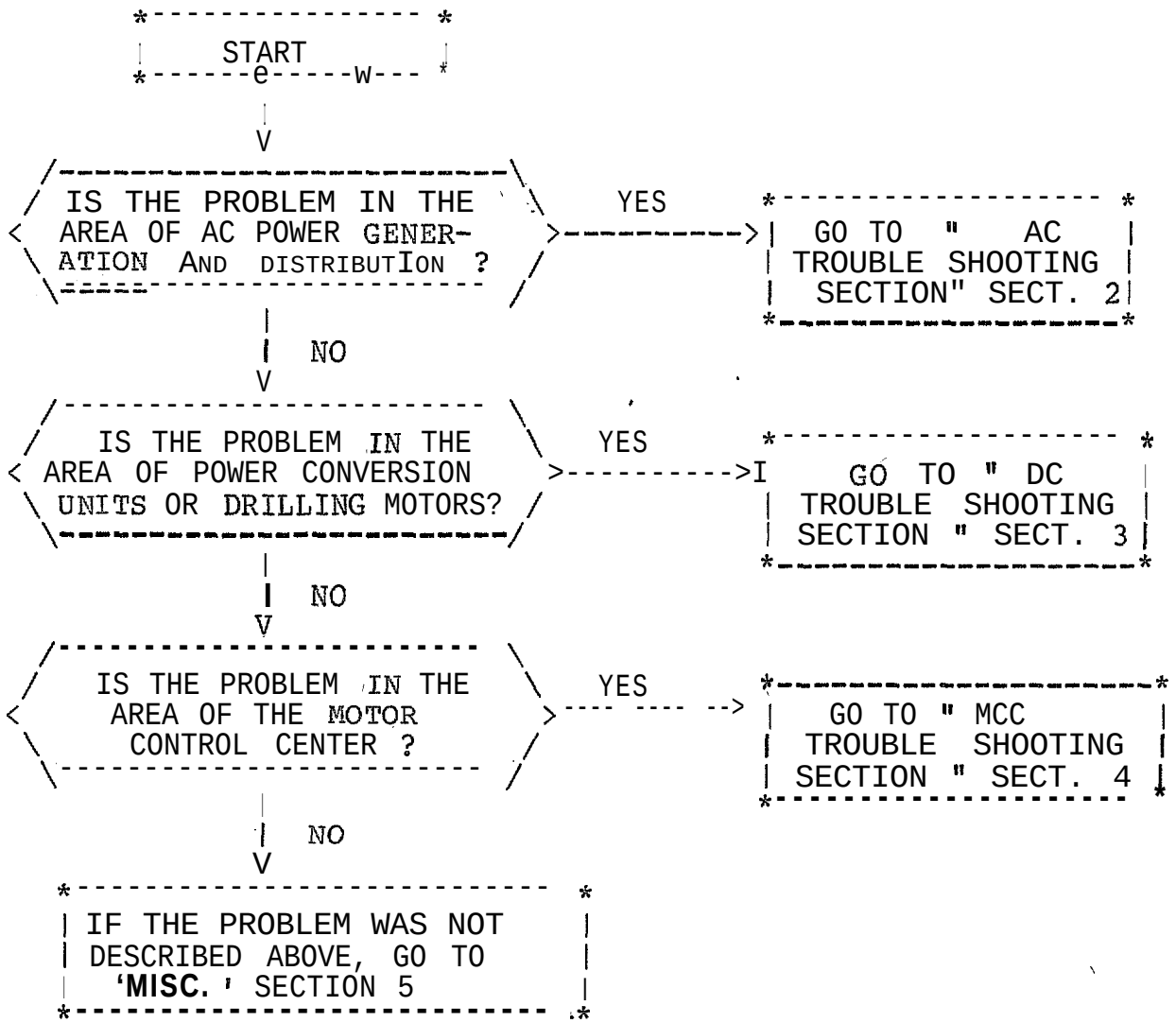

(A)
THE PRECEDING SYMBOLS WERE 'OFF PAGE" CONNECTOR.. UNLESS A' PAGE NUMBER IS GIVEN, THEY WILL REPRESENT A CONTINUATION TO THE NEXT PAGE.
INCLUDING THIS BLOCK, HAVE THERE BEEN TWO (2) DECISION BLOCKS ?
NO
GO BACK TO START
YES
YOU UNDERSTAND. PROCEED TO PROBLEM' DETERMINATION

PROBLEM DETERMINATION .

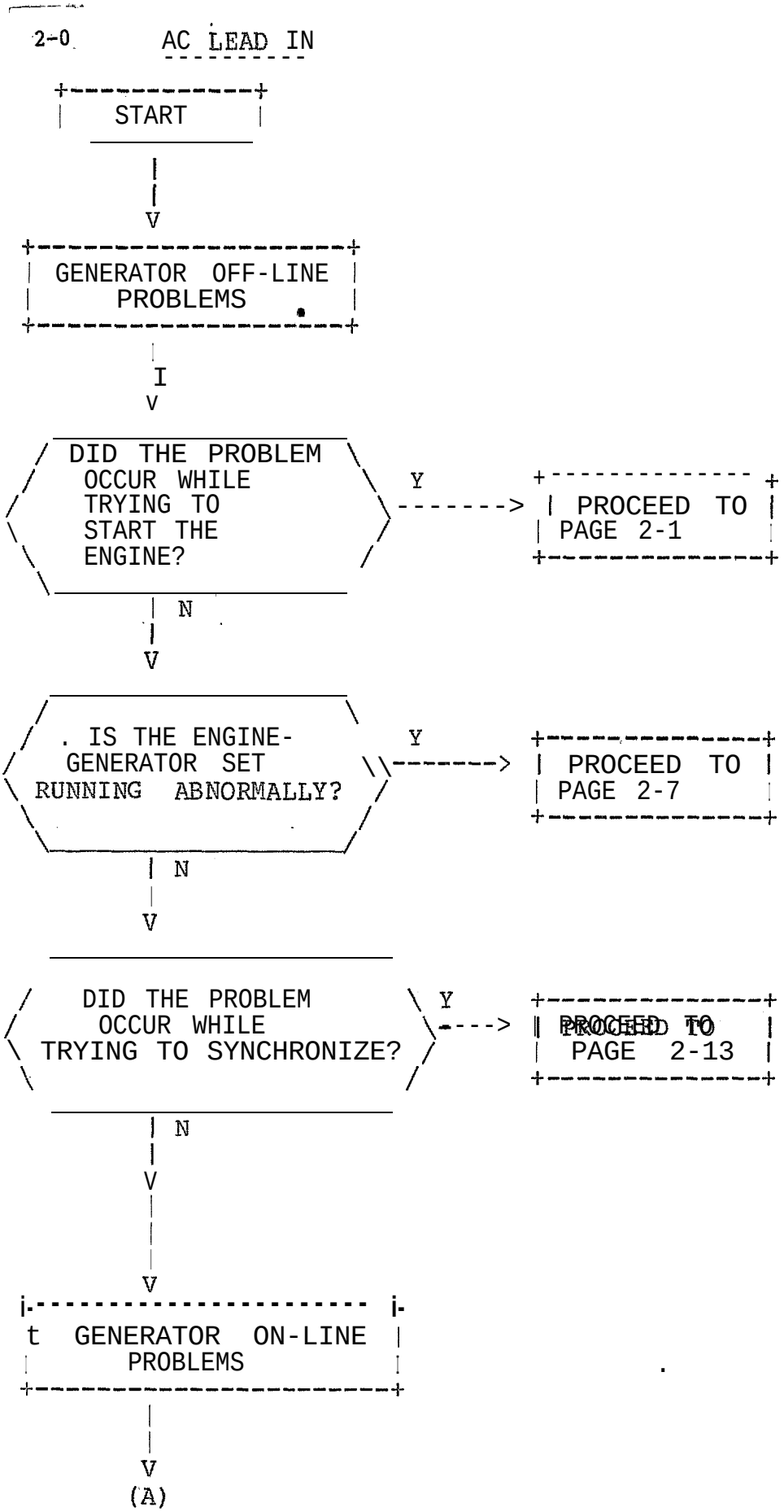
THE DRTLLIING RIG ELECTRICAL SYSTEM CAN BE BROKEN DOWN INTO THREE (3) BASIC CATEGORIES.

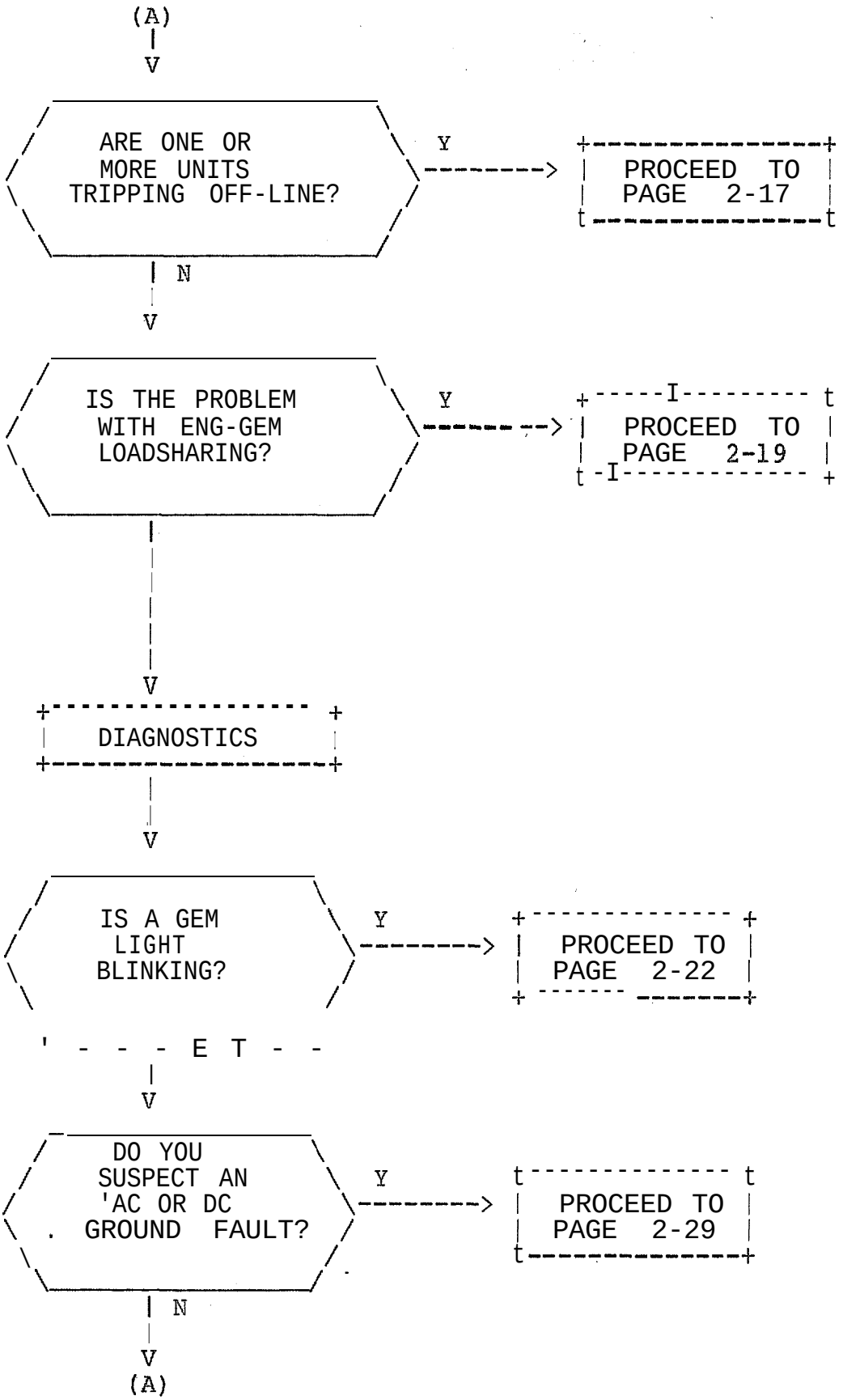
- 600 VOLT AC POWER GENERATION AND DISTRIBUTION (ENGINE-- GENERATOR CONTROL)
- DC POWER (SCR POWER CONVERTERS AND DRILLING MOTORS)
- AUXILIARY MOTOR FUNCTIONS (MOTOR CONTROL CENTER DISTRIBUTION)

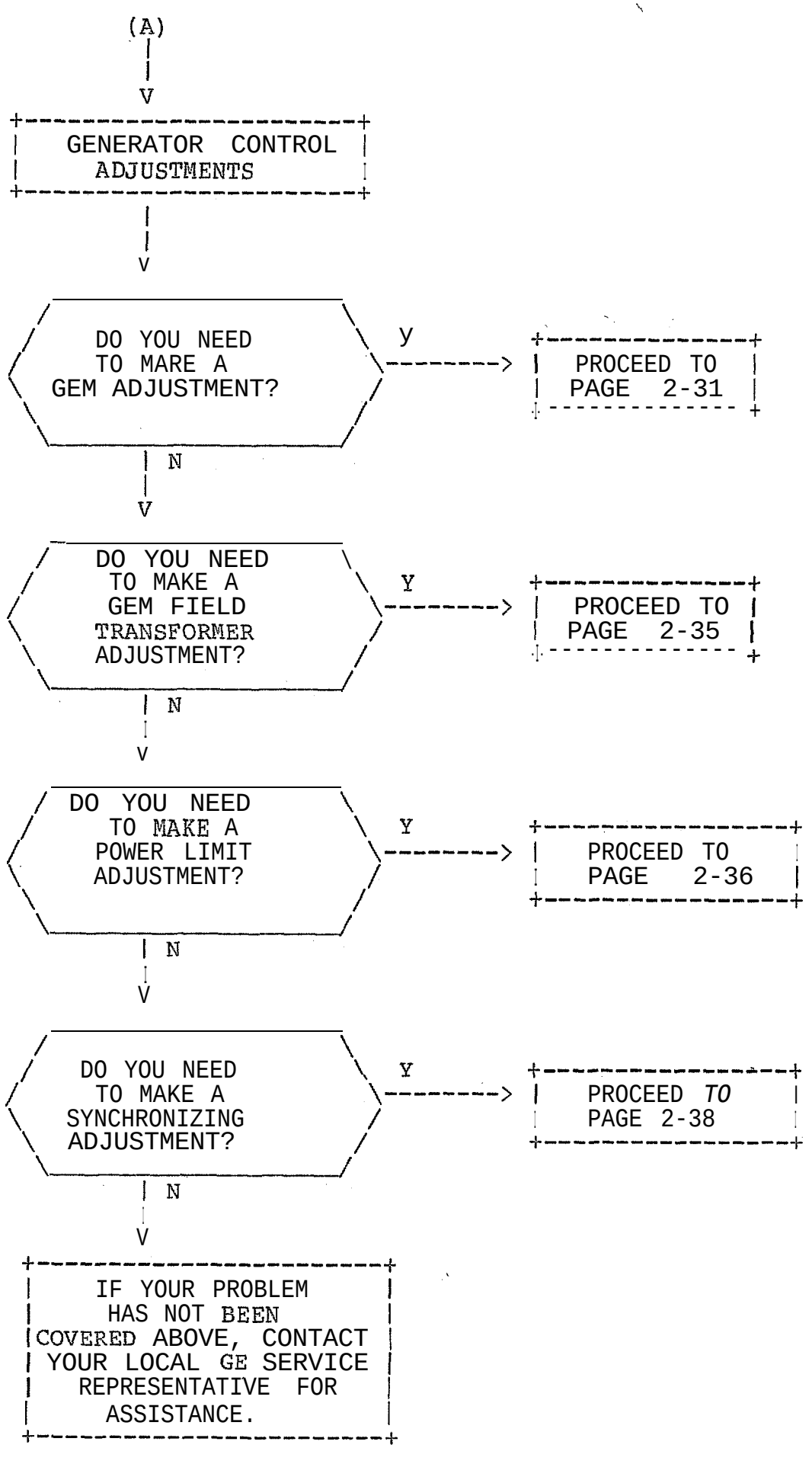
IN THE SAME SENSE, PROBLEMS CAN BE CATEGORIZED INTO THE SAME AREAS. EACH CATEGORY HAS MULTIPLE AREAS UNDER EACH TOPIC AND THE AREAS SOMETIMES OVERLAP (SUCH AS THE INTERACTION OF THE DC AND AC CONTROL MODULES TO LIMIT LOADING OF THE ENGINE-GENERATORS), THESE AREAS WILL BE TAKEN INTO ACCOUNT IN THE COURSE OF THE FLOWCHART TROUBLESHOOTING GUIDE, USING THE FOLLOWING FLOWCHART, NARROW THE PROBLEM DOWN TO A SPECIFIC AREA,



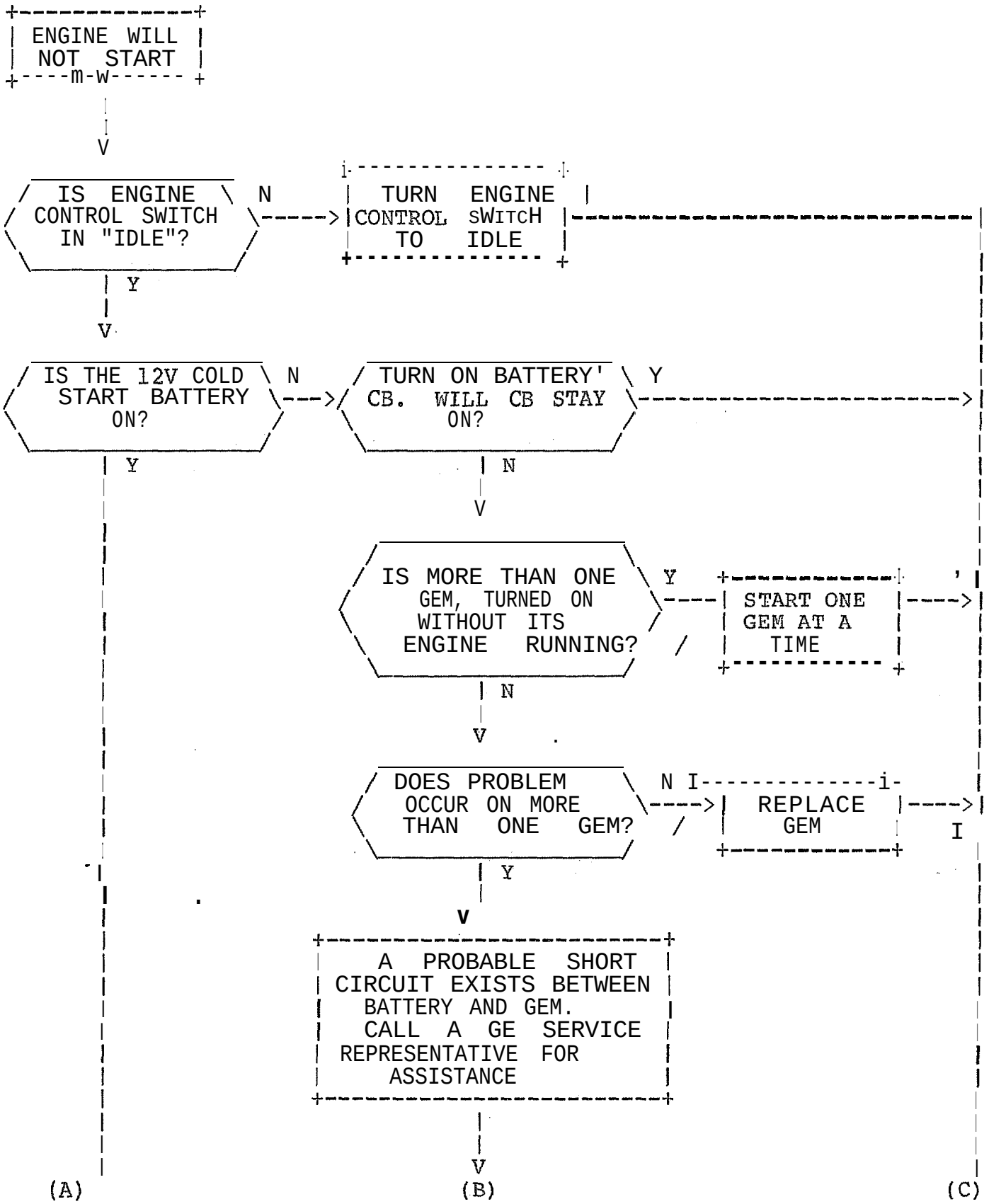
2-0 AC LEAD IN

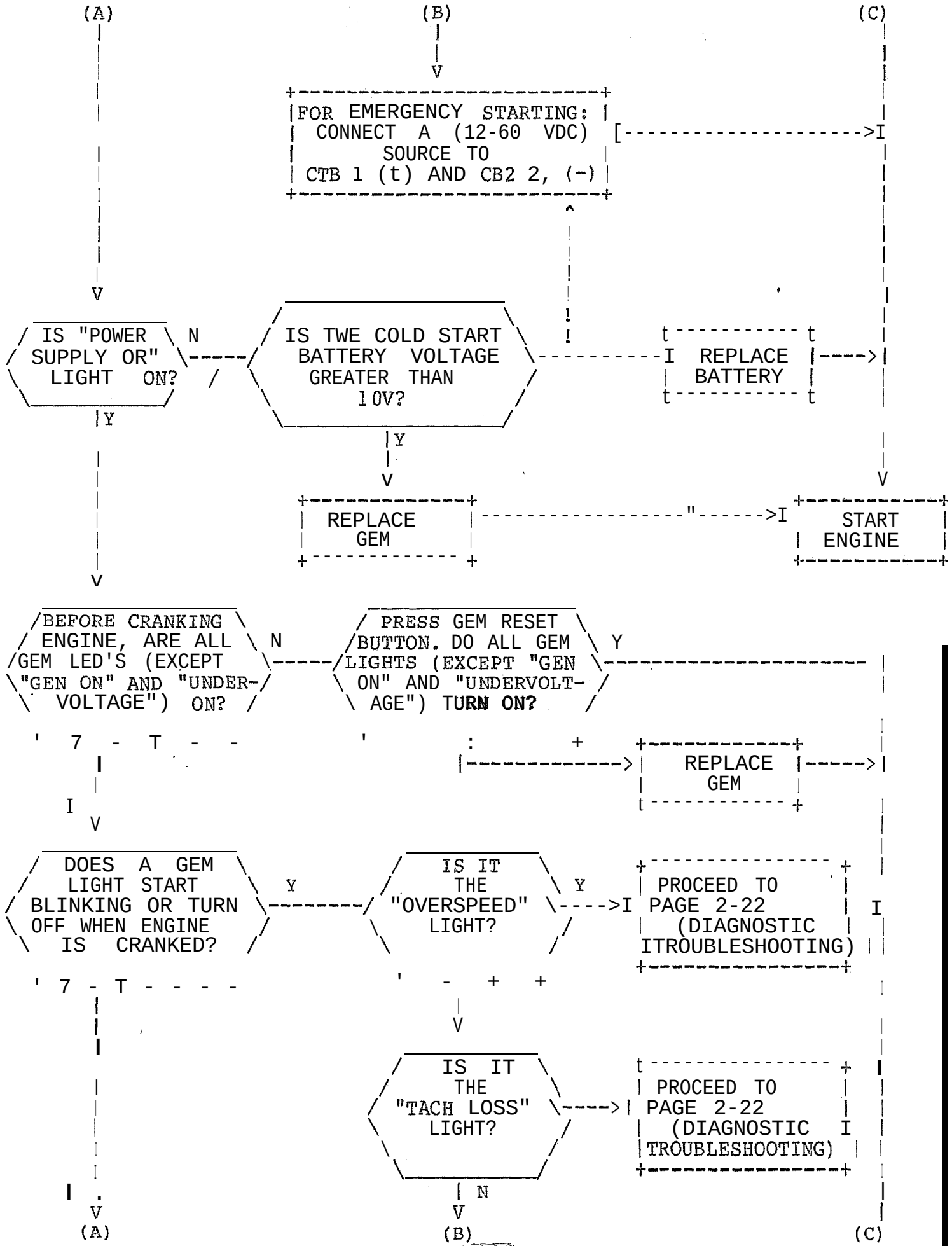

START
GENERATOR OFF-LINE PROBLEMS
DID THE PROBLEM OCCUR WHILE TRYING TO START THE ENGINE?
Y
PROCEED TO PAGE 2-1
N
. IS THE ENGINE-GENERATOR SET RUNNING ABNORMALLY?
Y
PROCEED TO PAGE 2-7
N
DID THE PROBLEM OCCUR WHILE TRYING TO SYNCHRONIZE?
Y
PROCEED TO PAGE 2-13
N
GENERATOR ON-LINE PROBLEMS
(A)

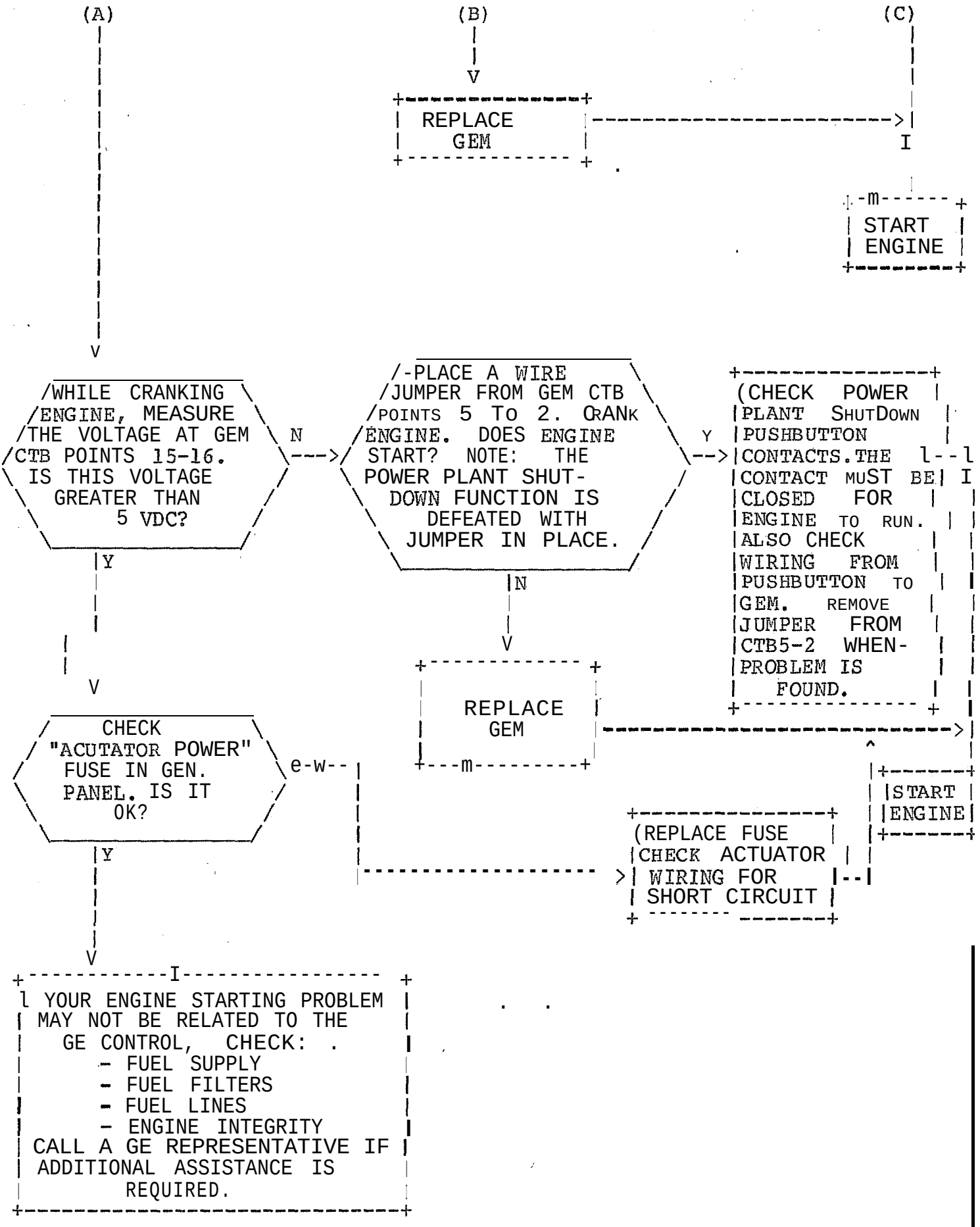

(A)
ARE ONE OR MORE UNITS TRIPPING OFF-LINE?
Y
PROCEED TO PAGE 2-17
N
IS THE PROBLEM WITH ENG-GEM LOADSHARING?
Y
PROCEED TO PAGE 2-19
DIAGNOSTICS
IS A GEM LIGHT BLINKING?
Y
PROCEED TO PAGE 2-22
E T
DO YOU SUSPECT AN 'AC OR DC GROUND FAULT?
Y
PROCEED TO PAGE 2-29
N
(A)


(A)
GENERATOR CONTROL ADJUSTMENTS
DO YOU NEED TO MARE A GEM ADJUSTMENT?
Y
PROCEED TO PAGE 2-31
N
DO YOU NEED TO MAKE A GEM FIELD TRANSFORMER ADJUSTMENT?
Y
PROCEED TO PAGE 2-35
N
DO YOU NEED TO MAKE A POWER LIMIT ADJUSTMENT?
Y
PROCEED TO PAGE 2-36
N
DO YOU NEED TO MAKE A SYNCHRONIZING ADJUSTMENT?
Y
PROCEED TO PAGE 2-38
N
IF YOUR PROBLEM HAS NOT BEEN COVERED ABOVE, CONTACT YOUR LOCAL GE SERVICE REPRESENTATIVE FOR ASSISTANCE.

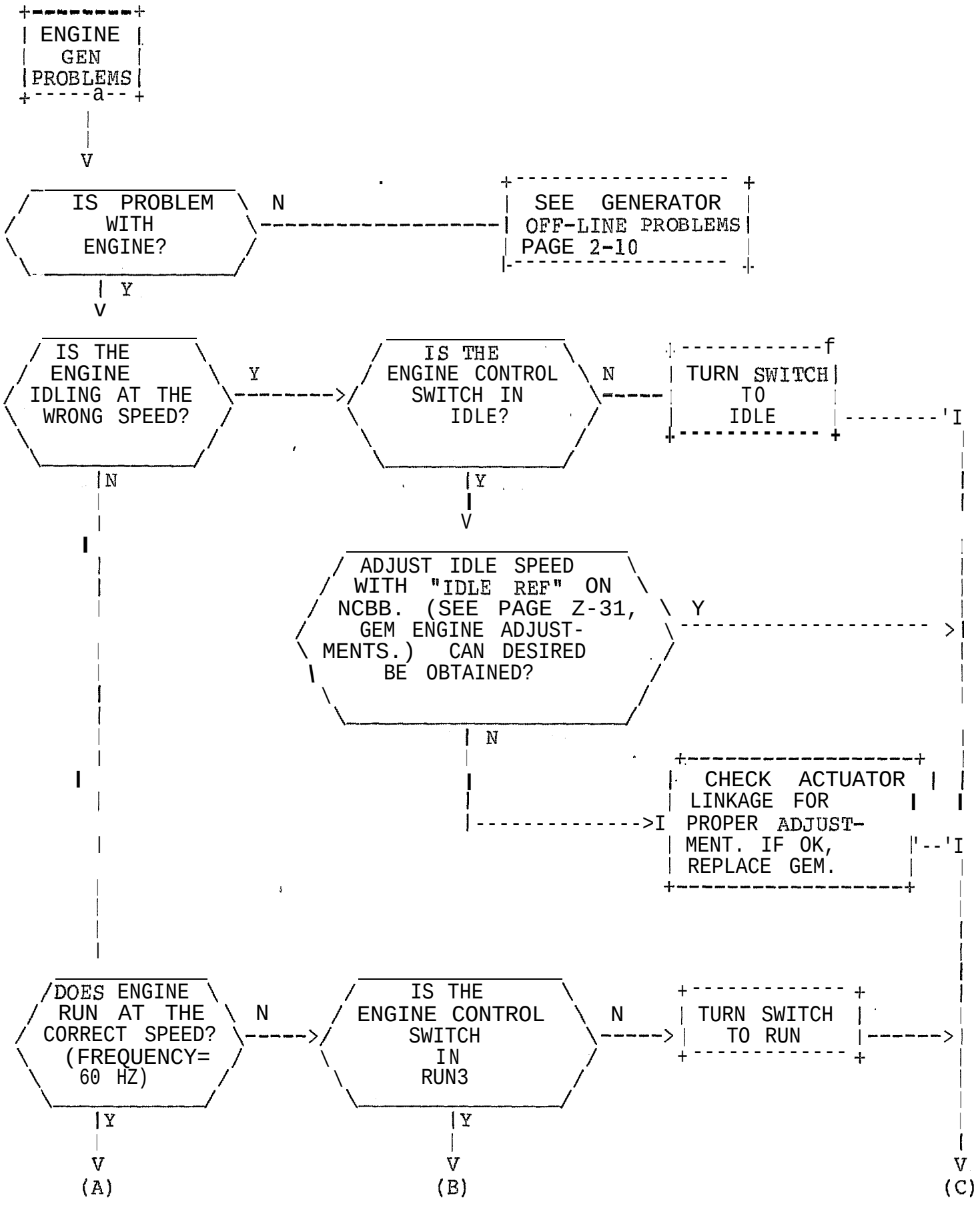
ENGINE STARTING

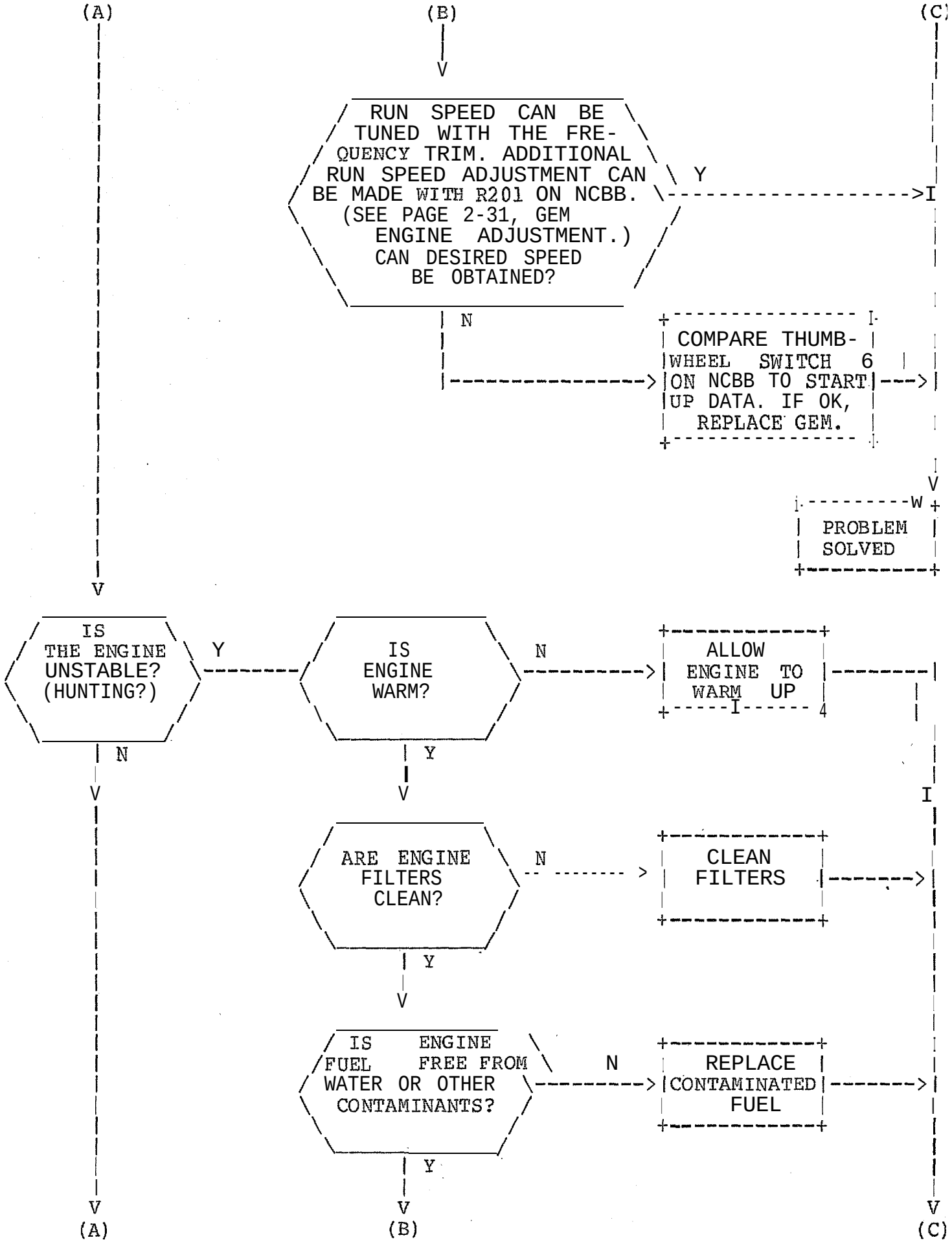


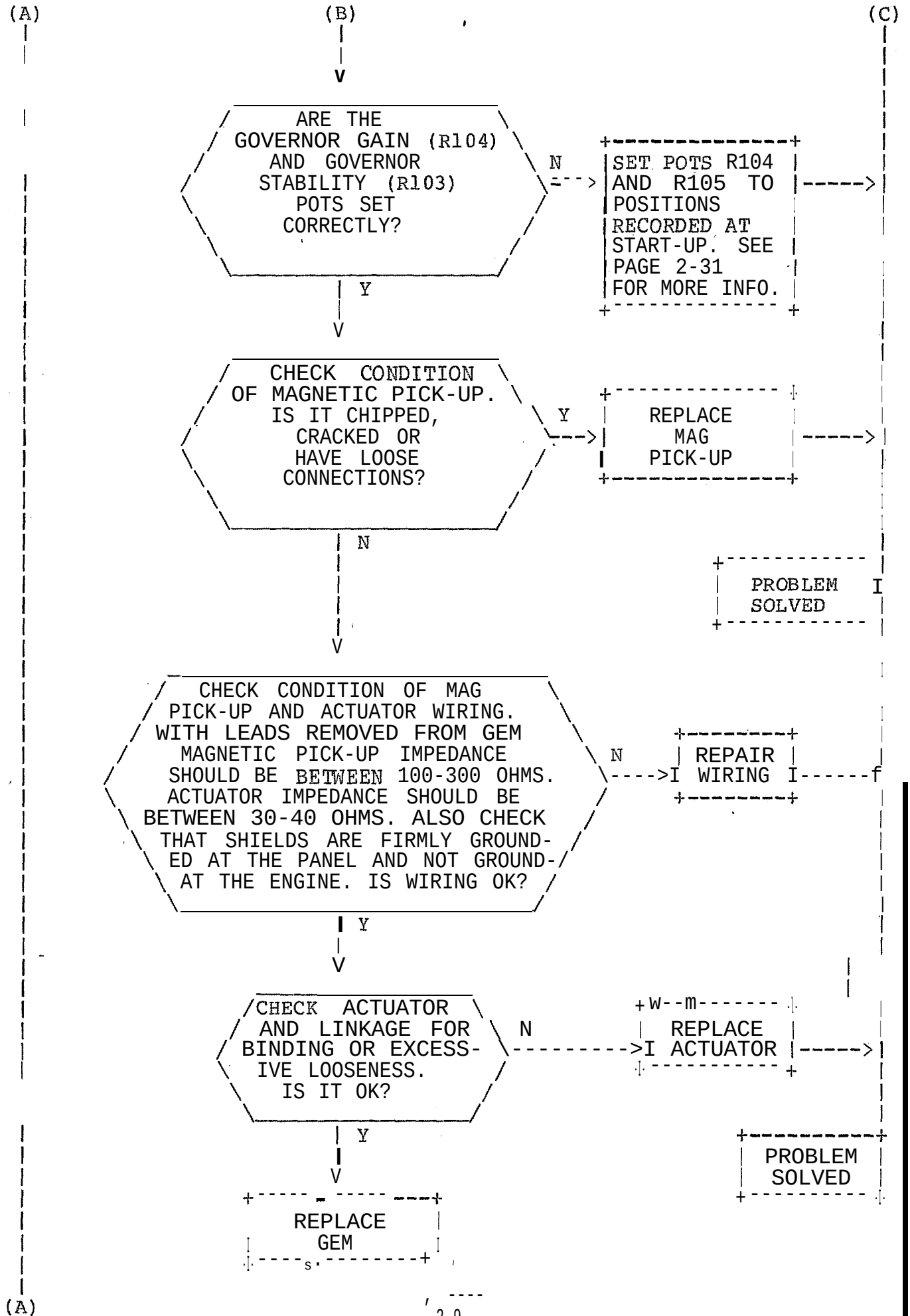

(A)
(B)
(C)
FOR EMERGENCY STARTING: CONNECT A (12-60 VDC) SOURCE TO CTB 1 (t) AND CB2 2, (-)
IS "POWER SUPPLY OR" LIGHT ON?
N
Y
IS TWE COLD START BATTERY VOLTAGE GREATER THAN 10V?
N
Y
REPLACE BATTERY
REPLACE GEM
START ENGINE
BEFORE CRANKING ENGINE, ARE ALL GEM LED'S (EXCEPT "GEN ON" AND "UNDERVOLTAGE") ON?
N
PRESS GEM RESET BUTTON. DO ALL GEM LIGHTS (EXCEPT "GEN ON" AND "UNDERVOLTAGE") TURN ON?
Y
REPLACE GEM
DOES A GEM LIGHT START BLINKING OR TURN OFF WHEN ENGINE IS CRANKED?
Y
IS IT THE "OVERSPEED" LIGHT?
Y
PROCEED TO PAGE 2-22 (DIAGNOSTIC TROUBLESHOOTING)
IS IT THE "TACH LOSS" LIGHT?
N
PROCEED TO PAGE 2-22 (DIAGNOSTIC TROUBLESHOOTING)
(A)
(B)
(C)

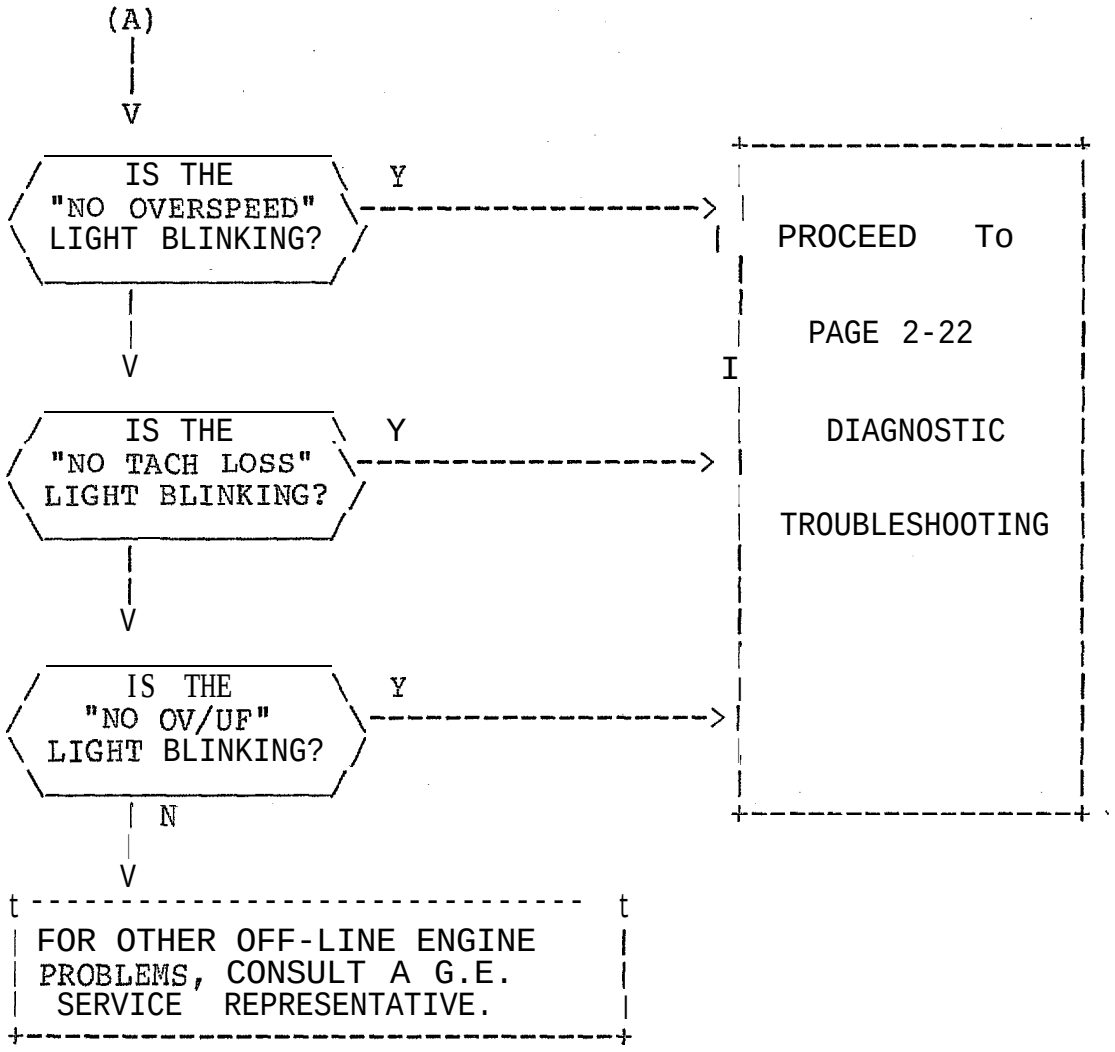

(A)
(B)
(C)
REPLACE GEM
START ENGINE
WHILE CRANKING ENGINE, MEASURE THE VOLTAGE AT GEM CTB POINTS 15-16. IS THIS VOLTAGE GREATER THAN 5 VDC?
N
Y
PLACE A WIRE JUMPER FROM GEM CTB POINTS 5 TO 2. CRANK ENGINE. DOES ENGINE START? NOTE: THE POWER PLANT SHUT-DOWN FUNCTION IS DEFEATED WITH JUMPER IN PLACE.
CHECK POWER PLANT SHUTDOWN PUSHBUTTON CONTACTS. THE CONTACT MUST BE CLOSED FOR ENGINE TO RUN. ALSO CHECK WIRING FROM PUSHBUTTON TO GEM. REMOVE JUMPER FROM CTB5-2 WHEN-PROBLEM IS FOUND.
REPLACE GEM
START ENGINE
CHECK "ACUTATOR POWER" FUSE IN GEN. PANEL. IS IT OK?
e-w
REPLACE FUSE CHECK ACTUATOR WIRING FOR SHORT CIRCUIT
YOUR ENGINE STARTING PROBLEM MAY NOT BE RELATED TO THE GE CONTROL, CHECK:
- FUEL SUPPLY
- FUEL FILTERS
- FUEL LINES
- ENGINE INTEGRITY
CALL A GE REPRESENTATIVE IF ADDITIONAL ASSISTANCE IS REQUIRED.

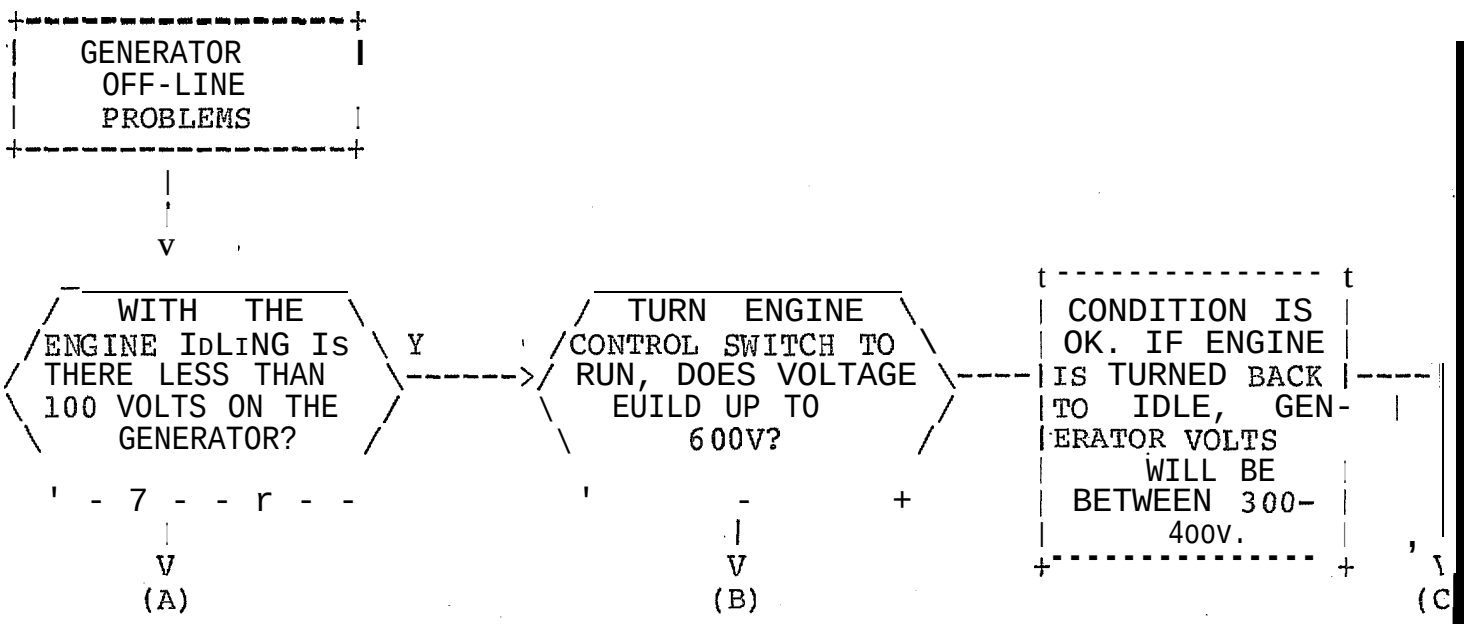
GENERATOR-ENGINE: OFF-LINE RUNNING PROBLEMS

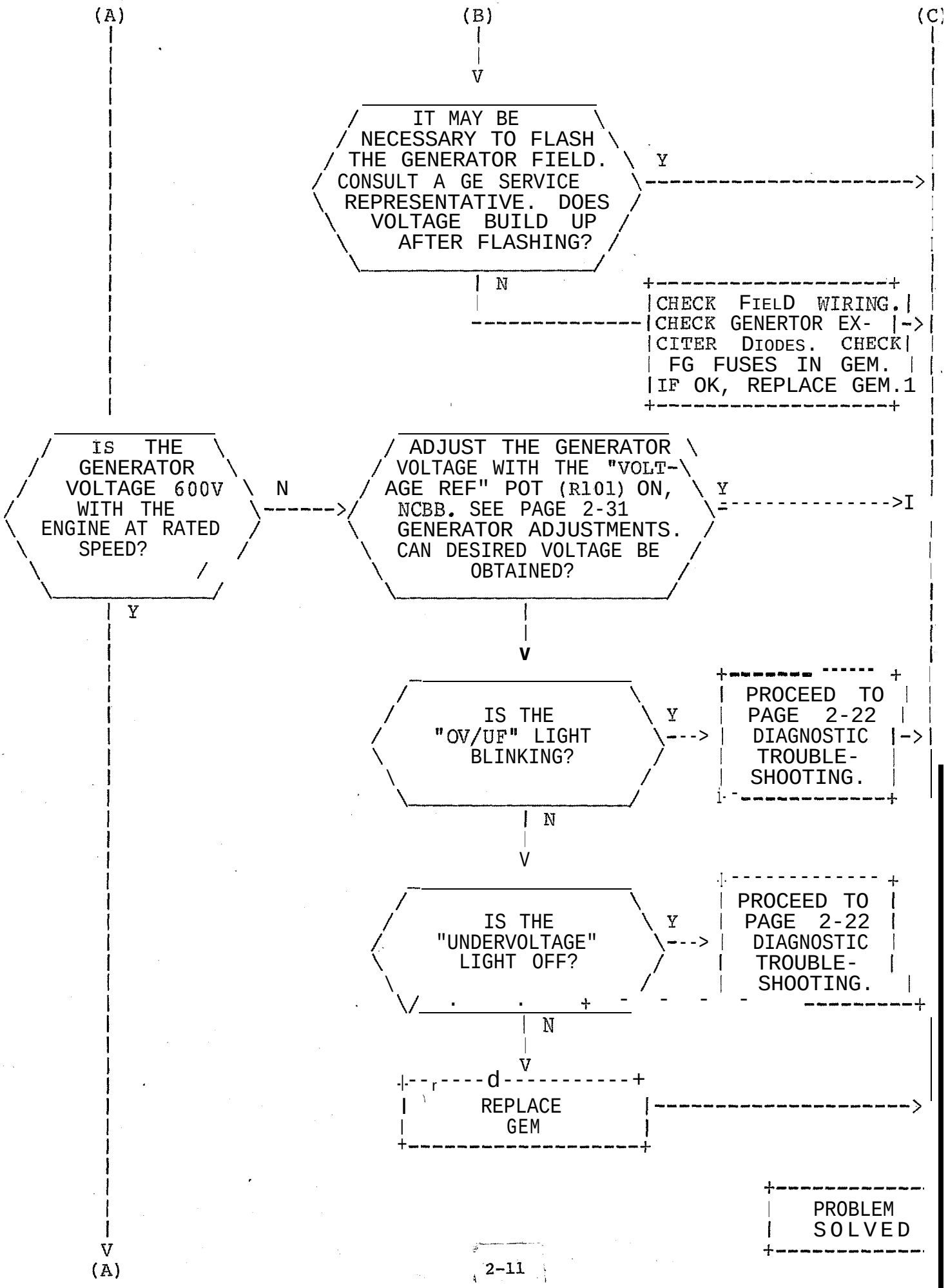


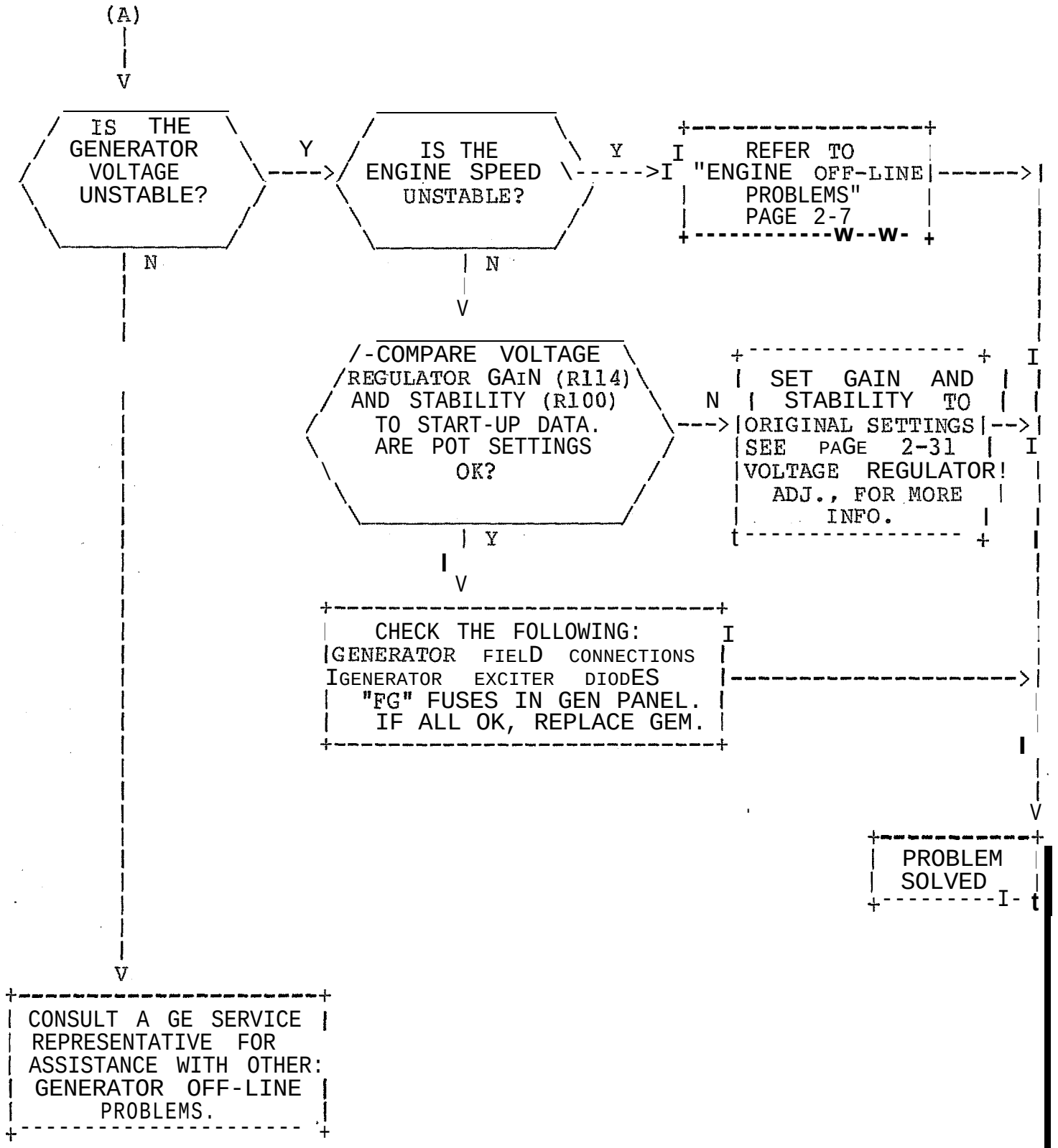

(A)
(B)
(C)
RUN SPEED CAN BE TUNED WITH THE FREQUENCY TRIM. ADDITIONAL RUN SPEED ADJUSTMENT CAN BE MADE WITH R201 ON NCBB. (SEE PAGE 2-31, GEM ENGINE ADJUSTMENT.) CAN DESIRED SPEED BE OBTAINED?
Y
N
COMPARE THUMBWHEEL SWITCH 6 ON NCBB TO START UP DATA. IF OK, REPLACE GEM.
PROBLEM SOLVED
IS THE ENGINE UNSTABLE? (HUNTING?)
Y
N
IS ENGINE WARM?
N
Y
ALLOW ENGINE TO WARM UP
ARE ENGINE FILTERS CLEAN?
N
Y
CLEAN FILTERS
IS ENGINE FUEL FREE FROM WATER OR OTHER CONTAMINANTS?
N
Y
REPLACE CONTAMINATED FUEL
(A)
(B)
(C)


(A)
(B)
(C)
ARE THE GOVERNOR GAIN (R104) AND GOVERNOR STABILITY (R103) POTS SET CORRECTLY?
N
SET POTS R104 AND R105 TO POSITIONS RECORDED AT START-UP. SEE PAGE 2-31 FOR MORE INFO.
Y
CHECK CONDITION OF MAGNETIC PICK-UP. IS IT CHIPPED, CRACKED OR HAVE LOOSE CONNECTIONS?
Y
REPLACE MAG PICK-UP
N
PROBLEM SOLVED
CHECK CONDITION OF MAG PICK-UP AND ACTUATOR WIRING. WITH LEADS REMOVED FROM GEM MAGNETIC PICK-UP IMPEDANCE SHOULD BE BETWEEN 100-300 OHMS. ACTUATOR IMPEDANCE SHOULD BE BETWEEN 30-40 OHMS. ALSO CHECK THAT SHIELDS ARE FIRMLY GROUND-ED AT THE PANEL AND NOT GROUND-AT THE ENGINE. IS WIRING OK?
N
REPAIR WIRING
Y
CHECK ACTUATOR AND LINKAGE FOR BINDING OR EXCESS-IVE LOOSENESS. IS IT OK?
N
REPLACE ACTUATOR
Y
REPLACE GEM
PROBLEM SOLVED
(A)

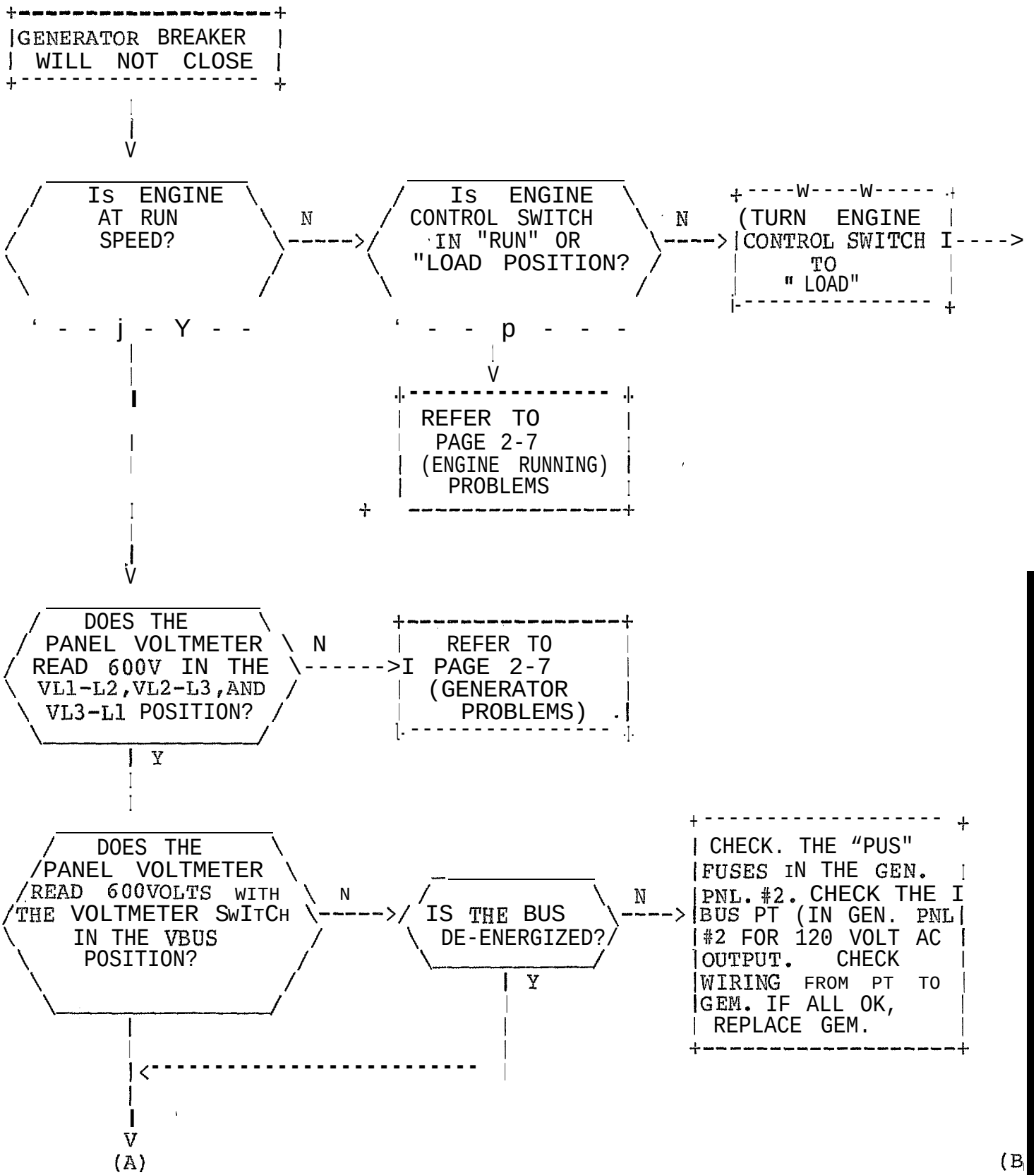

(A)
IS THE "NO OVERSPEED" LIGHT BLINKING?
Y
PROCEED To PAGE 2-22 DIAGNOSTIC TROUBLESHOOTING
IS THE "NO TACH LOSS" LIGHT BLINKING?
Y
IS THE "NO OV/UF" LIGHT BLINKING?
Y
N
FOR OTHER OFF-LINE ENGINE PROBLEMS, CONSULT A G.E. SERVICE REPRESENTATIVE.

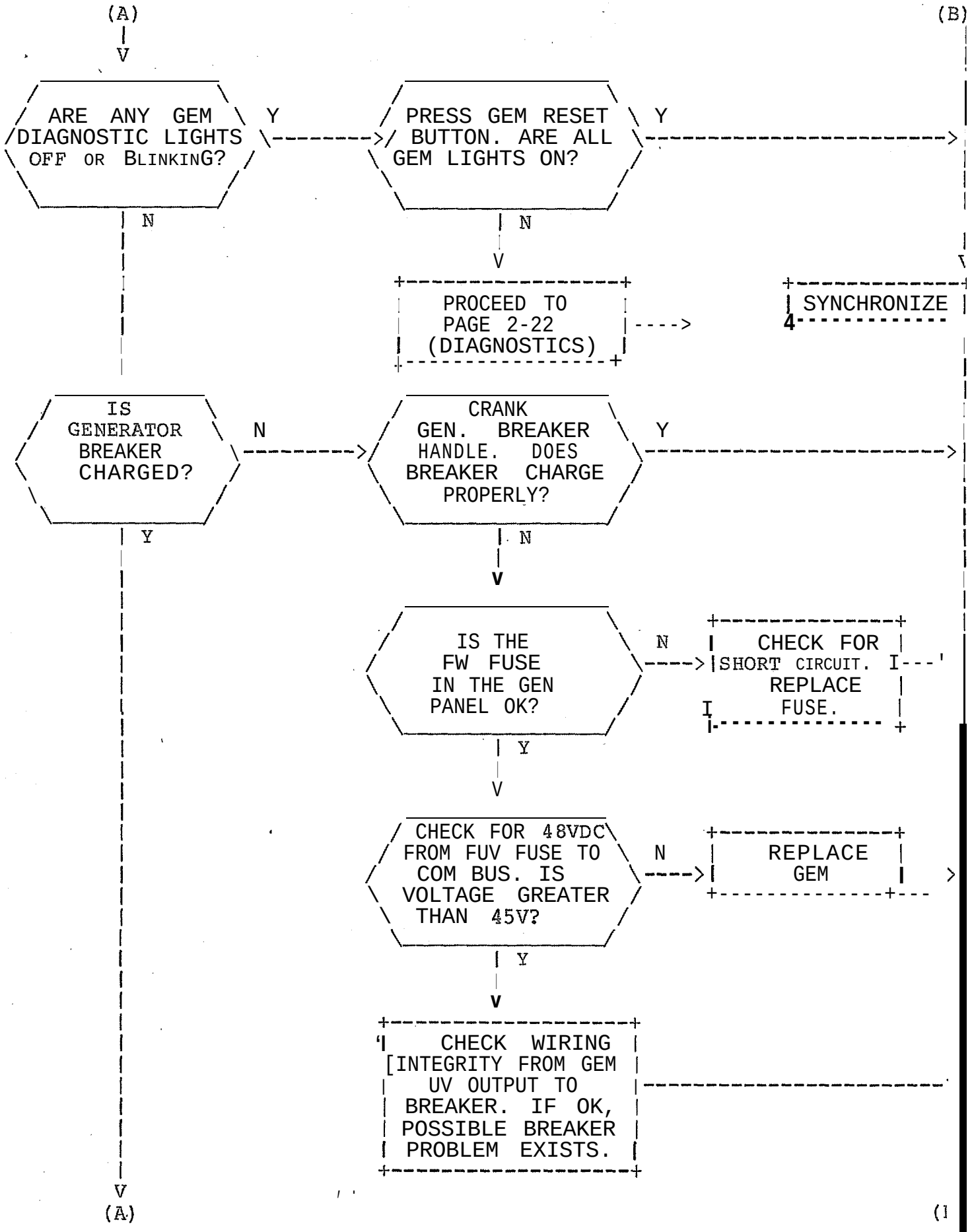

GENERATOR OFF-LINE PROBLEMS
WITH THE ENGINE IDLING IS THERE LESS THAN 100 VOLTS ON THE GENERATOR?
Y
TURN ENGINE CONTROL SWITCH TO RUN, DOES VOLTAGE EUILD UP TO 600V?
CONDITION IS OK. IF ENGINE IS TURNED BACK TO IDLE, GENERATOR VOLTS WILL BE BETWEEN 300-400v.
(A)
(B)
(C

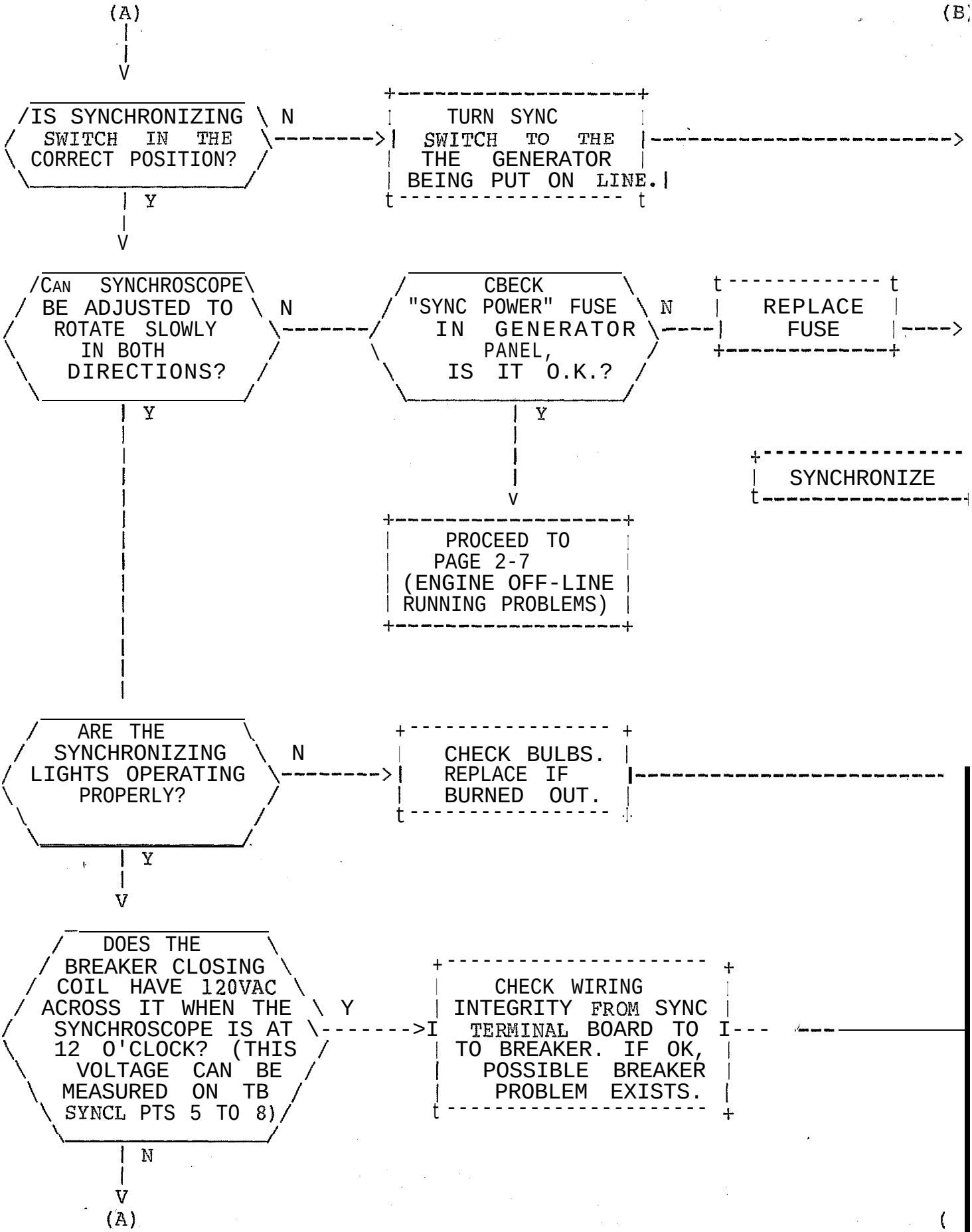

(A)
(B)
(C)
IT MAY BE NECESSARY TO FLASH THE GENERATOR FIELD. CONSULT A GE SERVICE REPRESENTATIVE. DOES VOLTAGE BUILD UP AFTER FLASHING?
Y
N
CHECK FIELD WIRING. CHECK GENERTOR EXCITER DIODES. CHECK FG FUSES IN GEM. IF OK, REPLACE GEM.1
IS THE GENERATOR VOLTAGE 600V WITH THE ENGINE AT RATED SPEED?
N
Y
ADJUST THE GENERATOR VOLTAGE WITH THE "VOLTAGE REF" POT (R101) ON, NCBB. SEE PAGE 2-31 GENERATOR ADJUSTMENTS. CAN DESIRED VOLTAGE BE OBTAINED?
Y
I
IS THE "OV/UF" LIGHT BLINKING?
Y
PROCEED TO PAGE 2-22 DIAGNOSTIC TROUBLE-SHOOTING.
N
IS THE "UNDERVOLTAGE" LIGHT OFF?
Y
PROCEED TO PAGE 2-22 DIAGNOSTIC TROUBLE-SHOOTING.
N
REPLACE GEM
PROBLEM SOLVED
V
(A)

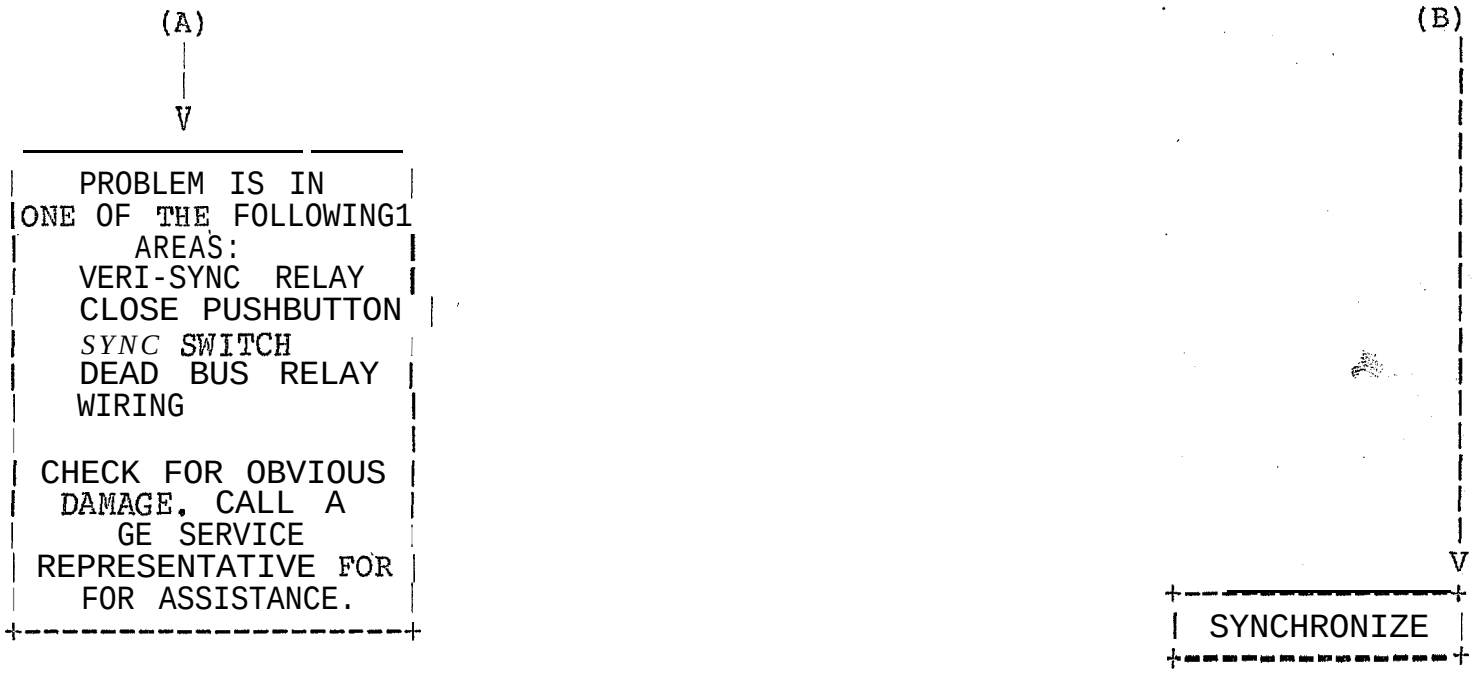

(A)
IS THE GENERATOR VOLTAGE UNSTABLE?
Y
IS THE ENGINE SPEED UNSTABLE?
Y
REFER TO "ENGINE OFF-LINE PROBLEMS" PAGE 2-7
N
N
COMPARE VOLTAGE REGULATOR GAIN (R114) AND STABILITY (R100) TO START-UP DATA. ARE POT SETTINGS OK?
N
SET GAIN AND STABILITY TO ORIGINAL SETTINGS SEE PAGE 2-31 VOLTAGE REGULATOR ADJ., FOR MORE INFO.
Y
CHECK THE FOLLOWING:
GENERATOR FIELD CONNECTIONS
GENERATOR EXCITER DIODES
"FG" FUSES IN GEN PANEL.
IF ALL OK, REPLACE GEM.
PROBLEM SOLVED
CONSULT A GE SERVICE REPRESENTATIVE FOR ASSISTANCE WITH OTHER: GENERATOR OFF-LINE PROBLEMS.

SYNCHRONIZING PROBLEMS

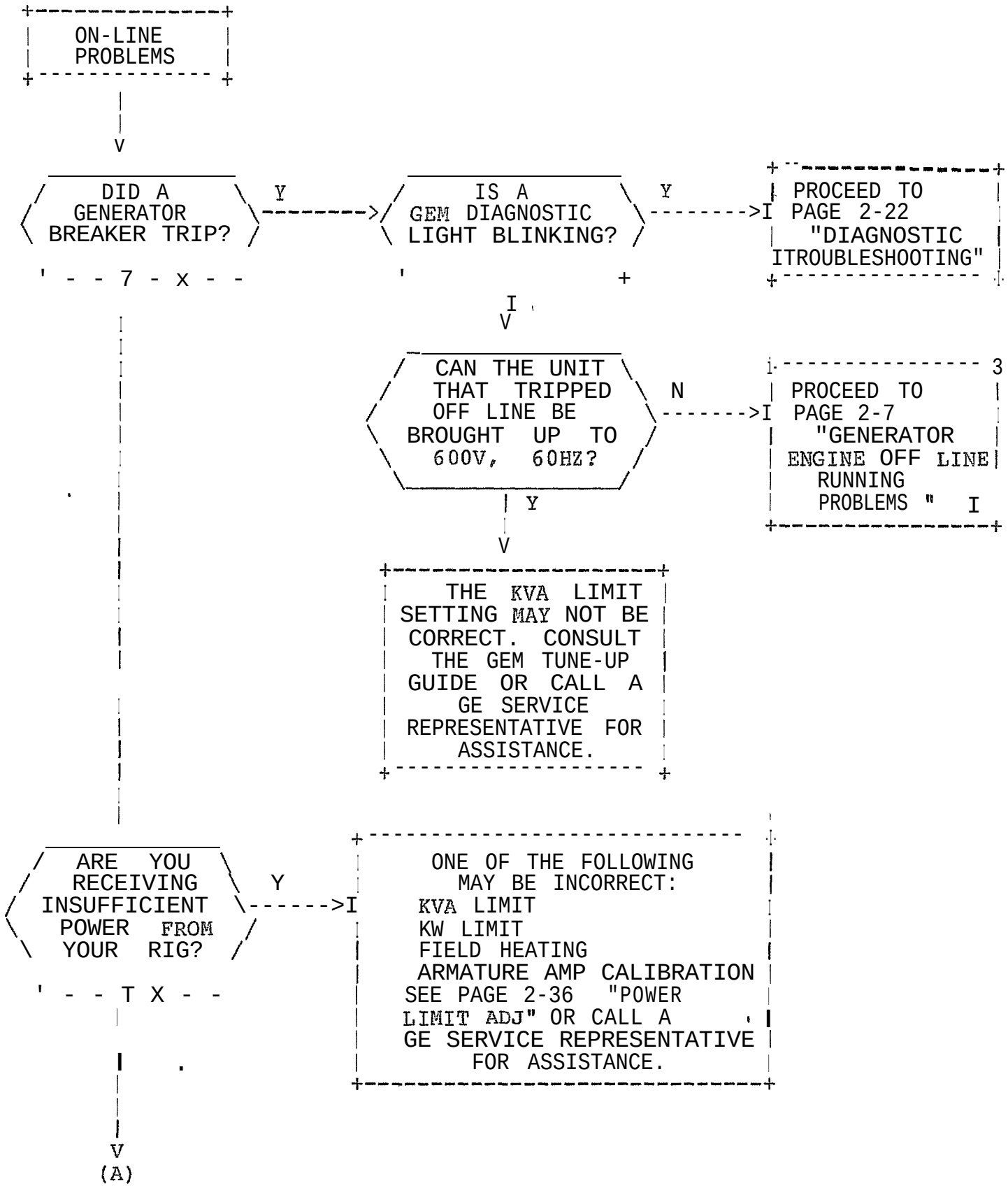


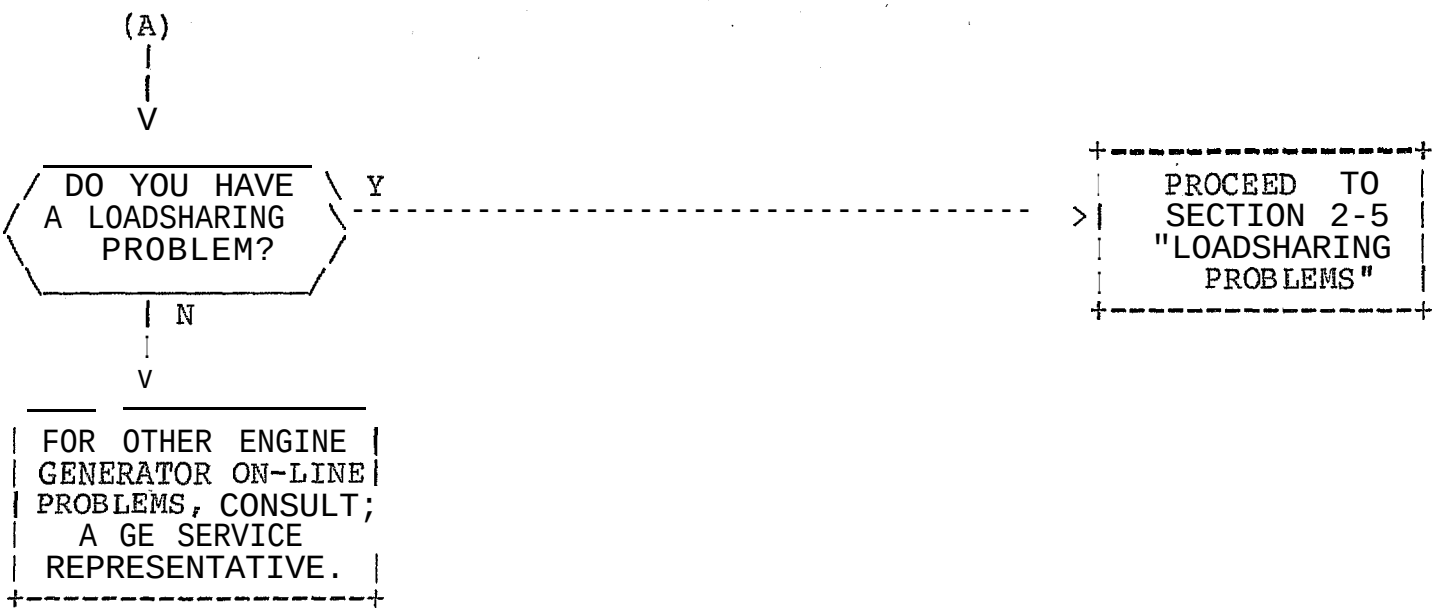

(A)
(B)
ARE ANY GEM DIAGNOSTIC LIGHTS OFF OR BLINKING?
Y
N
PRESS GEM RESET BUTTON. ARE ALL GEM LIGHTS ON?
Y
N
PROCEED TO PAGE 2-22 (DIAGNOSTICS)
SYNCHRONIZE
4
IS GENERATOR BREAKER CHARGED?
N
Y
CRANK GEN. BREAKER HANDLE. DOES BREAKER CHARGE PROPERLY?
Y
N
IS THE FW FUSE IN THE GEN PANEL OK?
N
Y
CHECK FOR SHORT CIRCUIT. REPLACE FUSE.
CHECK FOR 48VDC FROM FUV FUSE TO COM BUS. IS VOLTAGE GREATER THAN 45V?
N
Y
REPLACE GEM
CHECK WIRING INTEGRITY FROM GEM UV OUTPUT TO BREAKER. IF OK, POSSIBLE BREAKER PROBLEM EXISTS.
(A)
(I


(A)
(B)
IS SYNCHRONIZING SWITCH IN THE CORRECT POSITION?
N
TURN SYNC SWITCH TO THE THE GENERATOR BEING PUT ON LINE.
Y
CAN SYNCHROSCOPE BE ADJUSTED TO ROTATE SLOWLY IN BOTH DIRECTIONS?
N
CBECK "SYNC POWER" FUSE IN GENERATOR PANEL, IS IT O.K.?
N
REPLACE FUSE
Y
SYNCHRONIZE
PROCEED TO PAGE 2-7 (ENGINE OFF-LINE RUNNING PROBLEMS)
ARE THE SYNCHRONIZING LIGHTS OPERATING PROPERLY?
N
CHECK BULBS. REPLACE IF BURNED OUT.
Y
DOES THE BREAKER CLOSING COIL HAVE 120VAC ACROSS IT WHEN THE SYNCHROSCOPE IS AT 12 O'CLOCK? (THIS VOLTAGE CAN BE MEASURED ON TB SYNCL PTS 5 TO 8)
Y
CHECK WIRING INTEGRITY FROM SYNC TERMINAL BOARD TO TO BREAKER. IF OK, POSSIBLE BREAKER PROBLEM EXISTS.
N
(A)

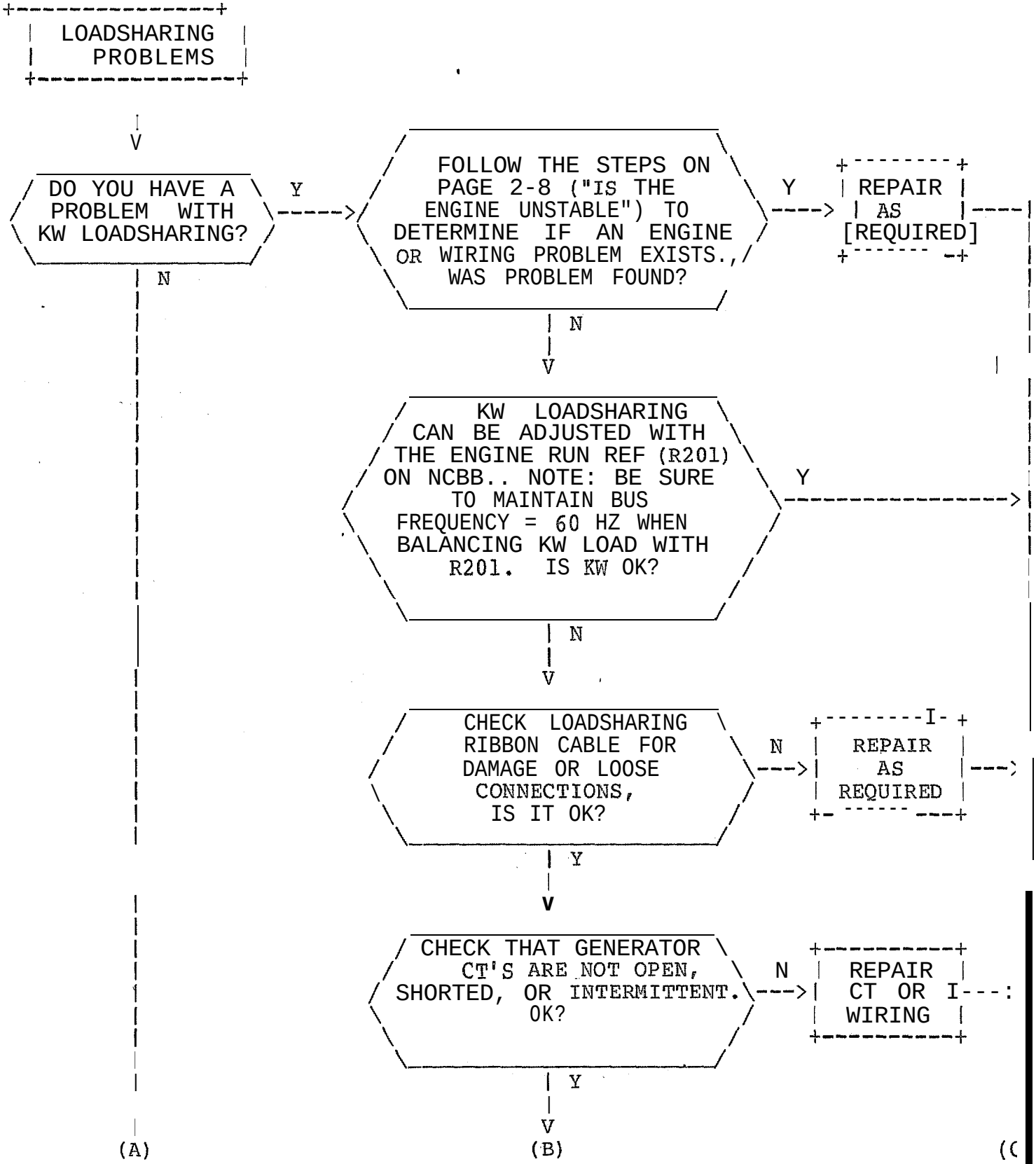

(A)
V
PROBLEM IS IN ONE OF THE FOLLOWING1 AREAS:
VERI-SYNC RELAY
CLOSE PUSHBUTTON
SYNC SWITCH
DEAD BUS RELAY
WIRING
CHECK FOR OBVIOUS DAMAGE. CALL A GE SERVICE REPRESENTATIVE FOR FOR ASSISTANCE.
(B)
V
SYNCHRONIZE

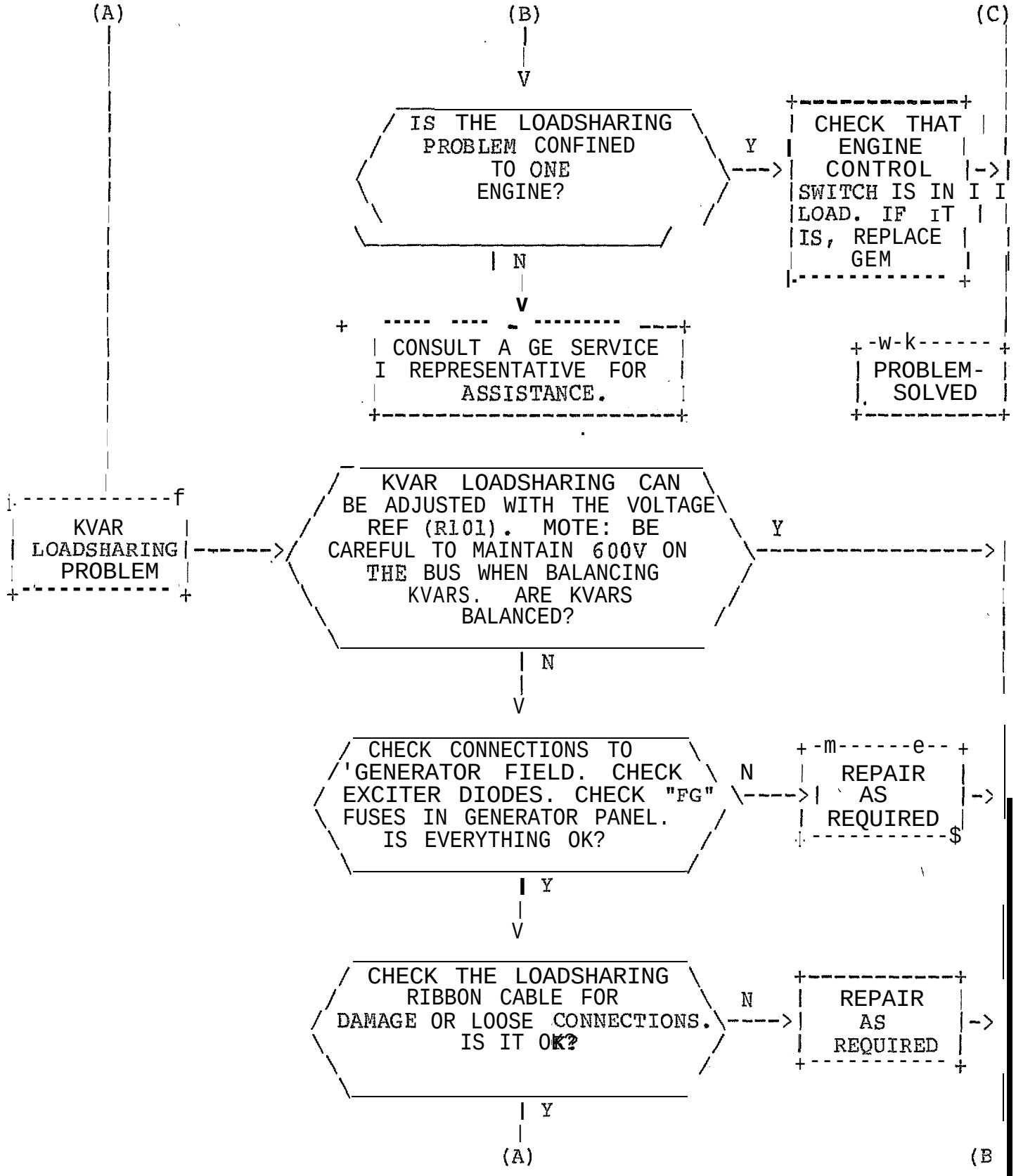
ENGINE-GENERATOR ON-LINE PROBLEMS

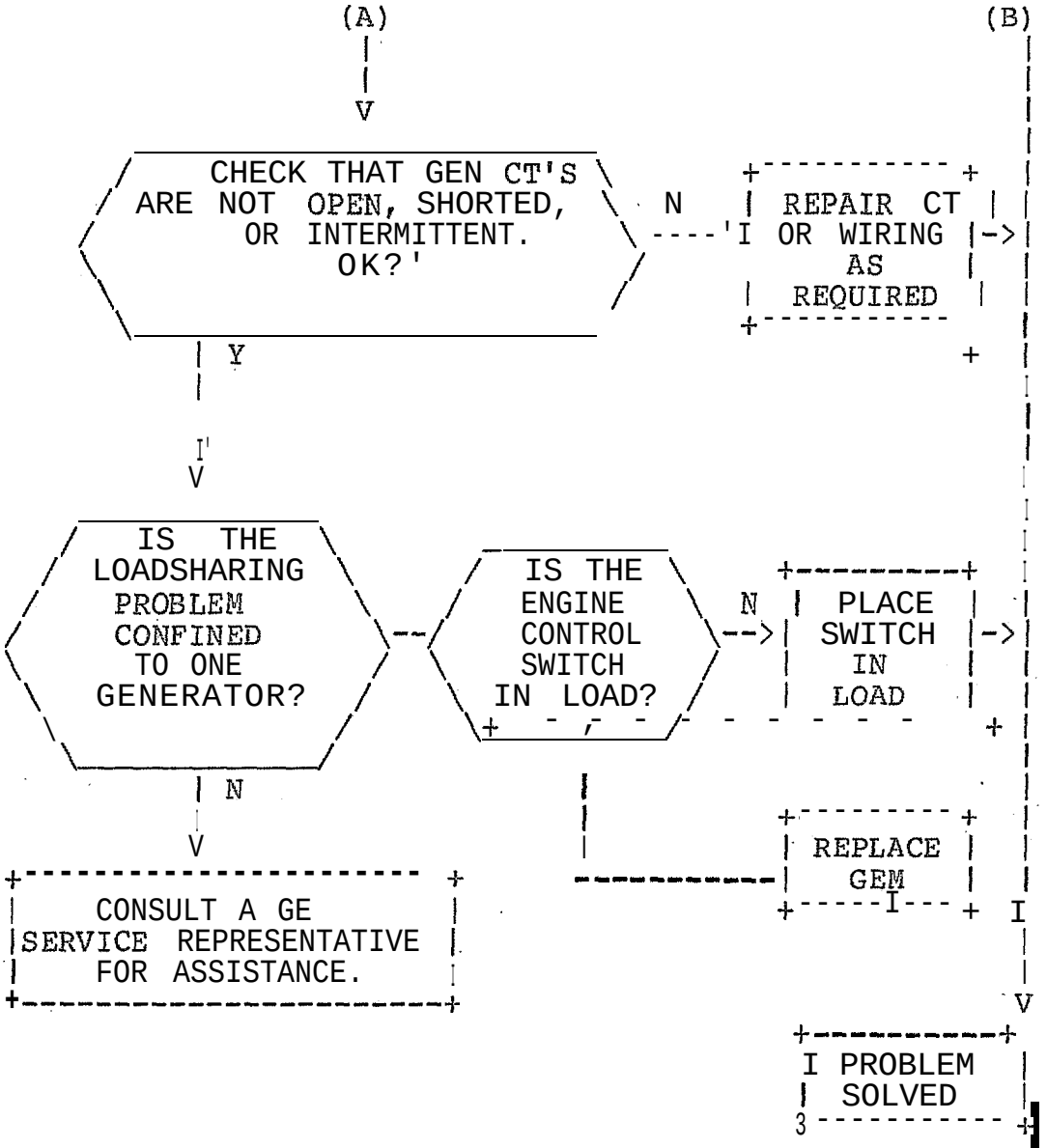


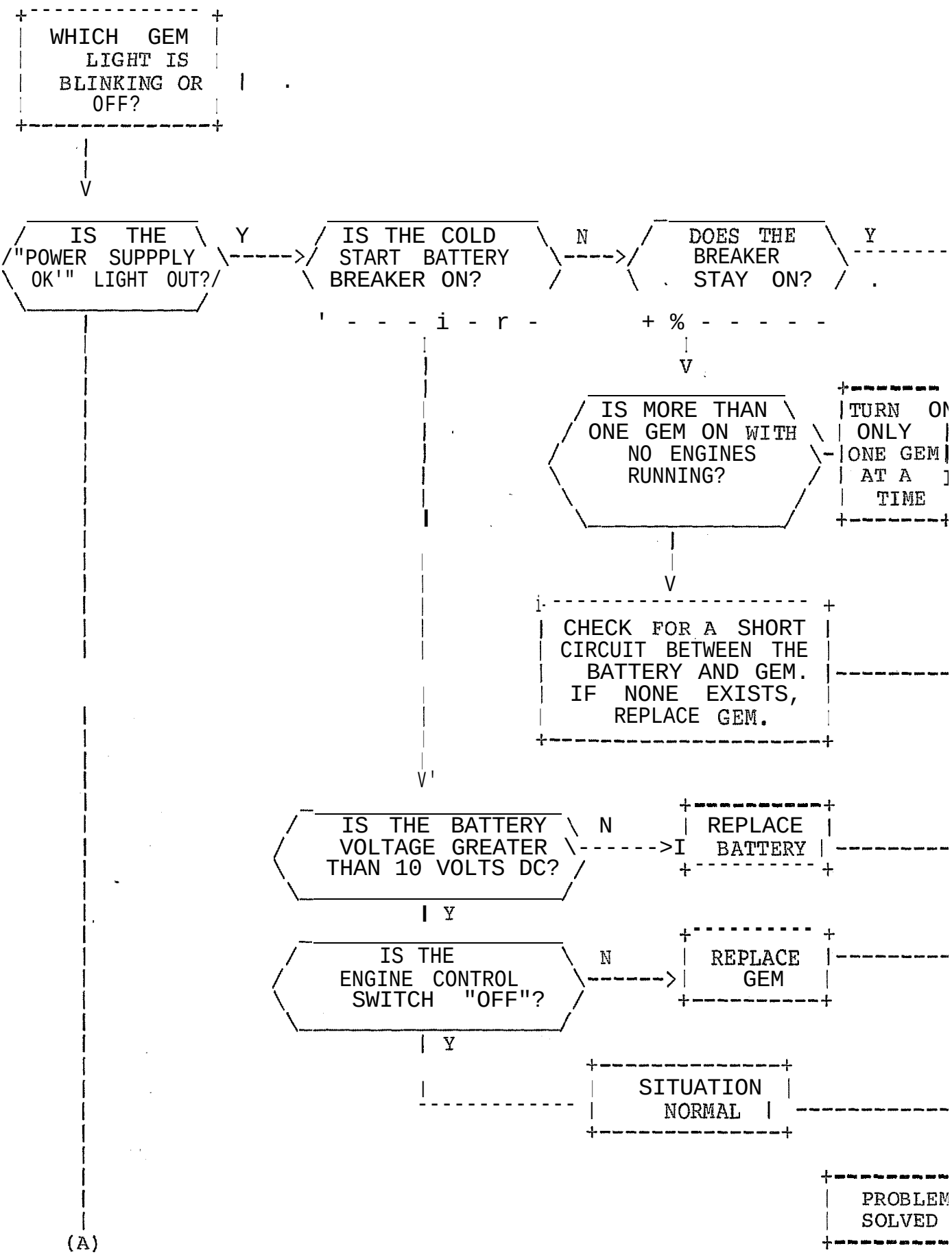

(A)
DO YOU HAVE A LOADSHARING PROBLEM?
Y
PROCEED TO SECTION 2-5 "LOADSHARING PROBLEMS"
N
FOR OTHER ENGINE GENERATOR ON-LINE PROBLEMS, CONSULT; A GE SERVICE REPRESENTATIVE.

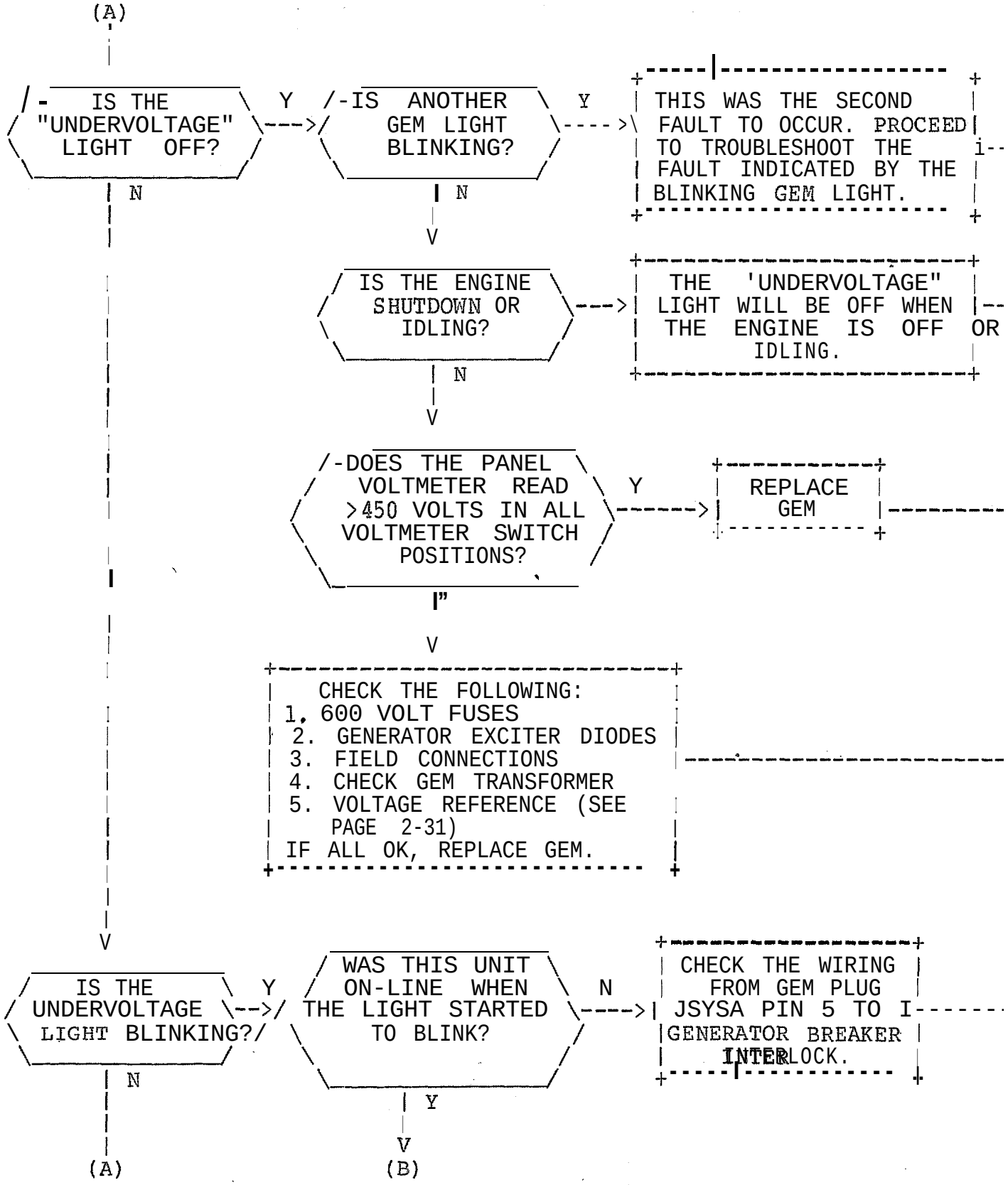
LOADSHARING PROBLEMS

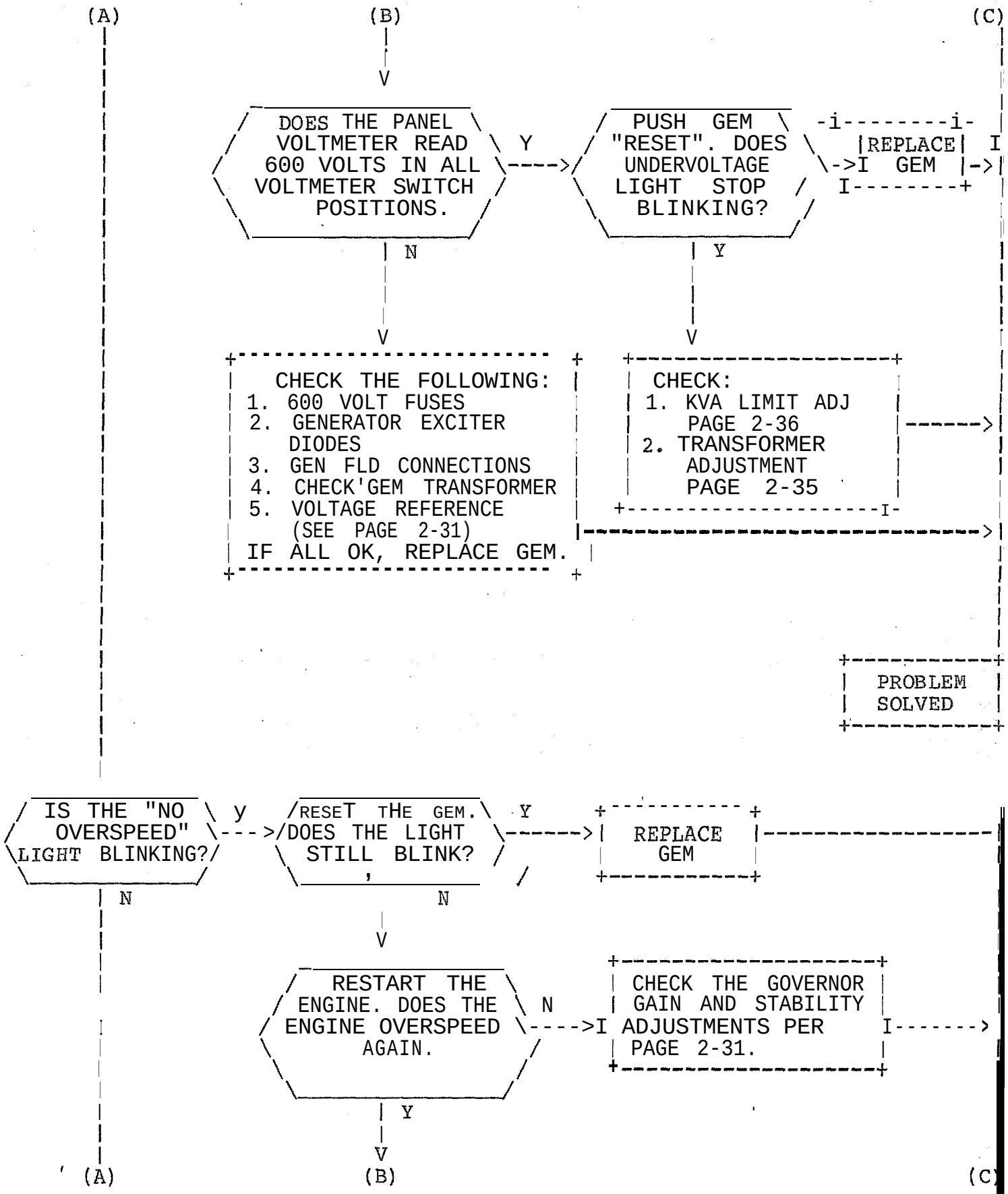


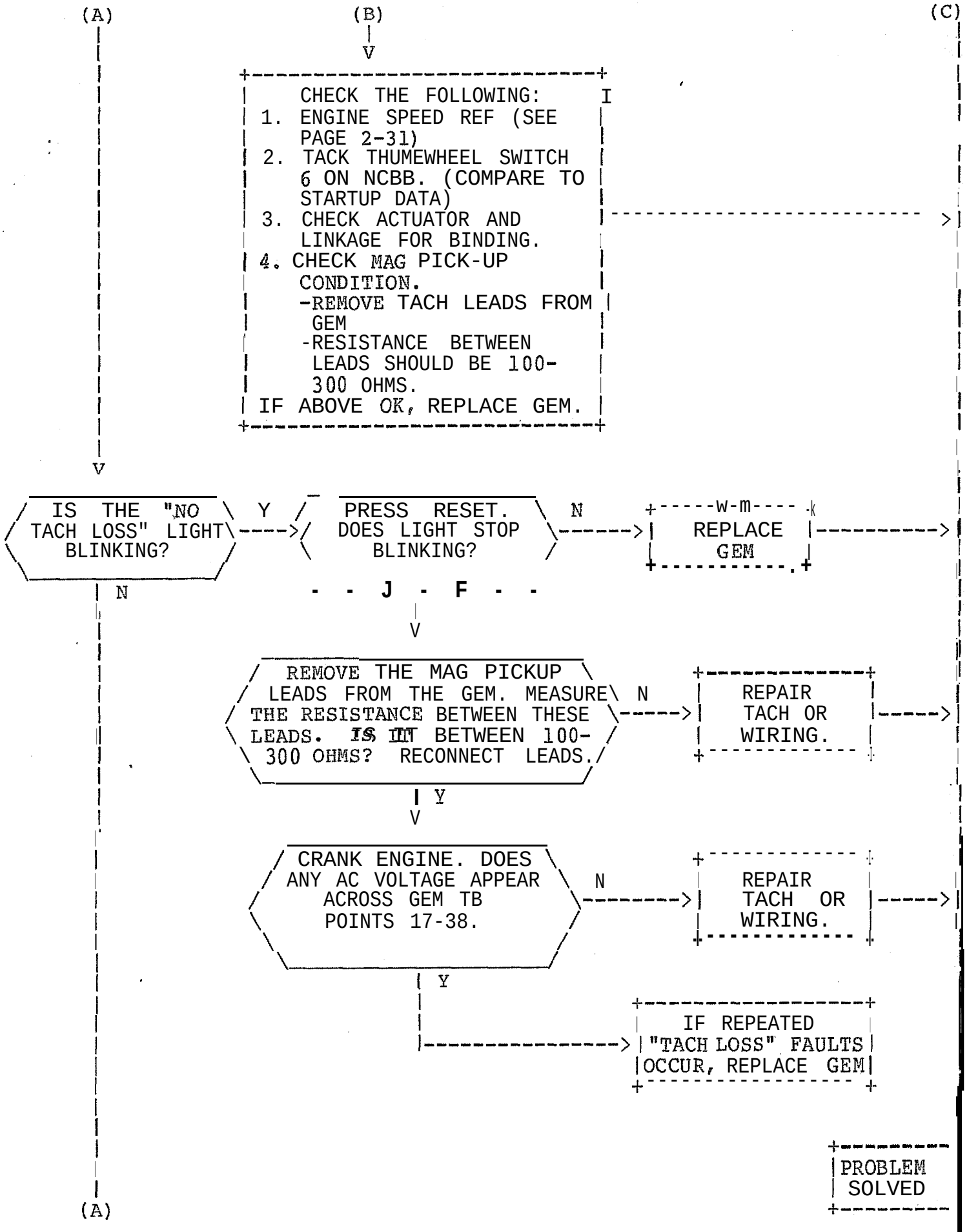

(A)
(B)
(C)
IS THE LOADSHARING PROBLEM CONFINED TO ONE ENGINE?
Y
CHECK THAT ENGINE CONTROL SWITCH IS IN I I LOAD. IF IT IS, REPLACE GEM
N
CONSULT A GE SERVICE I REPRESENTATIVE FOR ASSISTANCE.
PROBLEM- SOLVED
KVAR LOADSHARING PROBLEM
KVAR LOADSHARING CAN BE ADJUSTED WITH THE VOLTAGE REF (R101). MOTE: BE CAREFUL TO MAINTAIN 600V ON THE BUS WHEN BALANCING KVARS. ARE KVARS BALANCED?
Y
N
CHECK CONNECTIONS TO 'GENERATOR FIELD. CHECK EXCITER DIODES. CHECK "FG" FUSES IN GENERATOR PANEL. IS EVERYTHING OK?
N
REPAIR AS REQUIRED
Y
CHECK THE LOADSHARING RIBBON CABLE FOR DAMAGE OR LOOSE CONNECTIONS. IS IT OK?
N
REPAIR AS REQUIRED
Y
(A)
(B

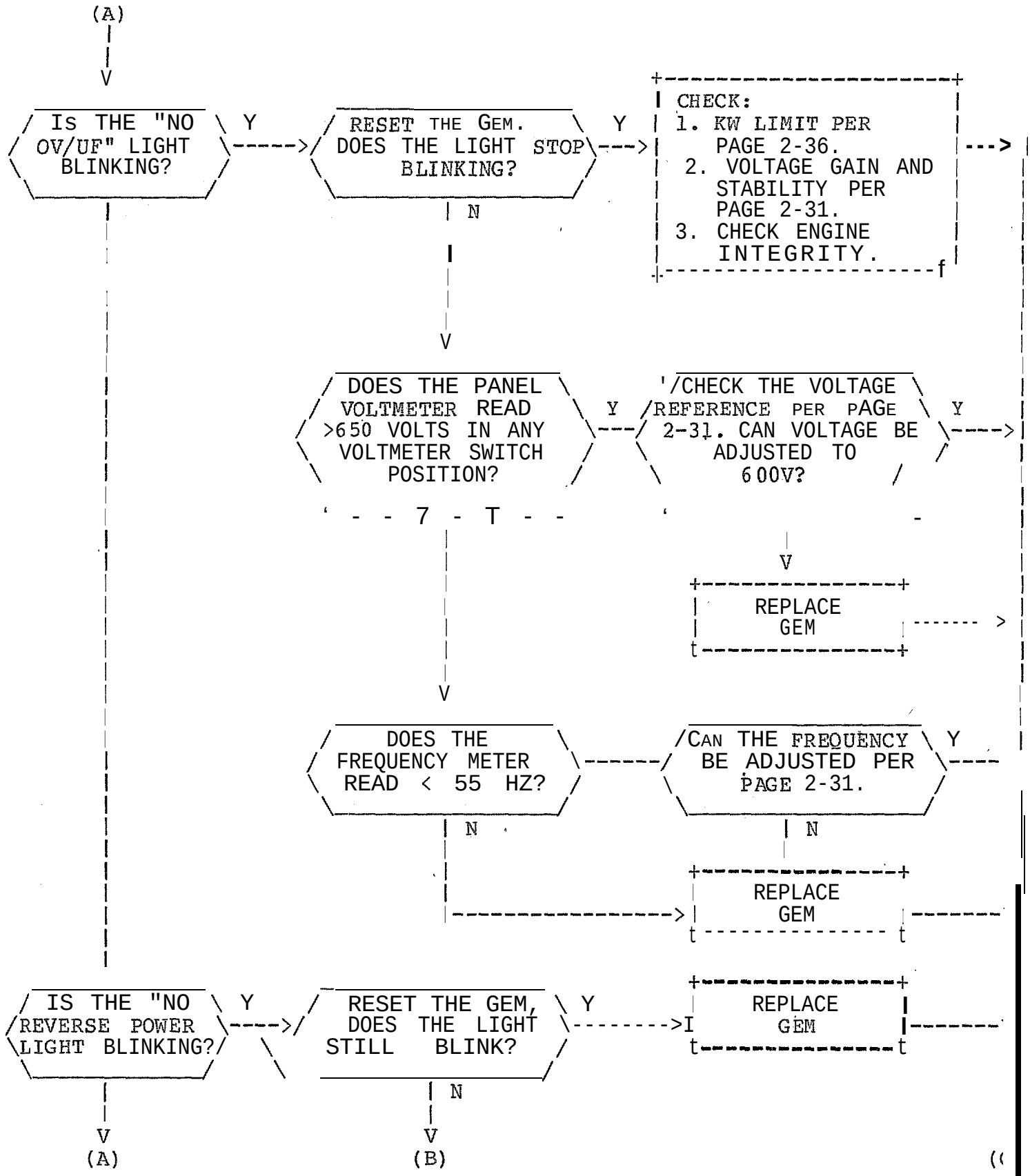

(A)
(B)
CHECK THAT GEN CT'S ARE NOT OPEN, SHORTED, OR INTERMITTENT. OK?
N
REPAIR CT OR WIRING AS REQUIRED
Y
IS THE LOADSHARING PROBLEM CONFINED TO ONE GENERATOR?
IS THE ENGINE CONTROL SWITCH IN LOAD?
N
PLACE SWITCH IN LOAD
N
REPLACE GEM
CONSULT A GE SERVICE REPRESENTATIVE FOR ASSISTANCE.
PROBLEM SOLVED

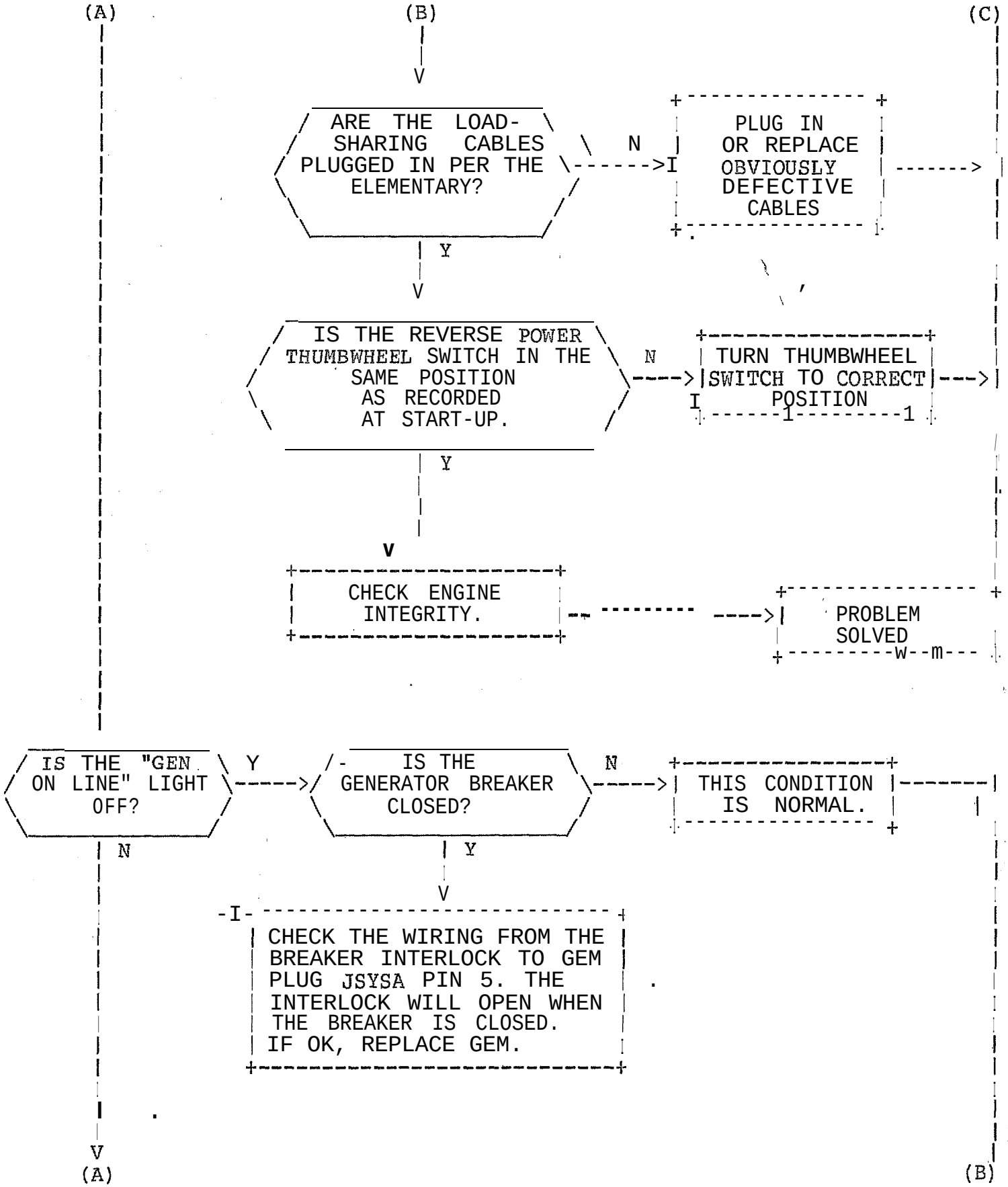

WHICH GEM LIGHT IS BLINKING OR OFF?
IS THE "POWER SUPPPLY OK'" LIGHT OUT?
Y
IS THE COLD START BATTERY BREAKER ON?
N
DOES THE BREAKER STAY ON?
Y
IS MORE THAN ONE GEM ON WITH NO ENGINES RUNNING?
TURN ON ONLY ONE GEM AT A TIME
CHECK FOR A SHORT CIRCUIT BETWEEN THE BATTERY AND GEM. IF NONE EXISTS, REPLACE GEM.
IS THE BATTERY VOLTAGE GREATER THAN 10 VOLTS DC?
N
REPLACE BATTERY
Y
IS THE ENGINE CONTROL SWITCH "OFF"?
N
REPLACE GEM
Y
SITUATION NORMAL
PROBLEM SOLVED
(A)

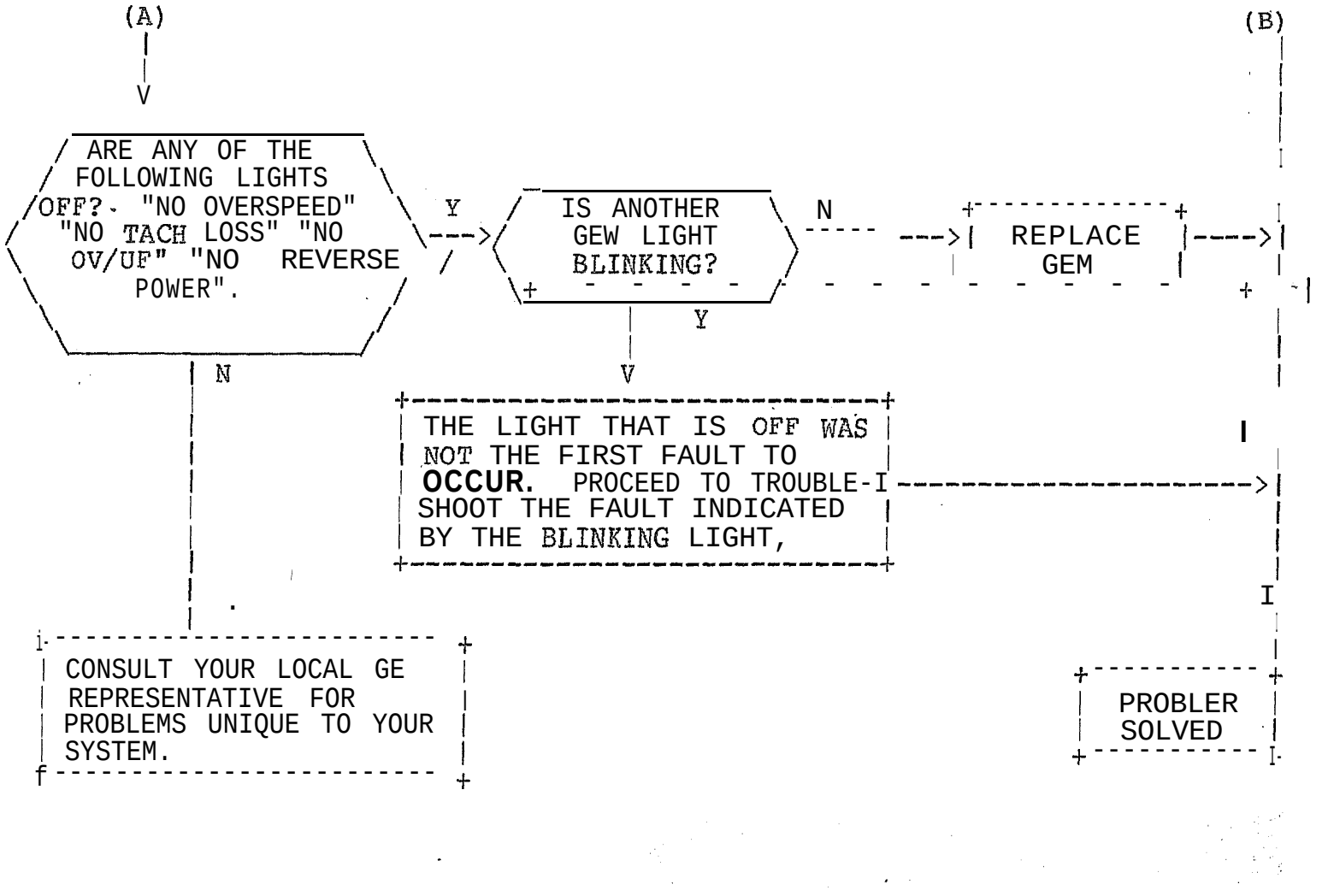

(A)
IS THE "UNDERVOLTAGE" LIGHT OFF?
Y
IS ANOTHER GEM LIGHT BLINKING?
Y
THIS WAS THE SECOND FAULT TO OCCUR. PROCEED TO TROUBLESHOOT THE FAULT INDICATED BY THE BLINKING GEM LIGHT.
N
N
IS THE ENGINE SHUTDOWN OR IDLING?
THE 'UNDERVOLTAGE" LIGHT WILL BE OFF WHEN THE ENGINE IS OFF OR IDLING.
N
DOES THE PANEL VOLTMETER READ >450 VOLTS IN ALL VOLTMETER SWITCH POSITIONS?
Y
REPLACE GEM
CHECK THE FOLLOWING:
1, 600 VOLT FUSES
2. GENERATOR EXCITER DIODES
3. FIELD CONNECTIONS
4. CHECK GEM TRANSFORMER
5. VOLTAGE REFERENCE (SEE PAGE 2-31)
IF ALL OK, REPLACE GEM.
IS THE UNDERVOLTAGE LIGHT BLINKING?
Y
WAS THIS UNIT ON-LINE WHEN THE LIGHT STARTED TO BLINK?
N
CHECK THE WIRING FROM GEM PLUG JSYSA PIN 5 TO I GENERATOR BREAKER INTERLOCK.
N
Y
(A)
(B)


(A)
(B)
(C)
DOES THE PANEL VOLTMETER READ 600 VOLTS IN ALL VOLTMETER SWITCH POSITIONS.
Y
PUSH GEM "RESET". DOES UNDERVOLTAGE LIGHT STOP BLINKING?
REPLACE GEM
N
Y
CHECK THE FOLLOWING:
1. 600 VOLT FUSES
2. GENERATOR EXCITER DIODES
3. GEN FLD CONNECTIONS
4. CHECK'GEM TRANSFORMER
5. VOLTAGE REFERENCE (SEE PAGE 2-31)
IF ALL OK, REPLACE GEM.
CHECK:
1. KVA LIMIT ADJ PAGE 2-36
2. TRANSFORMER ADJUSTMENT PAGE 2-35
PROBLEM SOLVED
IS THE "NO OVERSPEED" LIGHT BLINKING?
y
RESET THE GEM. DOES THE LIGHT STILL BLINK?
Y
REPLACE GEM
N
N
RESTART THE ENGINE. DOES THE ENGINE OVERSPEED AGAIN.
N
CHECK THE GOVERNOR GAIN AND STABILITY ADJUSTMENTS PER PAGE 2-31.
Y
(A)
(B)
(C)

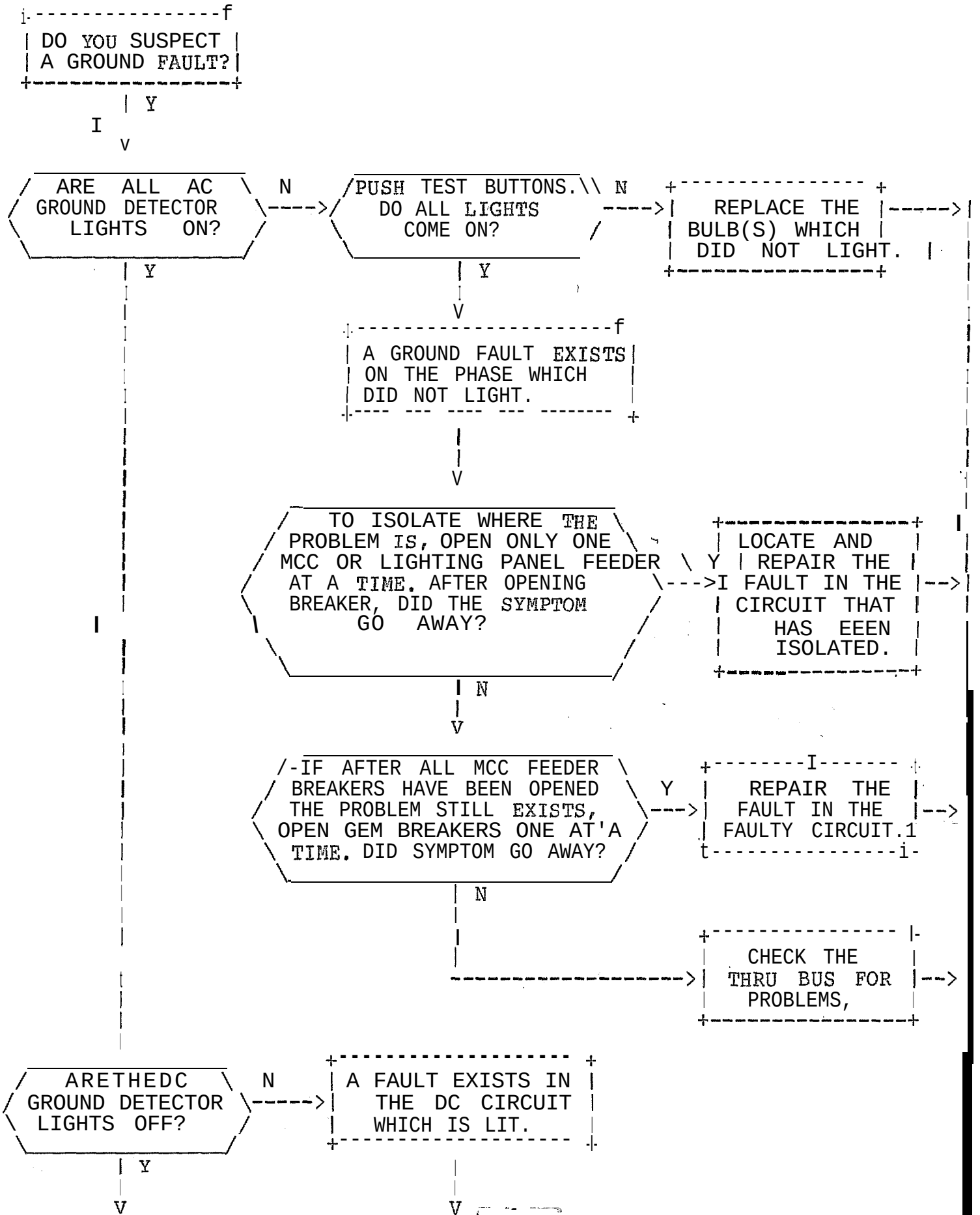

(A)
(B)
(C)
CHECK THE FOLLOWING:
1. ENGINE SPEED REF (SEE PAGE 2-31)
2. TACK THUMEWHEEL SWITCH 6 ON NCBB. (COMPARE TO STARTUP DATA)
3. CHECK ACTUATOR AND LINKAGE FOR BINDING.
4. CHECK MAG PICK-UP CONDITION.
-REMOVE TACH LEADS FROM GEM
-RESISTANCE BETWEEN LEADS SHOULD BE 100-300 OHMS.
IF ABOVE OK, REPLACE GEM.
I
IS THE "NO TACH LOSS" LIGHT BLINKING?
Y
PRESS RESET. DOES LIGHT STOP BLINKING?
N
REPLACE GEM
N
- - J - F - -
REMOVE THE MAG PICKUP LEADS FROM THE GEM. MEASURE THE RESISTANCE BETWEEN THESE LEADS. IS IT BETWEEN 100-300 OHMS? RECONNECT LEADS.
N
REPAIR TACH OR WIRING.
Y
CRANK ENGINE. DOES ANY AC VOLTAGE APPEAR ACROSS GEM TB POINTS 17-38.
N
REPAIR TACH OR WIRING.
Y
IF REPEATED "TACH LOSS" FAULTS OCCUR, REPLACE GEM
PROBLEM SOLVED
(A)

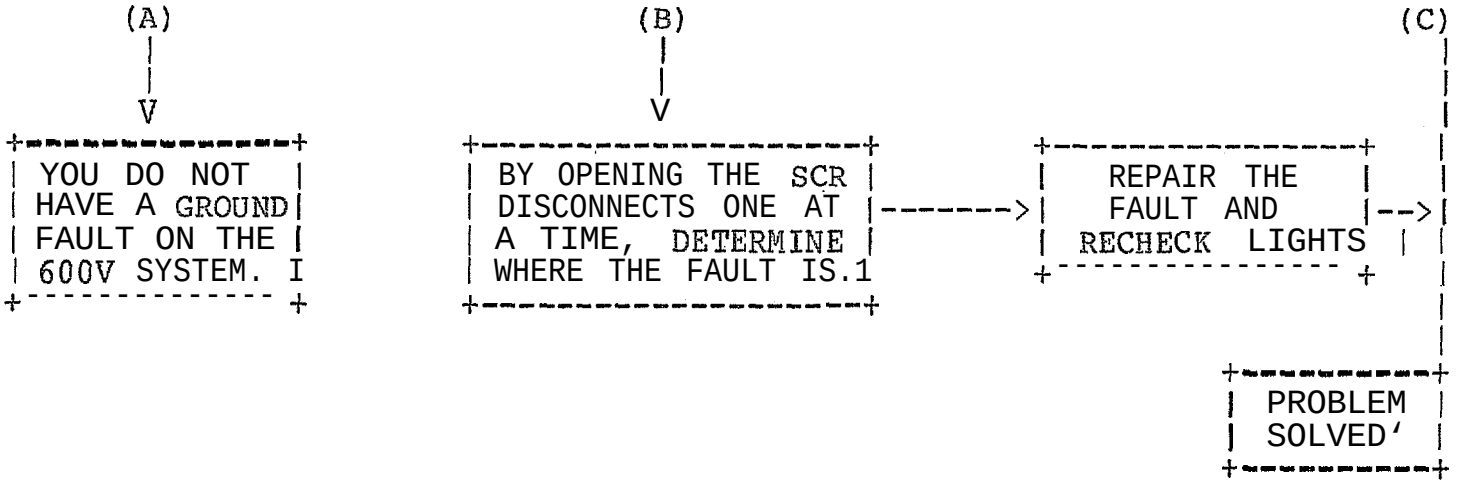

(A)
Is THE "NO OV/UF" LIGHT BLINKING?
Y
RESET THE GEM. DOES THE LIGHT STOP BLINKING?
Y
N
CHECK:
1. KW LIMIT PER PAGE 2-36.
2. VOLTAGE GAIN AND STABILITY PER PAGE 2-31.
3. CHECK ENGINE INTEGRITY.
DOES THE PANEL VOLTMETER READ >650 VOLTS IN ANY VOLTMETER SWITCH POSITION?
Y
CHECK THE VOLTAGE REFERENCE PER PAGE 2-31. CAN VOLTAGE BE ADJUSTED TO 600V?
Y
REPLACE GEM
DOES THE FREQUENCY METER READ < 55 HZ?
N
CAN THE FREQUENCY BE ADJUSTED PER PAGE 2-31.
Y
N
REPLACE GEM
IS THE "NO REVERSE POWER LIGHT BLINKING?
Y
RESET THE GEM, DOES THE LIGHT STILL BLINK?
Y
N
REPLACE GEM
V
(A)
V
(B)


(A)
(B)
(C)
ARE THE LOAD-SHARING CABLES PLUGGED IN PER THE ELEMENTARY?
N
PLUG IN OR REPLACE OBVIOUSLY DEFECTIVE CABLES
Y
IS THE REVERSE POWER THUMBWHEEL SWITCH IN THE SAME POSITION AS RECORDED AT START-UP.
N
TURN THUMBWHEEL SWITCH TO CORRECT POSITION
Y
CHECK ENGINE INTEGRITY.
PROBLEM SOLVED
IS THE "GEN ON LINE" LIGHT OFF?
Y
IS THE GENERATOR BREAKER CLOSED?
N
THIS CONDITION IS NORMAL.
N
Y
CHECK THE WIRING FROM THE BREAKER INTERLOCK TO GEM PLUG JSYSA PIN 5. THE INTERLOCK WILL OPEN WHEN THE BREAKER IS CLOSED. IF OK, REPLACE GEM.
(A)
(B)


(A)
(B)
ARE ANY OF THE FOLLOWING LIGHTS OFF? "NO OVERSPEED" "NO TACH LOSS" "NO OV/UF" "NO REVERSE POWER".
Y
IS ANOTHER GEW LIGHT BLINKING?
N
REPLACE GEM
Y
N
THE LIGHT THAT IS OFF WAS NOT THE FIRST FAULT TO OCCUR. PROCEED TO TROUBLE-SHOOT THE FAULT INDICATED BY THE BLINKING LIGHT,
CONSULT YOUR LOCAL GE REPRESENTATIVE FOR PROBLEMS UNIQUE TO YOUR SYSTEM.
PROBLER SOLVED

AC GROUND FAULT DETECTION




(A)
(B)
(C)
YOU DO NOT HAVE A GROUND FAULT ON THE 600V SYSTEM. I
BY OPENING THE SCR DISCONNECTS ONE AT A TIME, DETERMINE WHERE THE FAULT IS.1
REPAIR THE FAULT AND RECHECK LIGHTS
PROBLEM SOLVED'

GEM ADJUSTMENTS

1. ENGINE ADJUSTMENTS

ALL GEM ADJUSTMENTS ARE LOCATED ON THE FRONT DOOR (DS3800NCBB) OF THE GEM. REFER TO FIGURE 1 TO HELP LOCATE A SPECIFIC ADJUSTMENT.

NOTE: CASUAL TUNE-UP OF THE GEM IS NOT REQUIRED. THIS SECTION IS INCLUDED ONLY TO HELP YOU SOLVE GEM PROBLEMS.

PLEASE KEEP IN MIND THAT REPLACEMENT MODULES CAN BE ADJUSTED BY SIMPLY COMPARING POTS AND SWITCHES TO THE ORIGINAL MODULE OR THE START-UP DATA.

(A) ----> INDICATES SWITCH OR POT REFERRED TO IN FOLLOWING ADJUSTMENT SECTION.

(A) IDLE SPEED - TURN ENGINE CONTROL SWITCH TO IDLE. ADJUST "IDLE REF" POT R102 FOR DESIRED SPEED,

(B) RUN SPEED (FREQUENCY) - TURN ENGINE CONTROL SWITCH TO RUN. GENERATOR BREAKER MUST BE OPEN. TURN FREQUENCY "TRIM POT'" TO MIDDLE POSITION, TURN "FREQUENCY SWITCH" TO DESIRED POSITION. ADJUST POT R105 SO THAT FREQUENCY METER READS 60 HZ;

(C) GOVERNOR STABILITY - ADJUST POT R103 TO POSITION 10. NO FURTHER ADJUSTMENT IS REQUIRED.

(D) GOVERNOR GAIN - TURN ENGINE CONTROL SWITCH TO RUN, GENERATOR BREAKER MUST BE OPEN. TURN POT R104 CLOCKWISE UNTIL ENGINE BEGINS TO GO UNSTABLE. NEXT, SLOWLY TURN POT R104 COUNTERCLOCKWISE JUST FAR ENOUGH SO THAT ENGINE IS STABLE. NOTE: ENGINE STABILITY CAN BE MONITORED BY WATCHING THE FREQUENCY METER, NEEDLE MOVEMENT SHOULD BE LESS THAN .1 HZ.

(E) TACH RANGE SWITCH #6 - SET SWITCH TO SAME POSITION AS RECORDED AT START-UP. NOTE: TURN ENGINE CONTROL SWITCH TO OFF WHEN MAKING THIS ADJUSTMENT.

(F) REVERSE POWER SWITCH #7 - THE REVERSE POWER SWITCH SETS THE LEVEL AND TIME DELAY REQUIRED TO TRIP THE GENERATOR BREAKER ON A REVERSE POWER FAULT. THE TYPICAL SETTING FOR THIS SWITCH IS POSITION "A", WHERE 10% OF REVERSE POWER WILL CAUSE GENERATOR BREAKER TO TRIP IN -3.5 SEC. NOTE: THE BREAKER WILL TRIP FASTER FOR THE SAME LEVEL OF REVERSE POWER AS THE THUMBWHEEL SWITCH IS MOVED TOWARD POSITION "0". IT WILL TAKE LONGER TO TRIP AS THE THUMBWHEEL SWITCH IS MOVED TOWARDS POSITION "F". NOTE: TURN ENGINE CONTROL SWITCH TO OFF WHEN MAKING THIS ADJUSTMENT.

(G) KW LOADSHARING - TURN THE ENGINE CONTROL SWITCH ON ALL ON-LINE UNITS TO "LOAD". ADJUST POT R105 TO BALANCE THE KW LOAD. BE CAREFUL TO MAINTAIN 60 HZ ON THE FREQUENCY METER WHEN MAKING THIS ADJUSTMENT.

2. GENERATOR ADJUSTMENTS

ALL GEM ADJUSTMENTS ARE LOCATED ON THE FRONT DOOR (DS3800NCBB) OF THE GEM. REFER TO FIGURE 1 TO HELP LOCATE A SPECIFIC ADJUSTMENT.

NOTE: CASUAL TUNE-UP OF THE GEM IS NOT REQUIRED. THIS SECTION IS INCLUDED OMLY TO HELP YOU SOLVE GEM PROBLEMS,

PLEASE KEEP IN MIND THAT REPLACEMENT MODULES CAN BE ADJUSTED BY SIMPLY COMPARING POTS AND SWITCHES TO THE ORIGINAL MODULE OR THE START-UP DATA.

(A) ----> INDICATES SWITCH OR POT REFERRED TO IN FOLLOWING ADJUSTMENT SECTION.

(A.) VOLTAGE REFERENCE - ENGINE MUST BE AT RATED SPEED. GENERATOR BREAKER MUST BE OPEN. ADJUST POT R101 FOR EXACTLY 600 VOLTS.

(B) VOLTAGE GAIN - ADJUST POT R114 TO THE VALUE RECORDED AT START-UP., TYPICAL SETTING = 40.

(C) VOLTAGE STABILITY - ADJUST POT R100 TO THE VALUE RECORDED AT START-UP, TYPICAL SETTING = 80.

(D) GENERATOR FIELD HEATING -

1. ENGINE MUST BE AT IDLE SPEED.
2. USE A DC VOLTMETER (ON THE 1 VOLT SCALE) TO MEASURE THE VOLTAGE DIRECTLY ACROSS R1 ON XPES.
3. DETERMINE THE GENERATOR FULL LOAD FIELD EXCITER CURRENT FROM THE GENERATOR NAMEPLATE.
4. CALCULATE:

$$VTP10 = \frac{52 \times \text{VOLTAGE ACROSS R1 ON XPES}}{\text{GENERATOR FULL LOAD FIELD CURRENT}}$$

5. OPEN THE GEM DOOR AND LOCATE TP10 ON THE HFPA CARD. MEASURE THE VOLTAGE AT TP10 WITH A DC VOLTMETER (10 VOLT SCALE). NOTE: THE COMMON PROBE OF THE VOLTMETER SHOULD BE ATTACHED TO CTB-2.
6. ADJUST THE FIELD HEATING GAIN (POT R113 ON NCBB) SO THAT THE VOLTAGE ON HFPA TP10 IS EQUIVALENT 'TO THE VALUE CALCULATED IN STEP 4.

(E) KVAR LOADSHARING - TURN THE ENGINE CONTROL SWITCH ON ALL ON-LINE UNITS TO "LOAD". ADJUST POT R101 TO BALANCE THE KVAR LOAD, BE CAREFUL TO MAINTAIN 600 VOLTS ON THE BUS WHEN MAKING THIS ADJUSTMENT.

3. METERING ADJUSTMENTS

ALL GEM ADJUSTMENTS ARE LOCATED ON THE FRONT DOOR-(DS3800NCBB) OF THE GEM, REFER TO FIGURE 1 TO HELP LOCATE A SPECIFIC ADJUSTMENT.

NOTE: CASUAL TUNE-UP OF THE GEM IS NOT REQUIRED. THIS SECTION IS INCLUDED ONLY TO HELP YOU SOLVE GEM PROBLEMS,

PLEASE KEEP IN MIND THAT REPLACEMENT MODULES CAN BE ADJUSTED BY SIMPLY COMPARING POTS AND SWITCHES TO THE ORIGINAL MODULE OR THE START-UP DATA,

(A) ----> INDICATES SWITCH OR POT REFERRED TO IN FOLLOWING ADJUSTMENT SECTION.

(A) PHASE 1 AMMETER GAIN - 1, PUT ONE GENERATOR ON LINE AND TURN THE ENGINE CONTROL SWITCH TO LOAD.

2. WITH AN AC VOLTMETER (1 VOLT SCALE), MEASURE THE VOLTAGE DIRECTLY ACROSS R3' ON XPES.

3. DETERMINE THE CT PRIMARY RATING FROM SHEET OF IN THE SYSTEM ELEMENTARY.

4. CALCULATE:

GEN PHASE 1 AMPS = (VOLTAGE ACROSS R3) X . {CT PRIMARY RATING)

5. PLACE AMMETER SWITCH IN "PHASE 1". ADJUST R106 SO THAT THE GENERATOR AMMETER READS THE VALUE CALCULATED IN STEP 4.

6. EXAMPLE: PUT ONE UNIT ON LINE
VOLTAGE ACROSS R3 = .2 VAC
CT RATIO = 1600/5
GEN PHASE 1 AMPS = .2 X 1600
= 320
TURN AMMETER SW TO PHASE 1.
ADJUST R106 SO THAT AMMETER READS 320 AMPS.

(B) PHASE 3 AMMETER GAIN - 1. PUT ONE GENERATOR ON LINE AND TURN THE ENGINE CONTROL SWITCH TO LOAD.

2. WITH AN AC VOLTMETER (1 VOLT SCALE) MEASURE THE VOLTAGE DIRECTLY ACROSS R4 ON XPES.

3. DETERMINE,THE CT PRIMARY RATING.

4. CALCULATE:

GEN PHASE 3AMPS = (VOLTAGE ACROSS R4) X (CT PRIMARY RATING)

5. PLACE AMMETER SWITCH IN "PHASE 3". ADJUST R107 SO THAT THE GENERATOR AMMETER' READS THE VALUE CALCULATED IN STEP 4.

(C) KW GAIN - 1. PUT ONE GENERATOR ON LINE.
TURN THE ENGINE CONTROL SWITCH TO LOAD.

2. RECORD GENERATOR AMPS.

3. RECORD GENERATOR VOLTS.

4. CALCULATE:

$$\text{GEN KVA} = \frac{\sqrt{3} \times (\text{GEN VOLTS}) \times (\text{GEN AMPS})}{1000}$$

5. RECORD GENERATOR KVARS,

6. CALCULATE:

$$\sqrt{(\text{GEN KVA})^2 - (\text{GEN KVARS})^2}$$

7. ADJUST POT R205 ON NCBB SO THAT KW METER READS THE VALUE CALCULATED IN STEP 6.

(D) FREQUENCY GAIN - ADJUSTMENT OF THIS POT IS NOT REQUIRED.

GEM TRANSFORMER FIELD OUTPUT ADJUSTMENT

NOTE :

REPLACEMENT TRANSFORMERS FOR THE GEM WILL HAVE A "PERSONALITY PLUG" WHICH DETERMINES GENERATOR FIELD EXCITER VOLTAGES. THE PLUG IS OF THE "MATE 'N' LOCH" TYPE AND IS LABELED JSV.

1. TRANSFORMER ADJUSTMENT

A. USE TABLE 1 BELOW TO CHOOSE THE APPROPRIATE PLUG FROM THE TWO SUPPLIED WITH THE REPLACEMENT TRANSFORMER.

PLUG TYPE	NUMBER OF WIRES IN MEW/OLD PLUG	NUMEER OF JUMPERS ON THE OLD STYLE TB	GENERATOR NAME-PLATE FIELD EXCITER FULL LOAD VOLTAGE
304A8679G1	6 WIRES	6 JUMPERS	LESS THAN 35V
304A8679G2	3WIRE'S	3 JUMPERS	GREATER THAN 35V BUT LESS THAN 70V

TABLE 1

B. INSERT THE CORRECT PERSONALITY PLUG INTO JSV.

C. EXAMPLE:

STEP 1 - OLD TRANSFORMER FAILS
STEP 2 - REMOVE AND CHECK THE TB JUMPERS, NOTING WHETHER THERE ARE 3 OR 6 JUMPERS.
STEP 3 - INSTALL REPLACEMENT TRANSFORMER.
STEP 4 - SELECT THE PLUG BASED ON THE NUMBER OF JUMPERS ON THE OLD TB. IN THIS' EXAMPLE, WE HAVE 3 JUMPERS . TABLE 1 SHOWS TO CHOOSE THE 304A8679G2 P L U G .
STEP 5 - DOUBLE CHECK AGAINST THE GENERATOR NAMEPLATE DATA. THE NAMEPLATE SHOWS FULL LOAD FIELD VOLTAGE TO BE 50V. TABLE 1 AGAIN SHOWS TO CHOOSE THE G2 (GROUP 2 PLUG).
STEP 6 - INSTALL THE PLUG,

2-10 3 POWER LIMIT ADJUSTMENTS

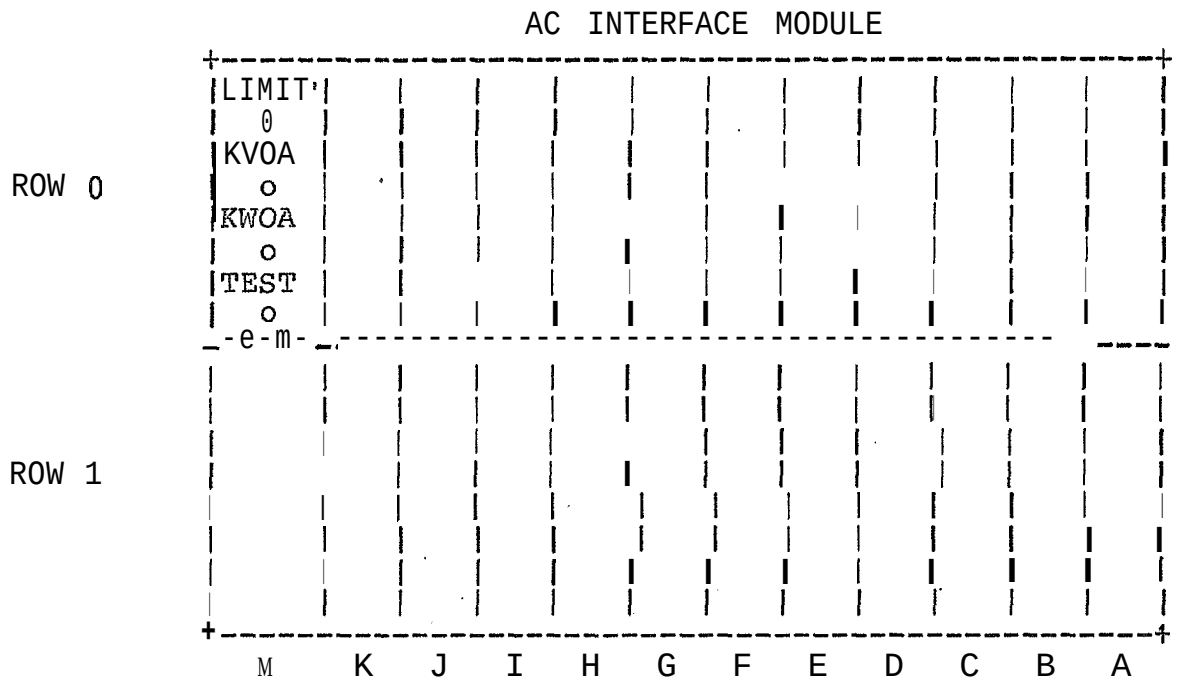


FIGURE 2

NOTE: TO MAKE THE ADJUSTMENTS IN THIS SECTION, YOU WILL NEED A TEST LEAD WITH A "BANANA PLUG" ON ONE END AND AN "E-Z CLIP" ON THE OTHER.

1. KW LIMIT'AND DRILLERS CONSOLE KW METER

   A) LOCATE THE AC INTERFACE MODULE SHOWN IN FIGURE 2.

   B) CLOSE ANY PCP BREAKER.

   C) REMOVE PLUG "JCX" FROM A GENERATOR MODULE WHICH IS NOT IN USE AND IS TURNED OFF. LET'S ASSUME THIS IS GEM #1.

   D) CONNECT YOUR TEST LEAD FROM THE "TEST"JACK ON THE AC INTERFACE MODULE TO TEST POINT 9 ON THE HFPA CARD IN GEM 1.

   E) ADJUST POT 4 (ON THE CARD IN SLOT OK IN THE AC INTERFACE MODULE) SO THAT THE KW METER ON GENERATOR PANEL #4 READS THE RATED KW OF ONE ENGINE,

   F) ADJUST THE KW BIAS (POT 1, CARD SLOT OK, AC INTERFACE MODULE) SO THAT THE "LIMIT" TEST JACK ON THE AC INTERFACE MODULE EQUALS -5.0 VDC.

   G) WITH NO KW LOAD ON THE SYSTEM, ADJUST POT 2 (CARD SLOT OK, AC INTERFACE MODULE) SO THAT THE KW METER ON THE DRILLERS CONSOLE READS EXACTLY ONE PIE DIVISION.

   H) REMOVE JUMPER FROM HFPA 9 ON GEM 1 AND REPLACE JCX PLUG.

2. KVA LIMIT

A) TURN ON ANY PCP.

B) CALCULATE THE KVA LIMIT SIGNAL EXPECTED FROM THE GEM MODULE WITH THE GEM AT RATED LOAD:

$$V_{KVA} = \frac{-1 \text{ VOLT}}{200 \text{ AMPS}} \times \text{RATED GENERATOR AMPS}$$

C) ADJUST THE VOLTAGE AT THE "TEST" JACK ON THE AC INTERFACE MODULE TO THE VALUE CALCULATED (VKVA) IN STEP A. THIS VALUE MUST BE NEGATIVE AND CAN BE OBTAINED BY ADJUSTING POT 4, CARD SLOT OK, IN THE AC INTERFACE MODULE

D) REMOVE THE WIRE CONNECTED TO THE JDC-4 "STAB ON'" ON ANY' GEM MODULE.

E) JUMPER THE "TEST" JACK VOLTAGE (STEP B) TO THE WIRE REMOVED IN STEP C.

F) ADJUST THE KVA BIAS (POT 3, CARD SLOT OK, AC INTERFACE MODULE) SO THAT THE "LIMIT" TEST JACK EQUALS -5.0 VDC.

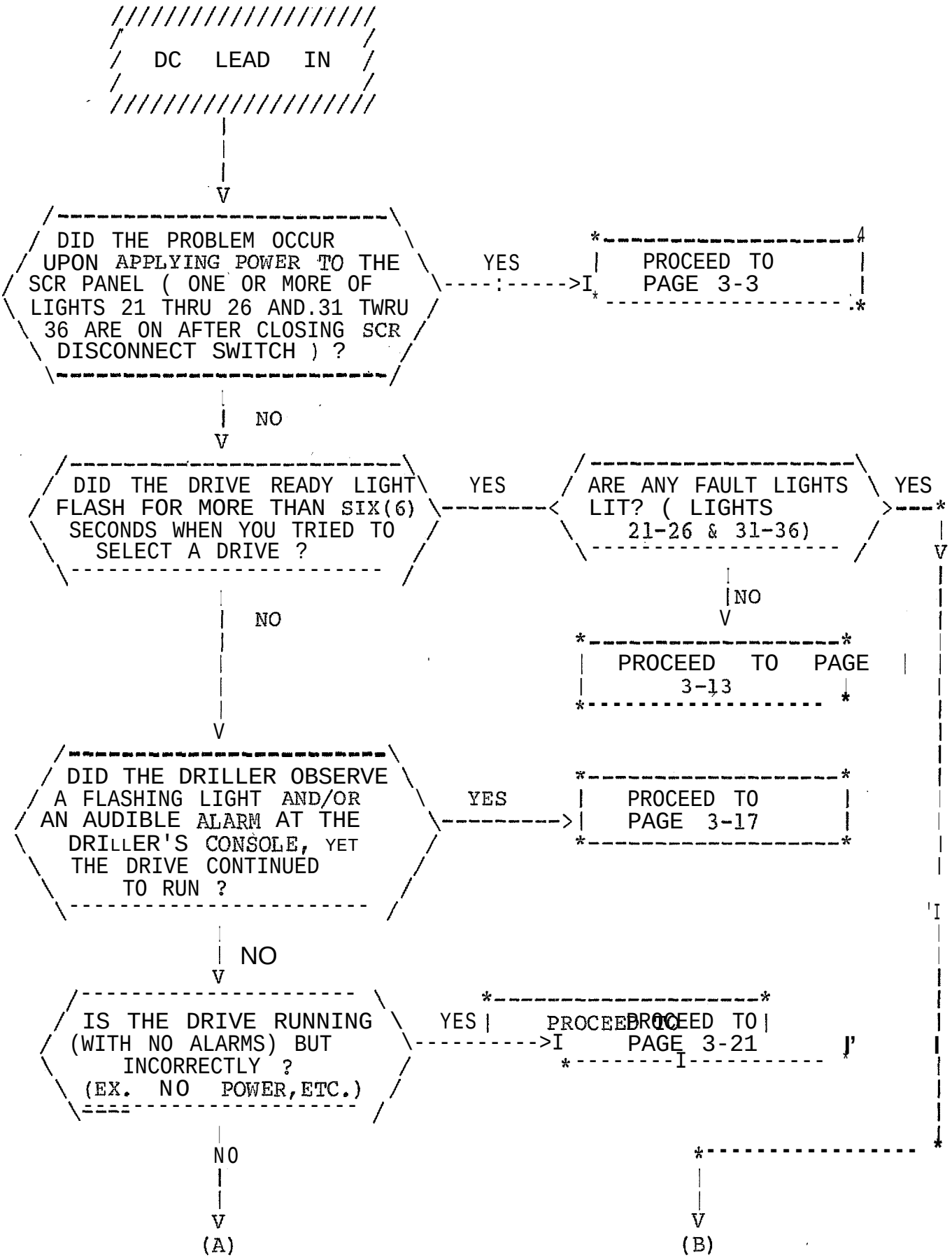
G) REMOVE JUMPER FROM THE "TEST"JACK AND RECONNECT THE ORIGINAL WIRE BACK TO THE JDC-4 STAB-ON.

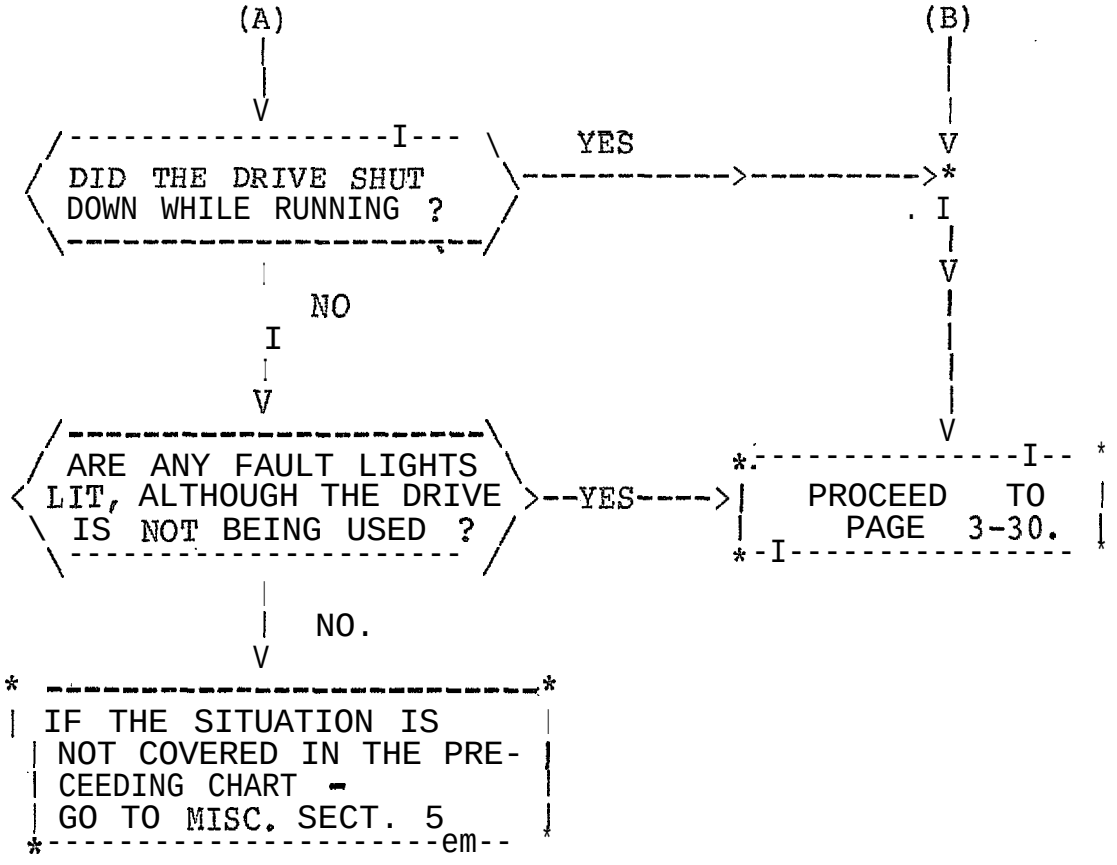
2-11 SYNCHRONIZING ADJUSTMENTS

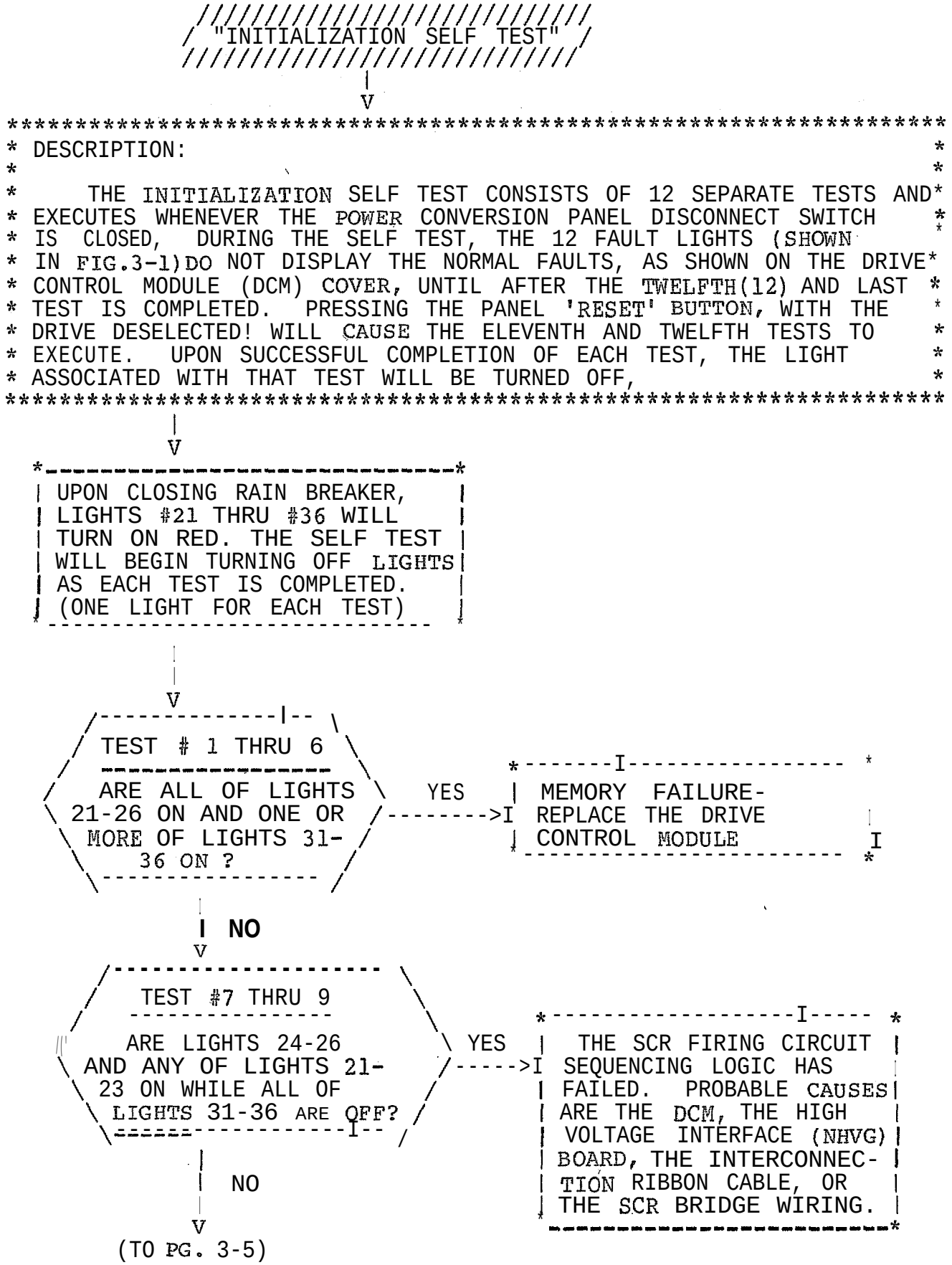
A. VERI-SYNC. RELAY - FOLD DOWN THE AC INTERFACE MODULE IN GENERATOR PANEL #2.

LOCATE THE VERI-SYNC RELAY (BEHIND THE AC INTERFACE MODULE).

TURN THE PHASE ANGLE ADJUSTMENT TO ITS MIDDLE POSITION (10 DEG. CLOSING ANGLE).


DC LEAD IN
DID THE PROBLEM OCCUR UPON APPLYING POWER TO THE SCR PANEL (ONE OR MORE OF LIGHTS 21 THRU 26 AND.31 TWRU 36 ARE ON AFTER CLOSING SCR DISCONNECT SWITCH) ?
YES
PROCEED TO PAGE 3-3
NO
DID THE DRIVE READY LIGHT FLASH FOR MORE THAN SIX(6) SECONDS WHEN YOU TRIED TO SELECT A DRIVE ?
YES
ARE ANY FAULT LIGHTS LIT? (LIGHTS 21-26 & 31-36)
YES
NO
PROCEED TO PAGE 3-13
NO
DID THE DRILLER OBSERVE A FLASHING LIGHT AND/OR AN AUDIBLE ALARM AT THE DRILLER'S CONSOLE, YET THE DRIVE CONTINUED TO RUN ?
YES
PROCEED TO PAGE 3-17
NO
IS THE DRIVE RUNNING (WITH NO ALARMS) BUT INCORRECTLY ? (EX. NO POWER,ETC.)
YES
PROCEED TO PAGE 3-21
NO
(A)
(B)


(A)
(B)
DID THE DRIVE SHUT DOWN WHILE RUNNING ?
YES
NO
ARE ANY FAULT LIGHTS LIT, ALTHOUGH THE DRIVE IS NOT BEING USED ?
YES
PROCEED TO PAGE 3-30.
NO.
IF THE SITUATION IS NOT COVERED IN THE PRE-CEEDING CHART - GO TO MISC. SECT. 5


"INITIALIZATION SELF TEST"
DESCRIPTION:
THE INITIALIZATION SELF TEST CONSISTS OF 12 SEPARATE TESTS AND EXECUTES WHENEVER THE POWER CONVERSION PANEL DISCONNECT SWITCH IS CLOSED, DURING THE SELF TEST, THE 12 FAULT LIGHTS (SHOWN IN FIG.3-1) DO NOT DISPLAY THE NORMAL FAULTS, AS SHOWN ON THE DRIVE CONTROL MODULE (DCM) COVER, UNTIL AFTER THE TWELFTH(12) AND LAST TEST IS COMPLETED. PRESSING THE PANEL 'RESET' BUTTON, WITH THE DRIVE DESELECTED! WILL CAUSE THE ELEVENTH AND TWELFTH TESTS TO EXECUTE. UPON SUCCESSFUL COMPLETION OF EACH TEST, THE LIGHT ASSOCIATED WITH THAT TEST WILL BE TURNED OFF,
UPON CLOSING RAIN BREAKER, LIGHTS #21 THRU #36 WILL TURN ON RED. THE SELF TEST WILL BEGIN TURNING OFF LIGHTS AS EACH TEST IS COMPLETED. (ONE LIGHT FOR EACH TEST)
TEST # 1 THRU 6
ARE ALL OF LIGHTS 21-26 ON AND ONE OR MORE OF LIGHTS 31-36 ON ?
YES
MEMORY FAILURE- REPLACE THE DRIVE CONTROL MODULE
NO
TEST #7 THRU 9
ARE LIGHTS 24-26 AND ANY OF LIGHTS 21-23 ON WHILE ALL OF LIGHTS 31-36 ARE OFF?
YES
THE SCR FIRING CIRCUIT SEQUENCING LOGIC HAS FAILED. PROBABLE CAUSES ARE THE DCM, THE HIGH VOLTAGE INTERFACE (NHVG) BOARD, THE INTERCONNECTION RIBBON CABLE, OR THE SCR BRIDGE WIRING.
NO
(TO PG. 3-5)

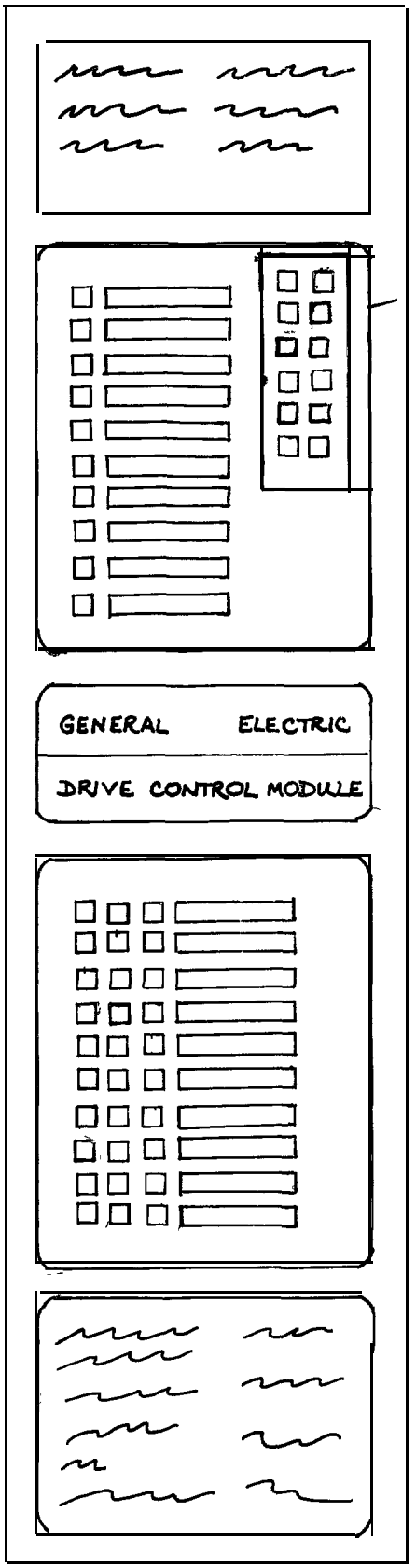
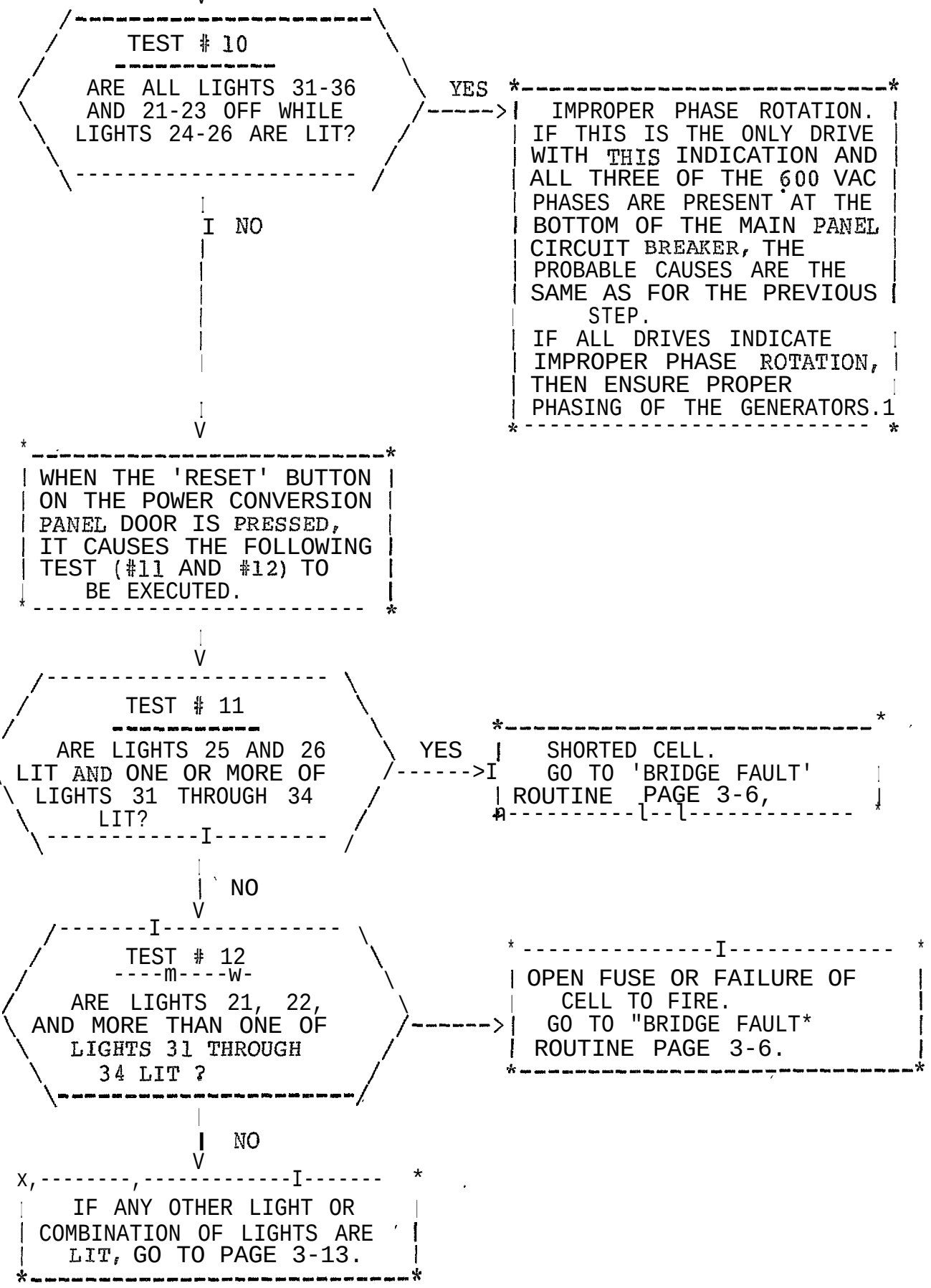


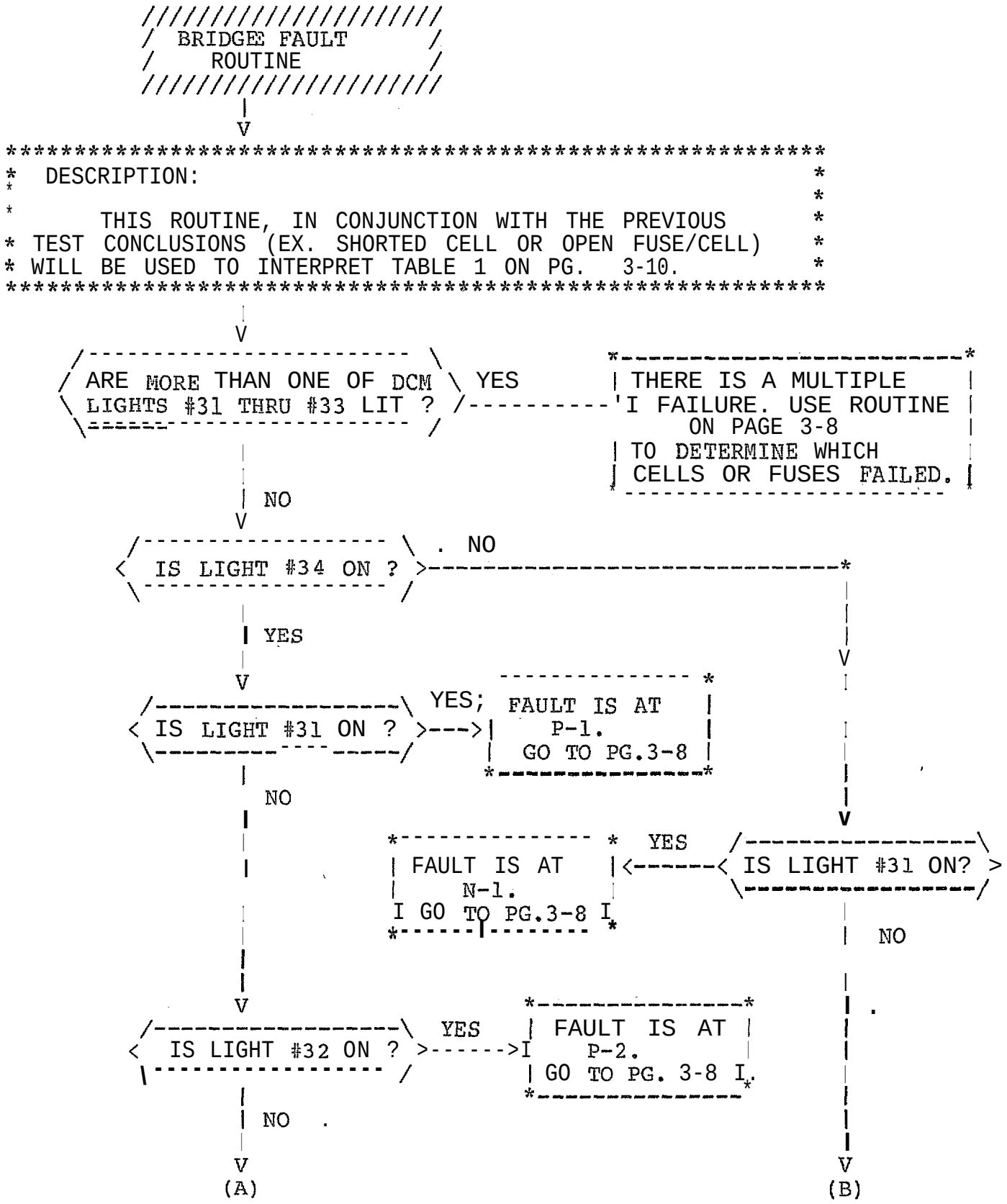
FIGURE 1

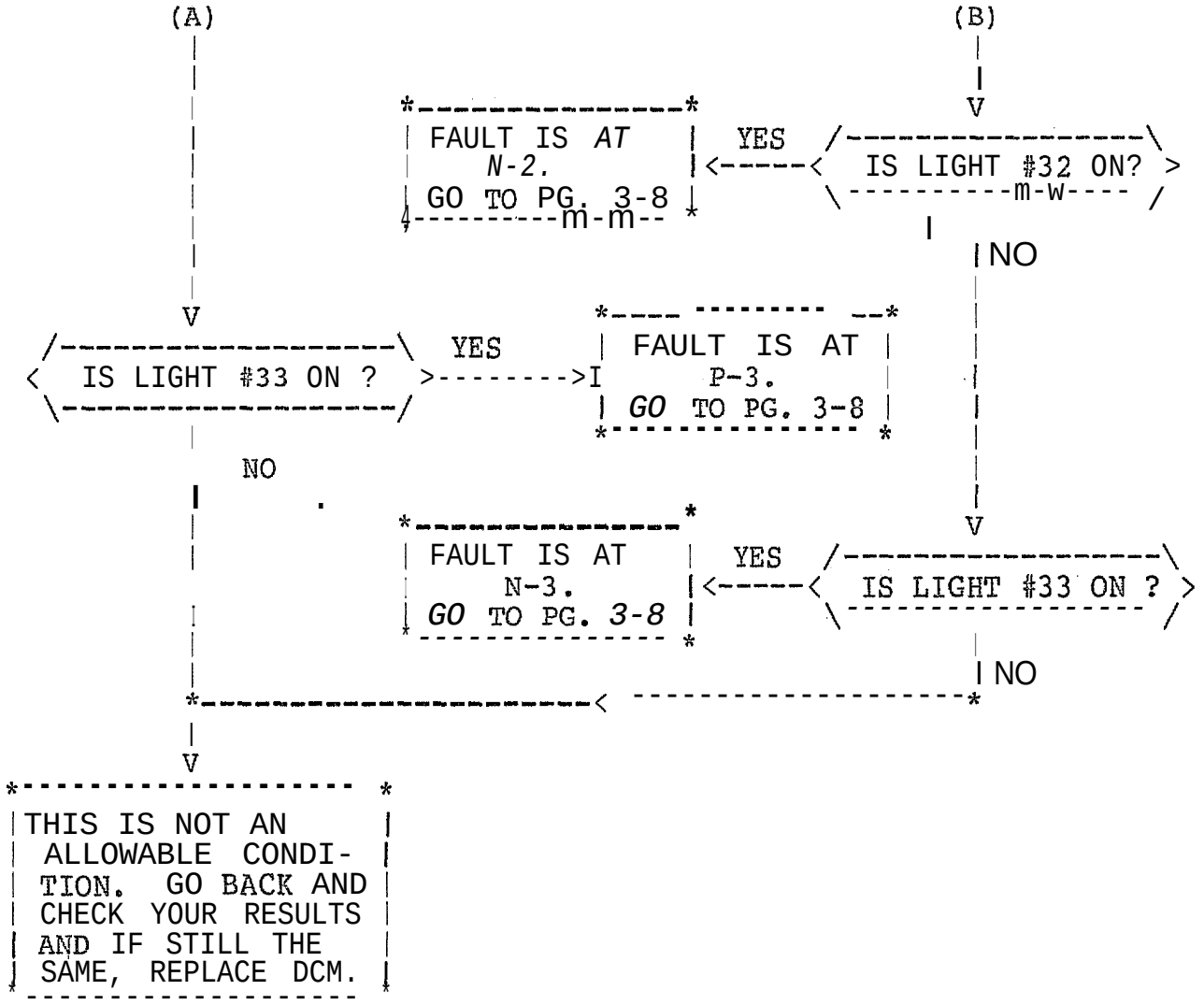
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

IV. 1	REV 4 REV. 5 PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
v. 2	ISSUED	FIRST MADE FOR			
iv. 3	MADE BY	I.C. NO.		CONT. ON SH.	SH. NO.

(FROM PG. 3-3)


TEST # 10
ARE ALL LIGHTS 31-36 AND 21-23 OFF WHILE LIGHTS 24-26 ARE LIT?
YES
IMPROPER PHASE ROTATION. IF THIS IS THE ONLY DRIVE WITH THIS INDICATION AND ALL THREE OF THE 600 VAC PHASES ARE PRESENT AT THE BOTTOM OF THE MAIN PANEL CIRCUIT BREAKER, THE PROBABLE CAUSES ARE THE SAME AS FOR THE PREVIOUS STEP.
IF ALL DRIVES INDICATE IMPROPER PHASE ROTATION, THEN ENSURE PROPER PHASING OF THE GENERATORS.1
NO
WHEN THE 'RESET' BUTTON ON THE POWER CONVERSION PANEL DOOR IS PRESSED, IT CAUSES THE FOLLOWING TEST (#11 AND #12) TO BE EXECUTED.
TEST # 11
ARE LIGHTS 25 AND 26 LIT AND ONE OR MORE OF LIGHTS 31 THROUGH 34 LIT?
YES
SHORTED CELL. GO TO 'BRIDGE FAULT' ROUTINE PAGE 3-6,
NO
TEST # 12
ARE LIGHTS 21, 22, AND MORE THAN ONE OF LIGHTS 31 THROUGH 34 LIT ?
OPEN FUSE OR FAILURE OF CELL TO FIRE. GO TO "BRIDGE FAULT* ROUTINE PAGE 3-6.
NO
IF ANY OTHER LIGHT OR COMBINATION OF LIGHTS ARE LIT, GO TO PAGE 3-13.


BRIDGE FAULT ROUTINE
DESCRIPTION:
THIS ROUTINE, IN CONJUNCTION WITH THE PREVIOUS TEST CONCLUSIONS (EX. SHORTED CELL OR OPEN FUSE/CELL) WILL BE USED TO INTERPRET TABLE 1 ON PG. 3-10.
ARE MORE THAN ONE OF DCM LIGHTS #31 THRU #33 LIT ?
YES
THERE IS A MULTIPLE I FAILURE. USE ROUTINE ON PAGE 3-8 TO DETERMINE WHICH CELLS OR FUSES FAILED.
NO
IS LIGHT #34 ON ?
NO
YES
IS LIGHT #31 ON ?
YES; FAULT IS AT P-1. GO TO PG.3-8
NO
IS LIGHT #31 ON?
YES
FAULT IS AT N-1. GO TO PG.3-8
NO
IS LIGHT #32 ON ?
YES
FAULT IS AT P-2. GO TO PG. 3-8
NO
(A)
(B)


(A)
(B)
IS LIGHT #32 ON?
YES
FAULT IS AT N-2. GO TO PG. 3-8
NO
IS LIGHT #33 ON ?
YES
FAULT IS AT P-3. GO TO PG. 3-8
NO
FAULT IS AT N-3. GO TO PG. 3-8
IS LIGHT #33 ON ?
YES
NO
THIS IS NOT AN ALLOWABLE CONDITION. GO BACK AND CHECK YOUR RESULTS AND IF STILL THE SAME, REPLACE DCM.

****
* TURN POWER OFF BEFORE ATTEMPTING ANY PART OF THE *
* FOLLOWING ROUTINE !!!!! *
****

DESCRIPTION:

USE THIS ROUTINE, IN CONJUNCTION WITH THE PREVIOUS SECTION AND TABLE 1,TO CHECK FOR FAILED FUSES AND CELLS IN THE POWER CONVERSION MODULE. PROCEED AS FOLLOWS:

1. OPEN THE POWER CONVERSION MODULE DOOR. LOOSEN THE CLIPS ON THE HINGED BAFFLE AT THE BOTTOM OF THE MODULE AND TILT BAFFLE FORWARD. DO NOT FORCE!!! (SEE FIGURE 2 FOR DETAILS)

2. SOME POWER CONVERSION MODULES HAVE A PROTECTIVE COVER BETWEEN THE BUS COMING OUT OF THE BOTTOM TO THE MAIN CIRCUIT BREAKER AND THE MODULE. IF THIS COVER IS PRESENT, REMOVE IT BY REMOVING THE FOUR(4) MOUNTING SCREWS (2 ON EACH SIDE OF COVER) AS SHOWN IN FIGURE 3-2.

3. FUSES AND CELLS WILL BE CHECKED USING AN OHMMETER.

4. CHECK FOR FAULTY FUSES AND CELLS USING TABLE 1 AND THE THE INFORMATION FOUND FROM THE PREVIOUS PAGE.
(REFER TO FIGURES 3-3 AND 3-4 ON PAGES 3-11 AND 3-12 FOR TEST POINT LOCATIONS)

5. IF ALL CELLS AND FUSES CHECK GOOD, THERE MAY BE A FAILURE IN THE ELECTRONICS OR SENSING FEEDBACK. AREAS OF PROBABLE CAUSE (IN ORDER OF LIKELIHOOD) ARE:

   A) DRIVE CONTROL MODULE
   B) HIGH VOLTAGE INTERFACE (NHVG) BOARD
   C) RIBBON CABLE AND POWER CONVERSION BRIDGE WIRING

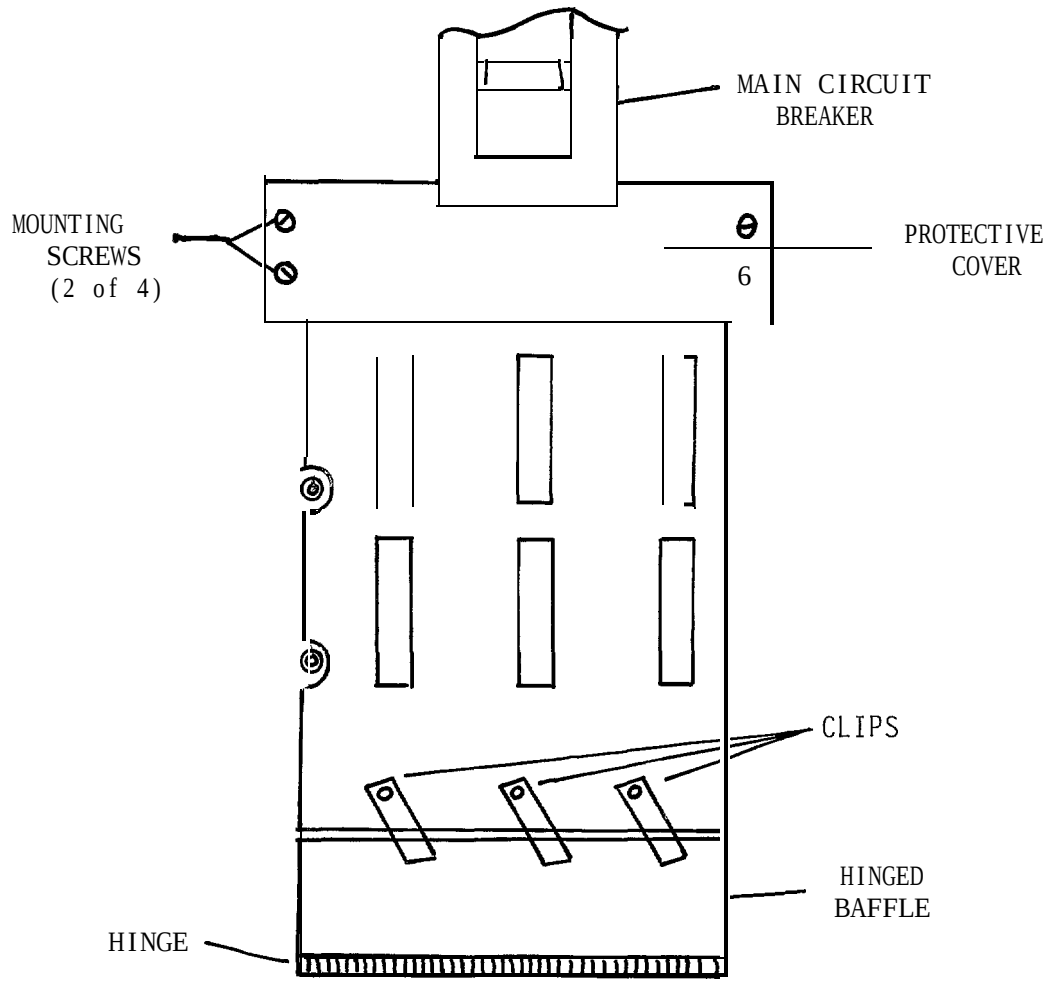


FIGURE 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

EV. 1	REV. 4 REV. 5 PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
iv. 2	ISSUED	FIRST MADE FOR			
EV. 3	MADE BY	I.C. NO.		CONT ON SH.	SH. NO.

FAULTY CELL OR FUSE	CONDITION: SHORTED CELL. SEE FIGS, 3-3 MEASURE BETWEEN TEST POINTS BELOW. *	OPEN FUSE CHECK. SEE FIGS. 3-4 MEASURE BETWEEN TEST POINTS BELOW. IF GOOD, READING SHOULD BE APPROXIMATELY ZERO(Q) OHMS, IF BAD, REPLACE THE FUSE.	OPEN CELL GATE, MEASURE AT'PLUGS BELOW ON HIGH VOLTAGE INTERFACE BOARD (NHVG). IF GOOD, READING SHOULD BE LESS THAN 100 OHMS. . IF BAD REPLACE THE SCR CELL.
P-l	TP 1 - 9	TP 5 -9	JE 1 - 2
P-2	TP 2 - 10	TP 5 - 10	JF 1 - 3
P-3	TP 3 - 11	TP 5 - 11	JG 2 - 3
N-l	TP 1 - 6	TP 4 - 6	JH l-2
PI-2	TP 2 - 7	TP4 -7	JJ 1 - 3
N-3	TP 3 - 8	TP4 -8	JK 2 - 3

TABLE 1

* A shorted cell will read zero ohms when the cell is 'at room temperature and proper mechanical pressure. A low reading with a volt-ohm-meter does not necessarily indicate a defective cell. Consult the factory for additional troubleshooting procedures if required.

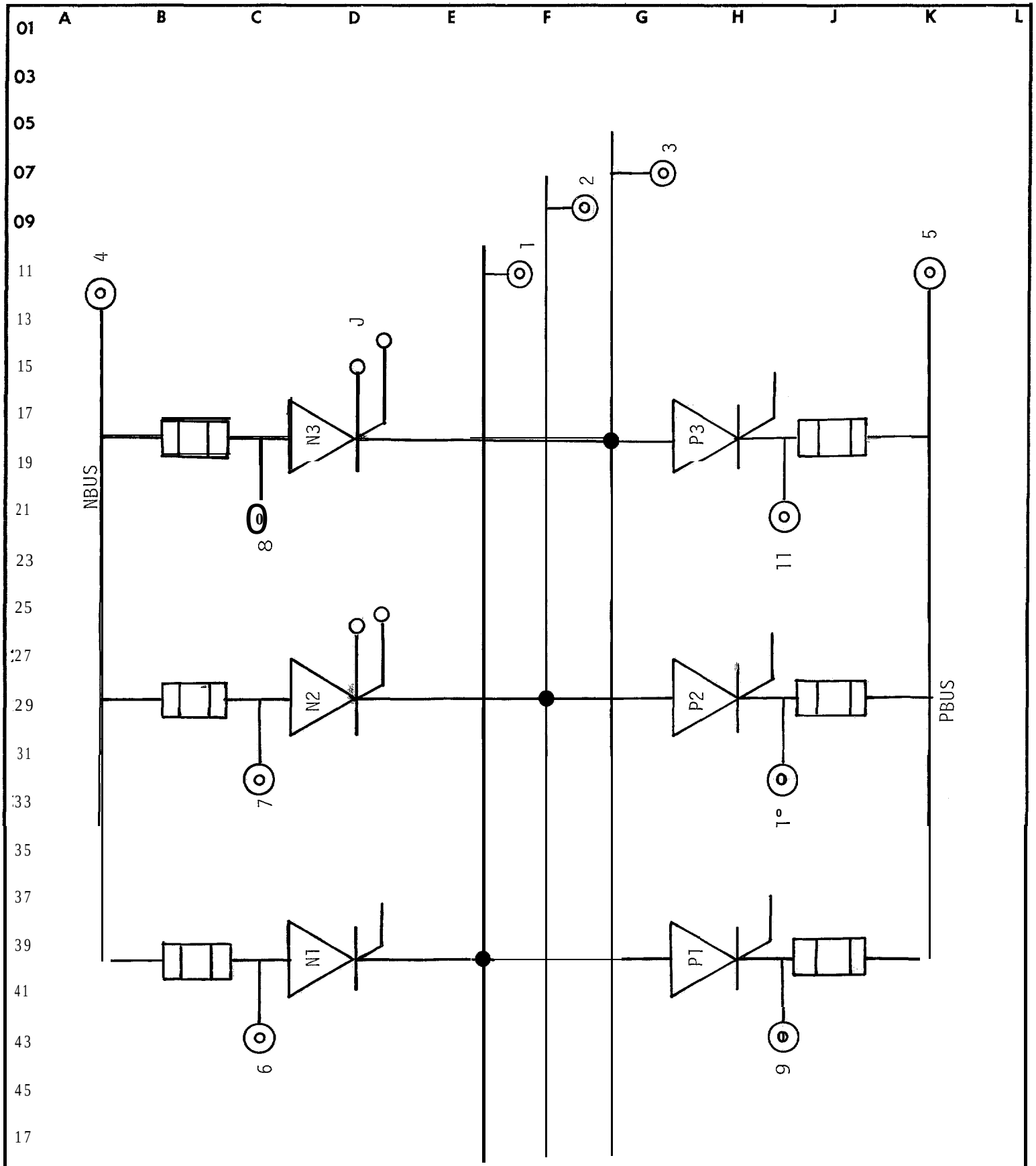


FIGURE 3

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

EV. 1	REV. 4 REV. 5 PRINTS TO	APPROVALS	ELEMENTARY DIAGRAM
EV. 2	ISSUED	FIRST MADE FOR	GENERAL ELECTRIC
EV. 3	MADE BY	I.C. NO.	CONT. CN SH. SH. NO.

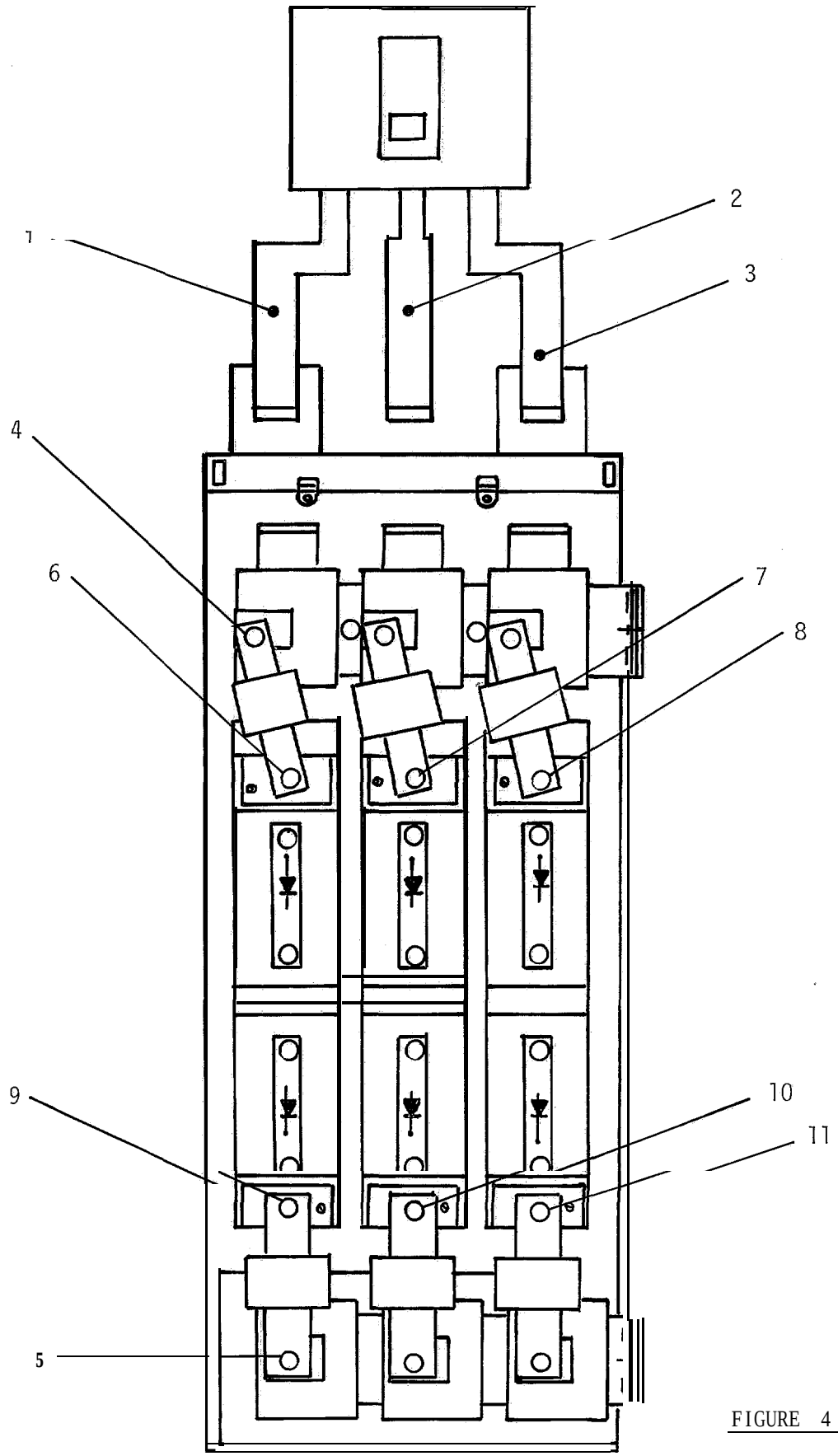
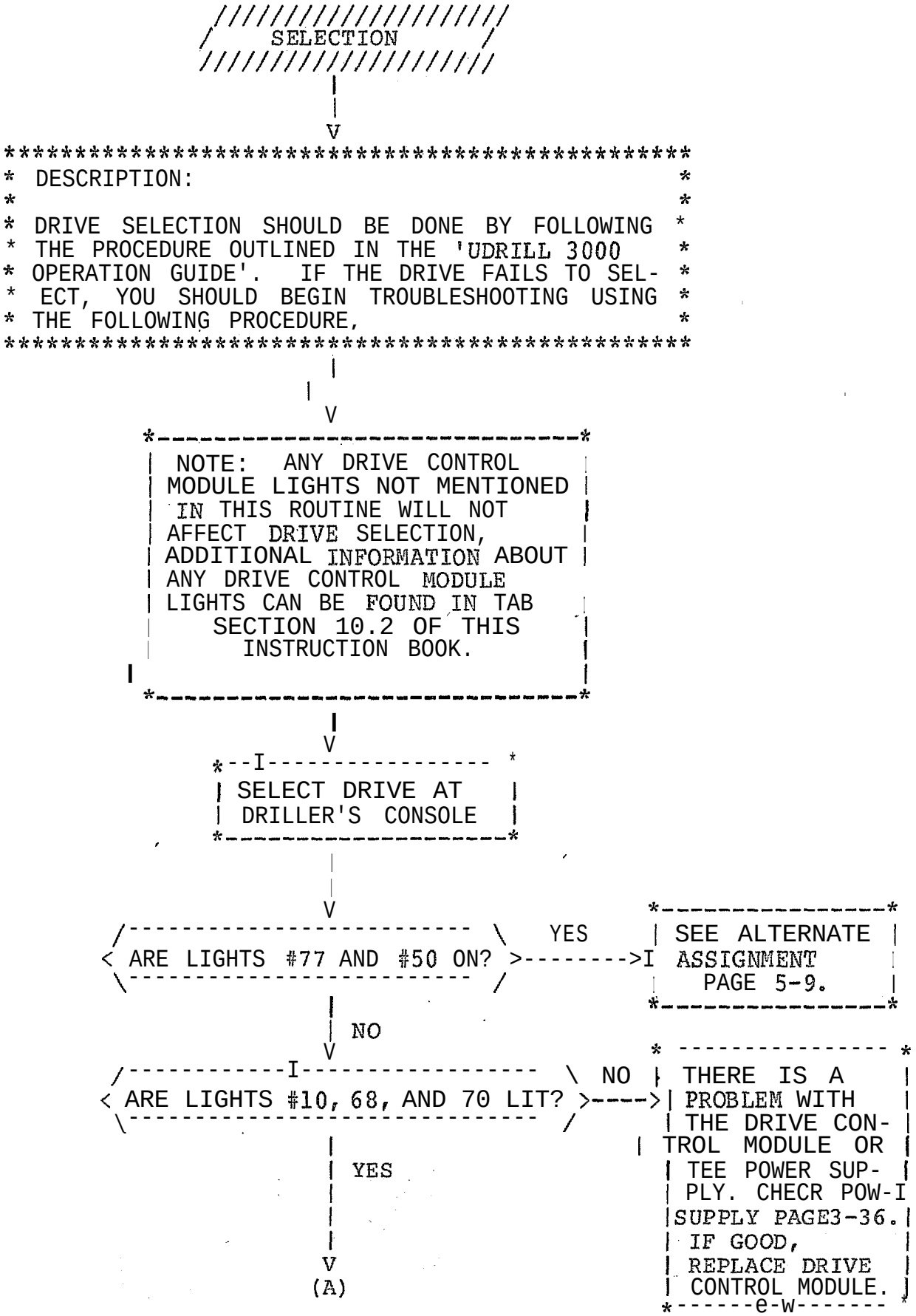
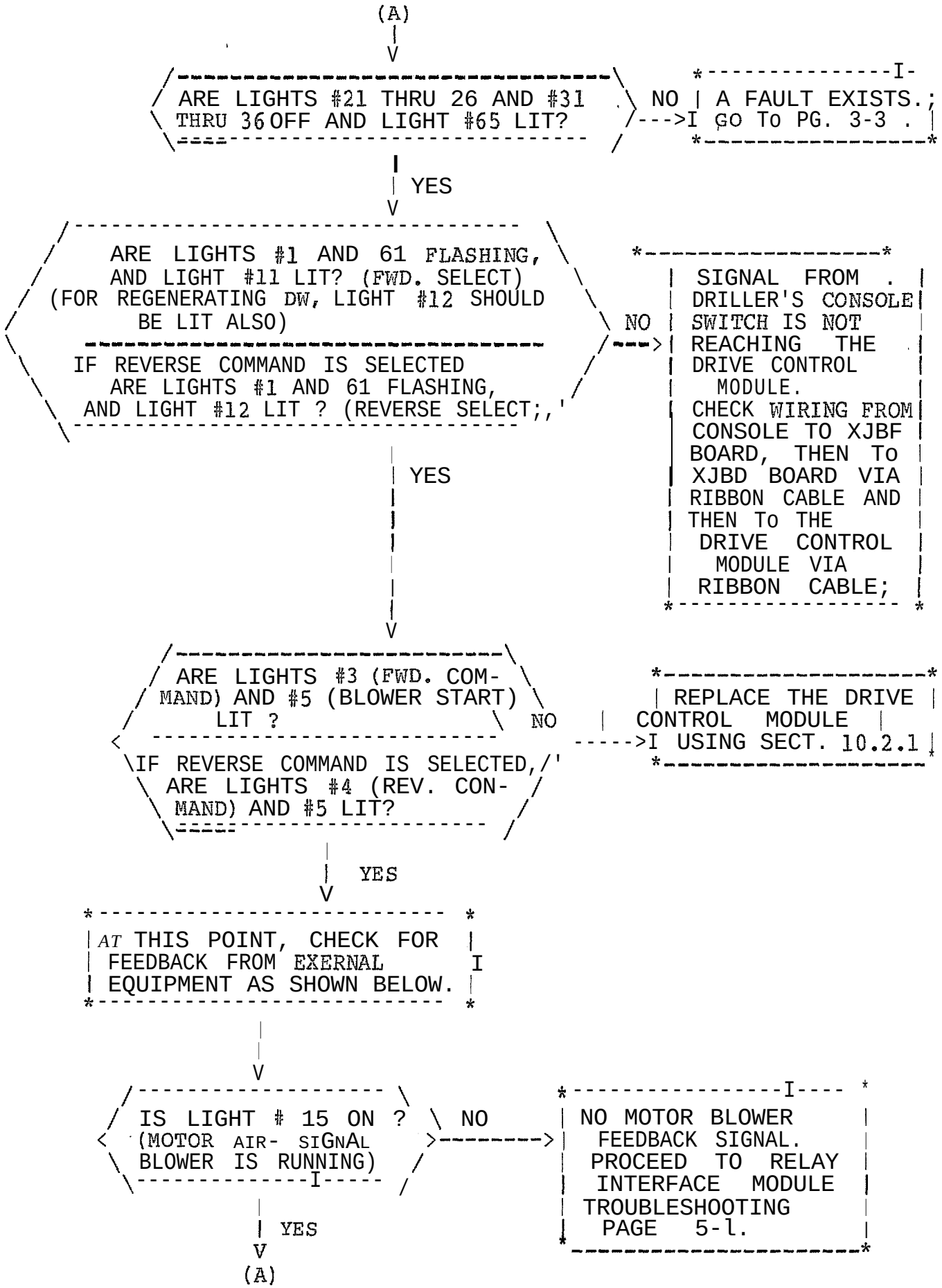


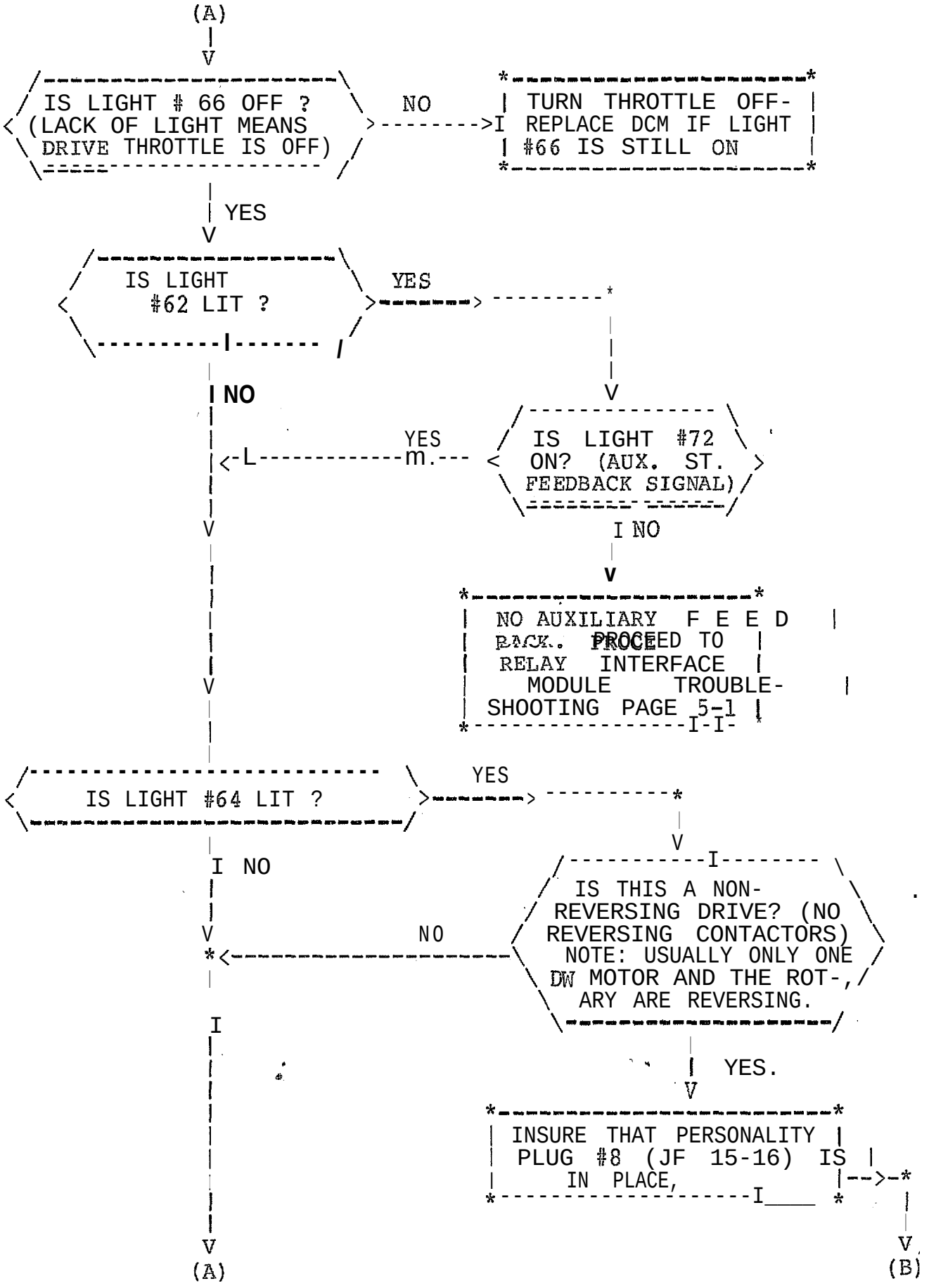
FIGURE 4

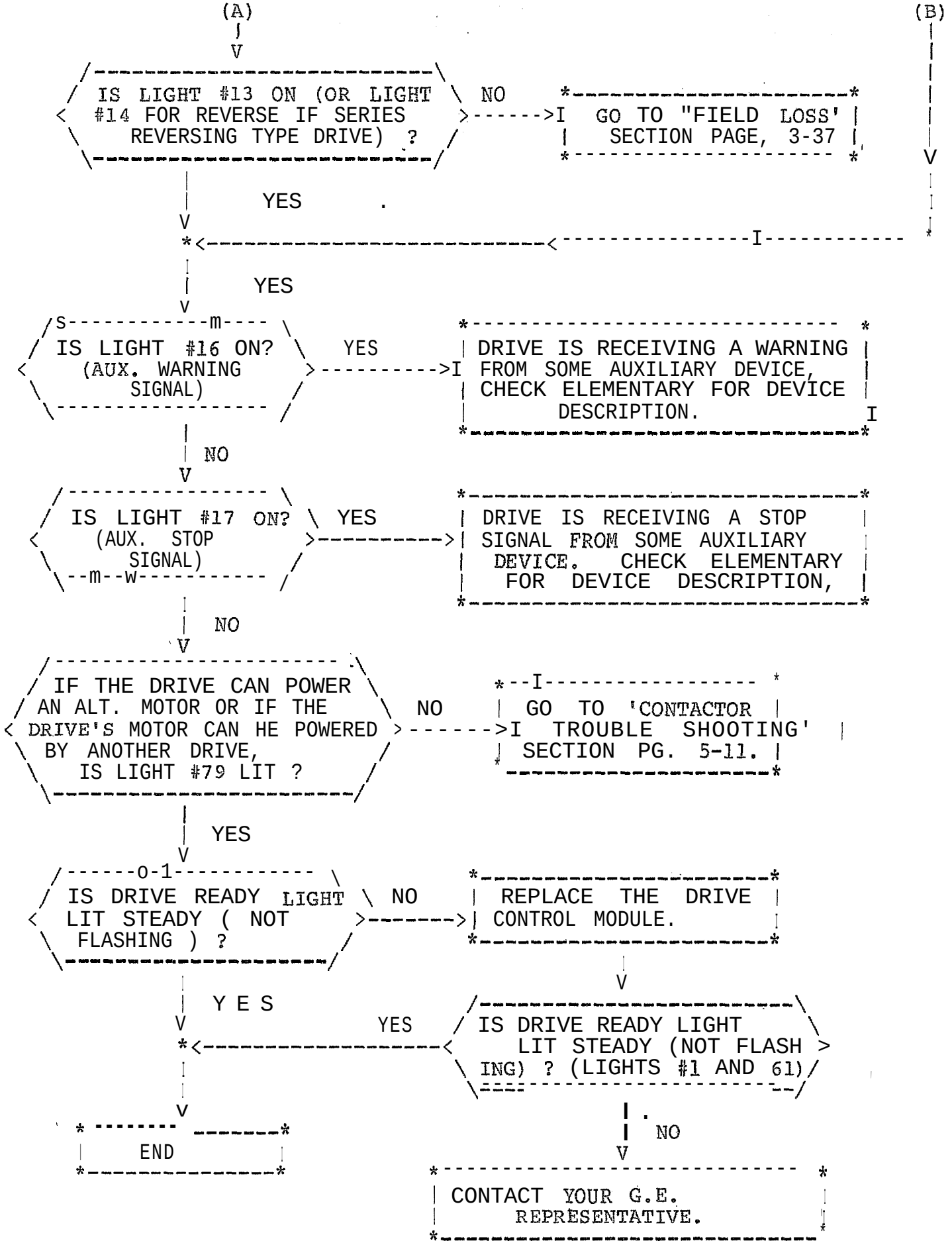
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

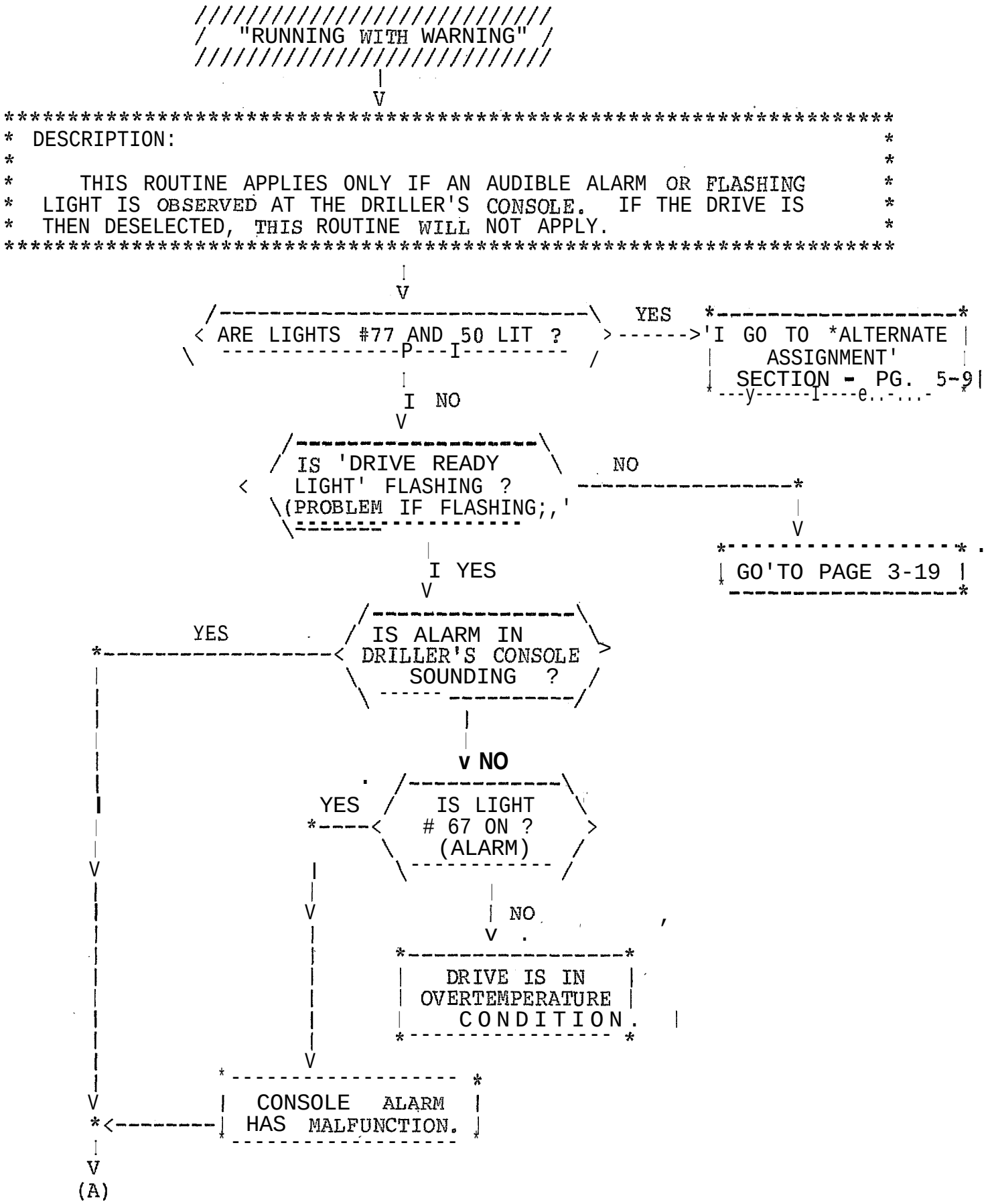
REV. 1	REV. 4 REV. 5 PRINTS TO	APPROVALS	ELEMENTARY DIAGRAM
REV. 2	ISSUED	FIRST MADE FOR	GENERAL ELECTRIC
REV. 3	MADE BY	I.C. NO:	CONT. CN SH. SH. NO.

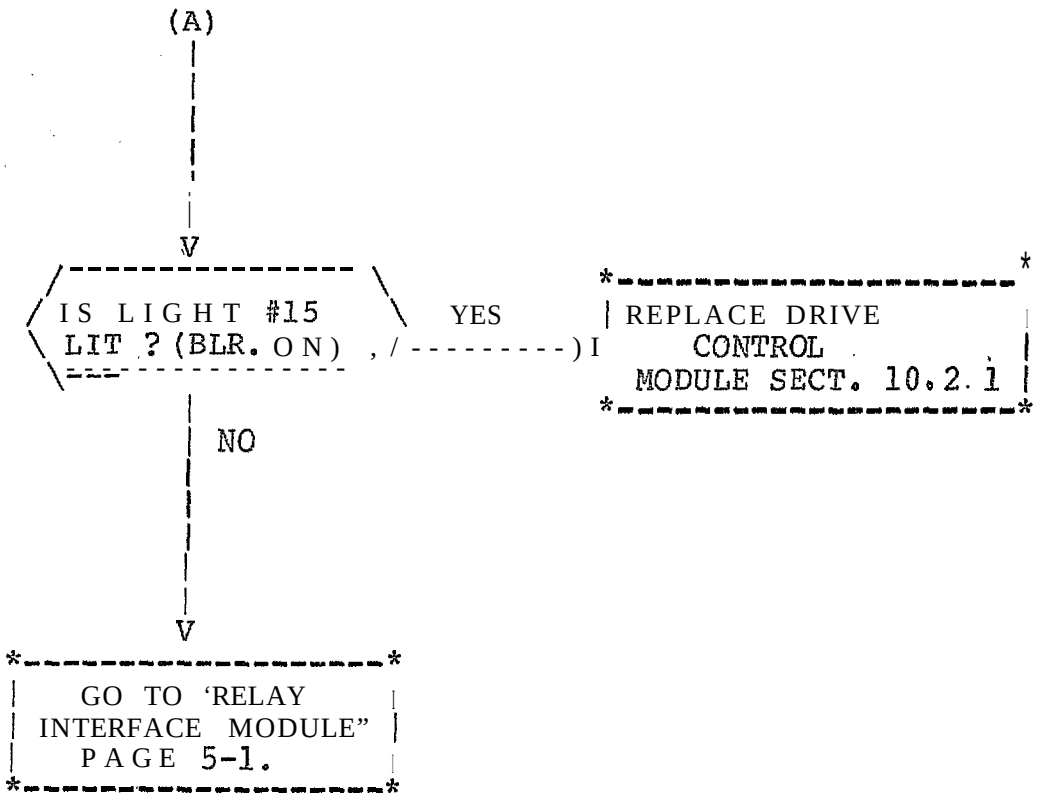

SELECTION
DESCRIPTION:
DRIVE SELECTION SHOULD BE DONE BY FOLLOWING THE PROCEDURE OUTLINED IN THE 'UDRILL 3000 OPERATION GUIDE'. IF THE DRIVE FAILS TO SELECT, YOU SHOULD BEGIN TROUBLESHOOTING USING THE FOLLOWING PROCEDURE,
NOTE: ANY DRIVE CONTROL MODULE LIGHTS NOT MENTIONED IN THIS ROUTINE WILL NOT AFFECT DRIVE SELECTION, ADDITIONAL INFORMATION ABOUT ANY DRIVE CONTROL MODULE LIGHTS CAN BE FOUND IN TAB SECTION 10.2 OF THIS INSTRUCTION BOOK.
SELECT DRIVE AT DRILLER'S CONSOLE
ARE LIGHTS #77 AND #50 ON?
YES
SEE ALTERNATE ASSIGNMENT PAGE 5-9.
NO
ARE LIGHTS #10, 68, AND 70 LIT?
NO
THERE IS A PROBLEM WITH THE DRIVE CONTROL MODULE OR TEE POWER SUPPLY. CHECR POW-I SUPPLY PAGE3-36. IF GOOD, REPLACE DRIVE CONTROL MODULE.
YES
(A)

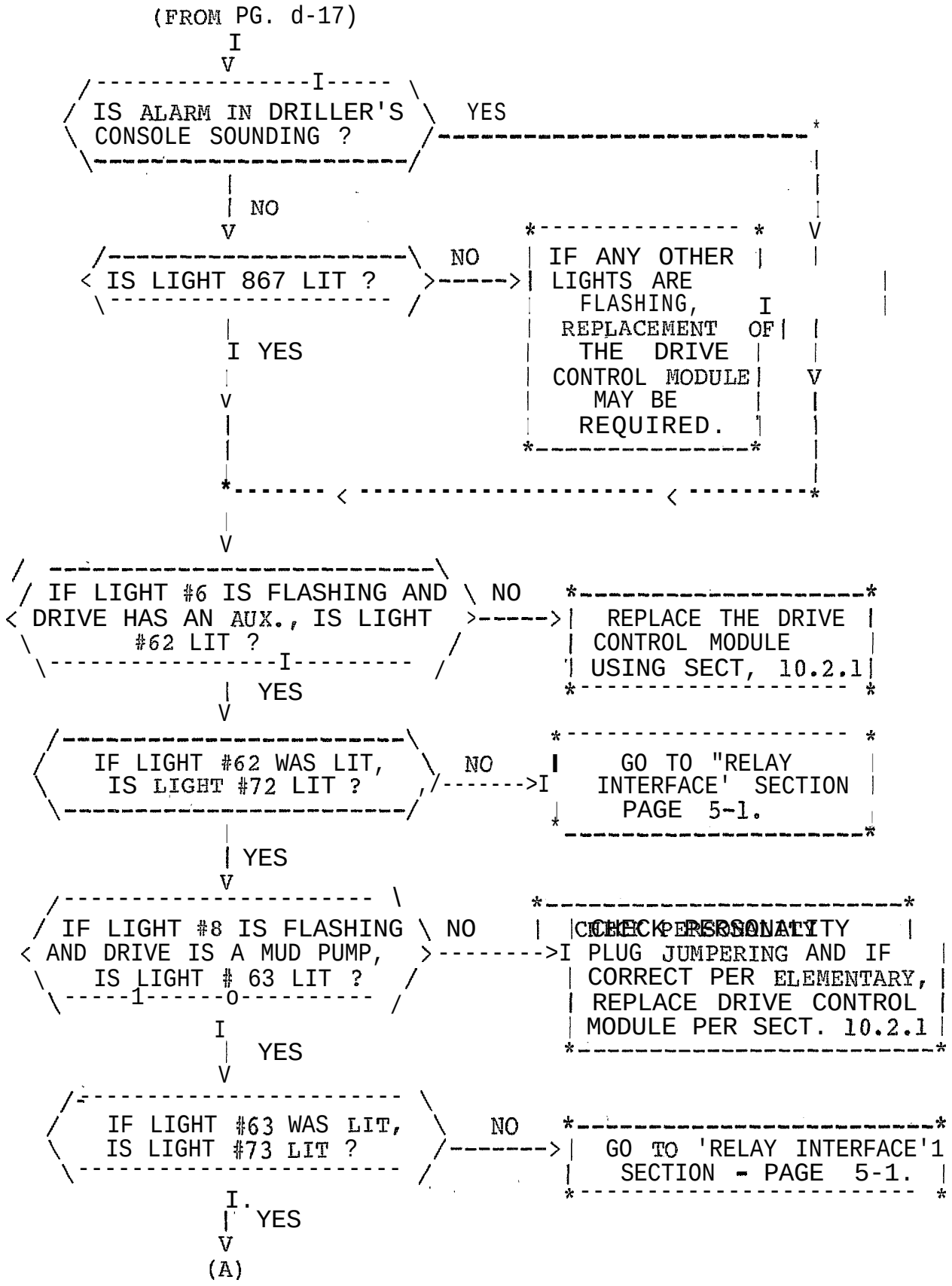

(A)
ARE LIGHTS #21 THRU 26 AND #31 THRU 36 OFF AND LIGHT #65 LIT?
NO
A FAULT EXISTS.; GO TO PG. 3-3
YES
ARE LIGHTS #1 AND 61 FLASHING, AND LIGHT #11 LIT? (FWD. SELECT)
(FOR REGENERATING DW, LIGHT #12 SHOULD BE LIT ALSO)
IF REVERSE COMMAND IS SELECTED ARE LIGHTS #1 AND 61 FLASHING, AND LIGHT #12 LIT ? (REVERSE SELECT;,'
NO
SIGNAL FROM DRILLER'S CONSOLE SWITCH IS NOT REACHING THE DRIVE CONTROL MODULE. CHECK WIRING FROM CONSOLE TO XJBF BOARD, THEN To XJBD BOARD VIA RIBBON CABLE AND THEN To THE DRIVE CONTROL MODULE VIA RIBBON CABLE;
YES
ARE LIGHTS #3 (FWD. COMMAND) AND #5 (BLOWER START) LIT ?
NO
REPLACE THE DRIVE CONTROL MODULE USING SECT. 10.2.1
IF REVERSE COMMAND IS SELECTED,/' ARE LIGHTS #4 (REV. COMMAND) AND #5 LIT?
YES
AT THIS POINT, CHECK FOR FEEDBACK FROM EXERNAL EQUIPMENT AS SHOWN BELOW.
IS LIGHT # 15 ON ? (MOTOR AIR- SIGNAL BLOWER IS RUNNING)
NO
NO MOTOR BLOWER FEEDBACK SIGNAL. PROCEED TO RELAY INTERFACE MODULE TROUBLESHOOTING PAGE 5-l.
YES
(A)

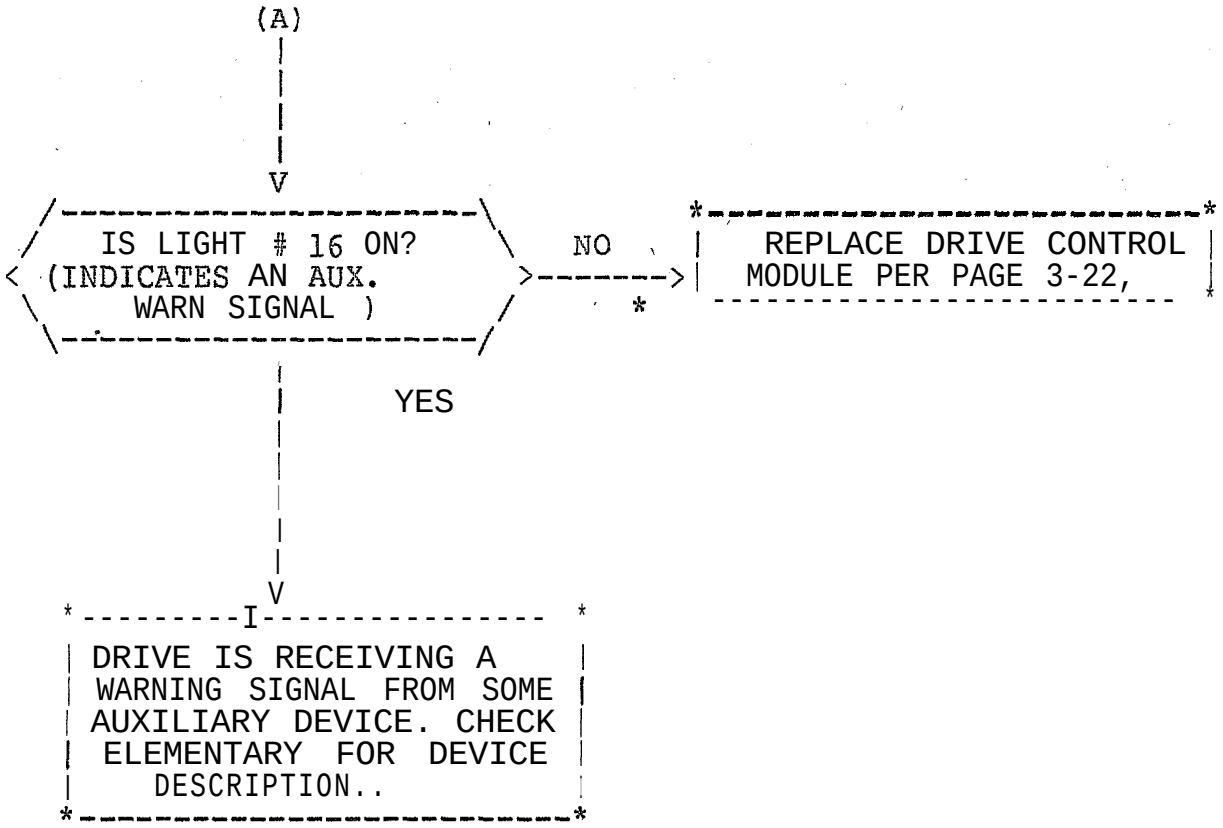

(A)
IS LIGHT # 66 OFF ? (LACK OF LIGHT MEANS DRIVE THROTTLE IS OFF)
NO
TURN THROTTLE OFF- REPLACE DCM IF LIGHT #66 IS STILL ON
YES
IS LIGHT #62 LIT ?
YES
NO
IS LIGHT #72 ON? (AUX. ST. FEEDBACK SIGNAL)
YES
NO
NO AUXILIARY F E E D BACK. PROCEED TO RELAY INTERFACE MODULE TROUBLE-SHOOTING PAGE 5-1
IS LIGHT #64 LIT ?
YES
NO
IS THIS A NON-REVERSING DRIVE? (NO REVERSING CONTACTORS) NOTE: USUALLY ONLY ONE DW MOTOR AND THE ROT-ARY ARE REVERSING.
NO
YES.
INSURE THAT PERSONALITY PLUG #8 (JF 15-16) IS IN PLACE,
(A)
(B)

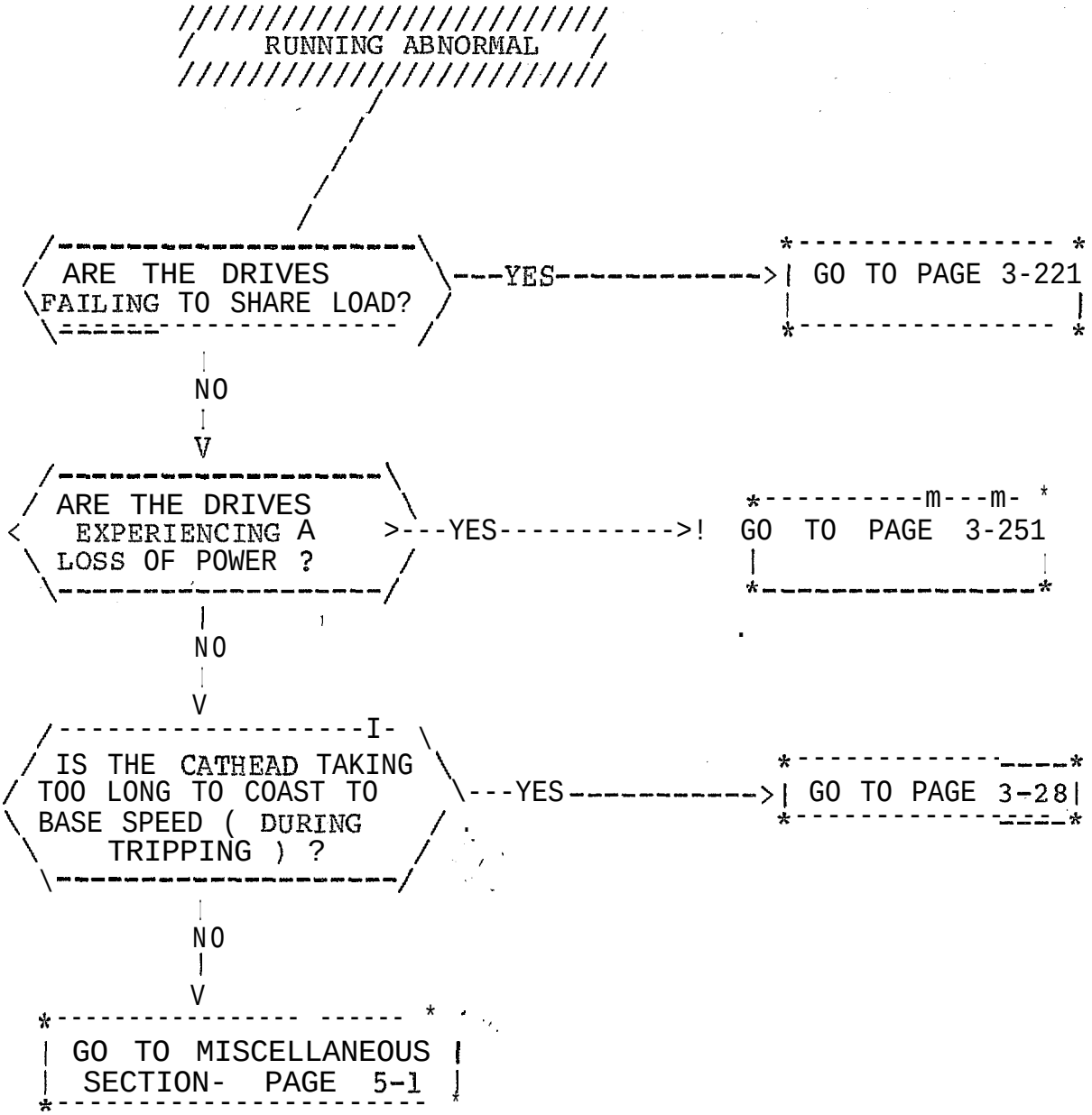

(A)
(B)
IS LIGHT #13 ON (OR LIGHT #14 FOR REVERSE IF SERIES REVERSING TYPE DRIVE) ?
NO
GO TO "FIELD LOSS' SECTION PAGE, 3-37
YES
YES
IS LIGHT #16 ON? (AUX. WARNING SIGNAL)
YES
DRIVE IS RECEIVING A WARNING FROM SOME AUXILIARY DEVICE, CHECK ELEMENTARY FOR DEVICE DESCRIPTION.
NO
IS LIGHT #17 ON? (AUX. STOP SIGNAL)
YES
DRIVE IS RECEIVING A STOP SIGNAL FROM SOME AUXILIARY DEVICE. CHECK ELEMENTARY FOR DEVICE DESCRIPTION,
NO
IF THE DRIVE CAN POWER AN ALT. MOTOR OR IF THE DRIVE'S MOTOR CAN HE POWERED BY ANOTHER DRIVE, IS LIGHT #79 LIT ?
NO
GO TO 'CONTACTOR TROUBLE SHOOTING' SECTION PG. 5-11.
YES
IS DRIVE READY LIGHT LIT STEADY (NOT FLASHING) ?
NO
REPLACE THE DRIVE CONTROL MODULE.
Y E S
YES
IS DRIVE READY LIGHT LIT STEADY (NOT FLASHING) ? (LIGHTS #1 AND 61)
NO
END
CONTACT YOUR G.E. REPRESENTATIVE.

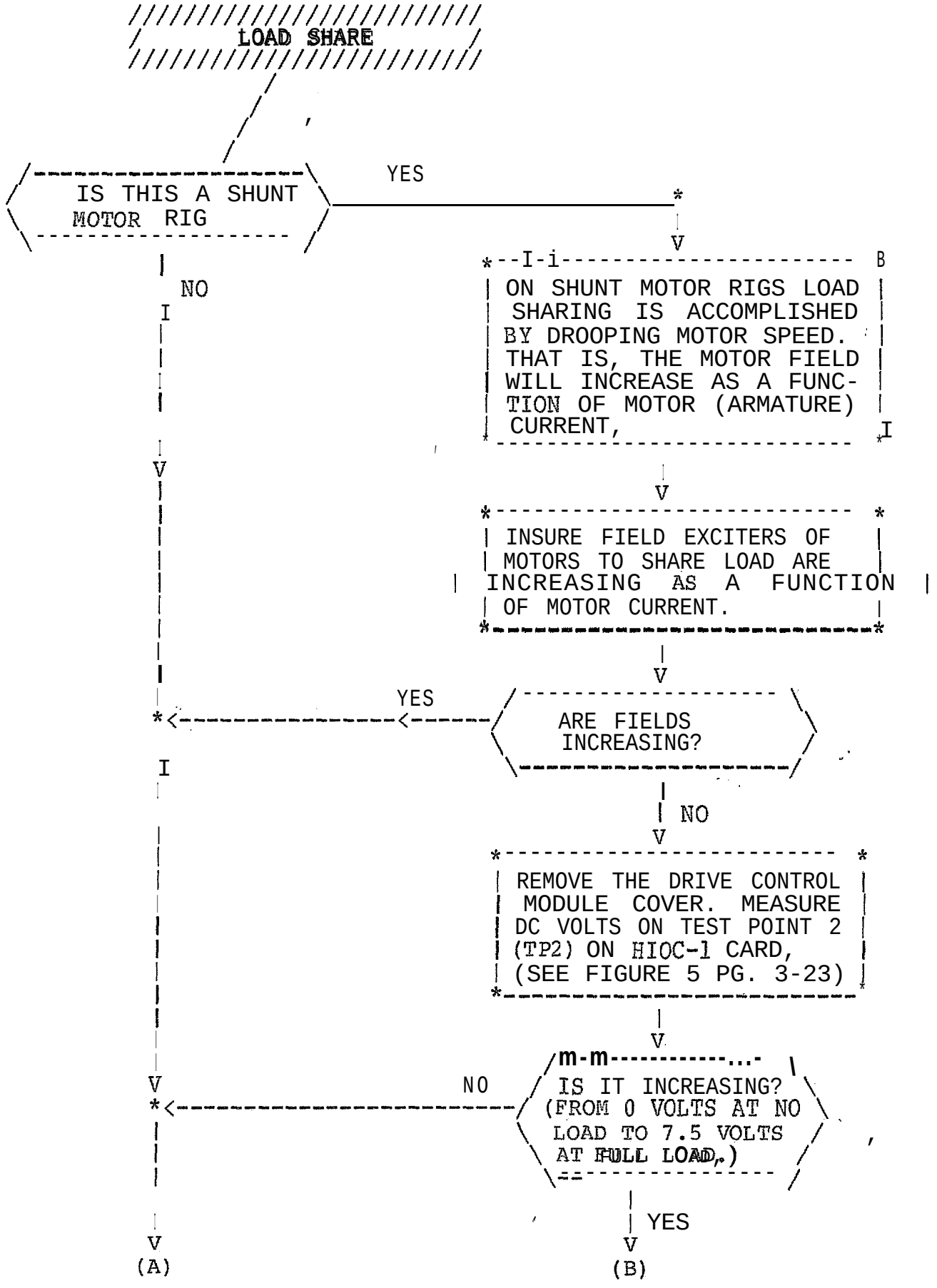

"RUNNING WITH WARNING"
DESCRIPTION:
THIS ROUTINE APPLIES ONLY IF AN AUDIBLE ALARM OR FLASHING LIGHT IS OBSERVED AT THE DRILLER'S CONSOLE. IF THE DRIVE IS THEN DESELECTED, THIS ROUTINE WILL NOT APPLY.
ARE LIGHTS #77 AND 50 LIT ?
YES
'I GO TO *ALTERNATE ASSIGNMENT' SECTION - PG. 5-9
NO
IS 'DRIVE READY LIGHT' FLASHING ? (PROBLEM IF FLASHING;,'
NO
GO'TO PAGE 3-19
YES
IS ALARM IN DRILLER'S CONSOLE SOUNDING ?
YES
IS LIGHT # 67 ON ? (ALARM)
NO
DRIVE IS IN OVERTEMPERATURE CONDITION.
CONSOLE ALARM HAS MALFUNCTION.
(A)


(A)
IS LIGHT #15 LIT ? (BLR. ON)
YES
REPLACE DRIVE CONTROL MODULE SECT. 10.2.1
NO
GO TO 'RELAY INTERFACE MODULE" PAGE 5-1.


(FROM PG. d-17)
IS ALARM IN DRILLER'S CONSOLE SOUNDING ?
YES
NO
IS LIGHT 867 LIT ?
NO
YES
IF ANY OTHER LIGHTS ARE FLASHING, REPLACEMENT OF THE DRIVE CONTROL MODULE MAY BE REQUIRED.
IF LIGHT #6 IS FLASHING AND DRIVE HAS AN AUX., IS LIGHT #62 LIT ?
NO
REPLACE THE DRIVE CONTROL MODULE USING SECT, 10.2.1
YES
IF LIGHT #62 WAS LIT, IS LIGHT #72 LIT ?
NO
GO TO "RELAY INTERFACE' SECTION PAGE 5-1.
YES
IF LIGHT #8 IS FLASHING AND DRIVE IS A MUD PUMP, IS LIGHT # 63 LIT ?
NO
CHECK PERSONALITY PLUG JUMPERING AND IF CORRECT PER ELEMENTARY, REPLACE DRIVE CONTROL MODULE PER SECT. 10.2.1
YES
IF LIGHT #63 WAS LIT, IS LIGHT #73 LIT ?
NO
GO TO 'RELAY INTERFACE'1 SECTION - PAGE 5-1.
YES
(A)


(A)
IS LIGHT # 16 ON? (INDICATES AN AUX. WARN SIGNAL)
NO
REPLACE DRIVE CONTROL MODULE PER PAGE 3-22,
YES
DRIVE IS RECEIVING A WARNING SIGNAL FROM SOME AUXILIARY DEVICE. CHECK ELEMENTARY FOR DEVICE DESCRIPTION..


RUNNING ABNORMAL
ARE THE DRIVES FAILING TO SHARE LOAD?
YES
GO TO PAGE 3-221
NO
ARE THE DRIVES EXPERIENCING A LOSS OF POWER ?
YES
GO TO PAGE 3-251
NO
IS THE CATHEAD TAKING TOO LONG TO COAST TO BASE SPEED (DURING TRIPPING) ?
YES
GO TO PAGE 3-28
NO
GO TO MISCELLANEOUS SECTION- PAGE 5-1


LOAD SHARE
IS THIS A SHUNT MOTOR RIG
YES
NO
ON SHUNT MOTOR RIGS LOAD SHARING IS ACCOMPLISHED BY DROOPING MOTOR SPEED. THAT IS, THE MOTOR FIELD WILL INCREASE AS A FUNCTION OF MOTOR (ARMATURE) CURRENT,
INSURE FIELD EXCITERS OF MOTORS TO SHARE LOAD ARE INCREASING AS A FUNCTION OF MOTOR CURRENT.
ARE FIELDS INCREASING?
YES
NO
REMOVE THE DRIVE CONTROL MODULE COVER. MEASURE DC VOLTS ON TEST POINT 2 (TP2) ON HIOC-1 CARD, (SEE FIGURE 5 PG. 3-23)
IS IT INCREASING? (FROM 0 VOLTS AT NO LOAD TO 7.5 VOLTS AT FULL LOAD.)
NO
YES
(A)
(B)

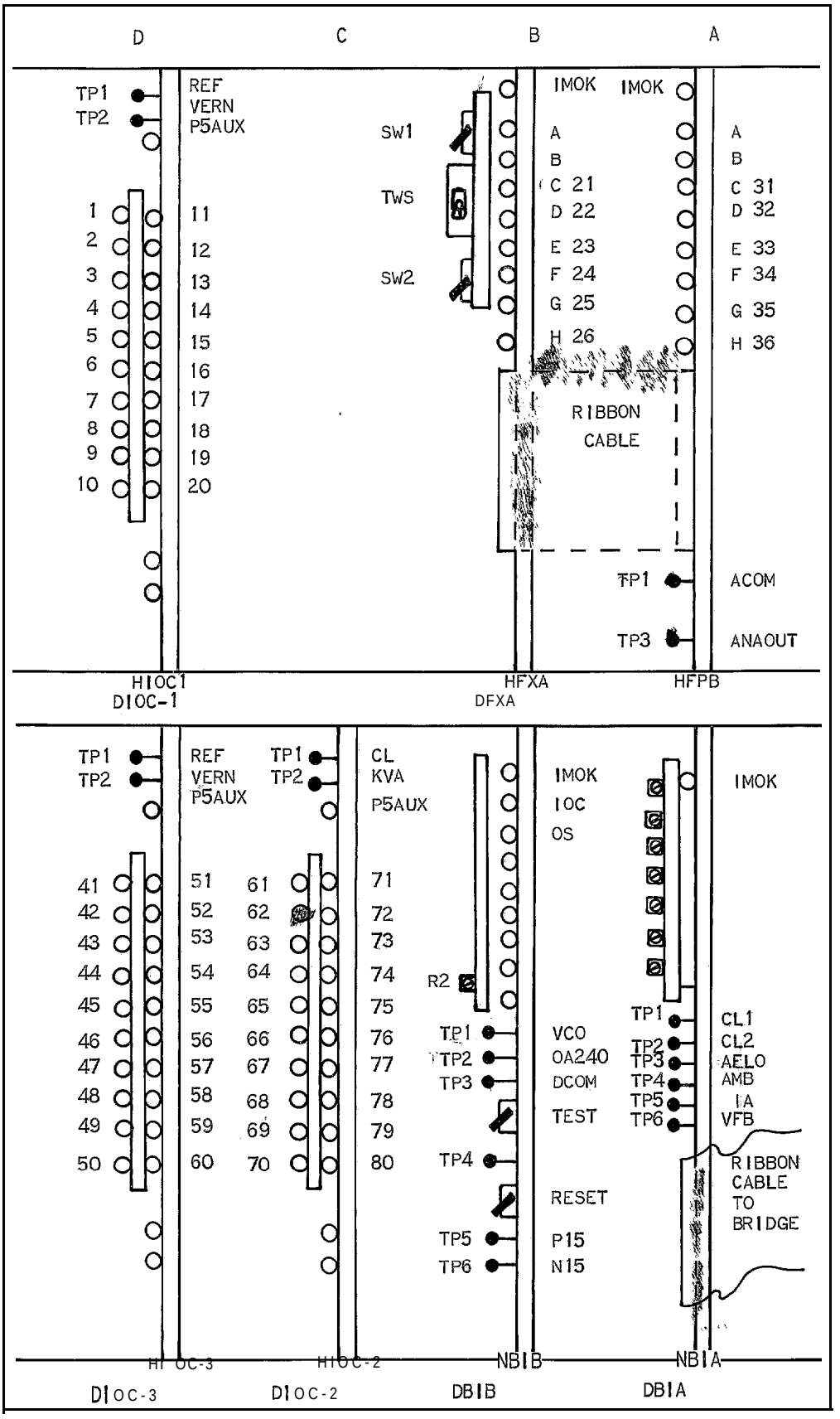
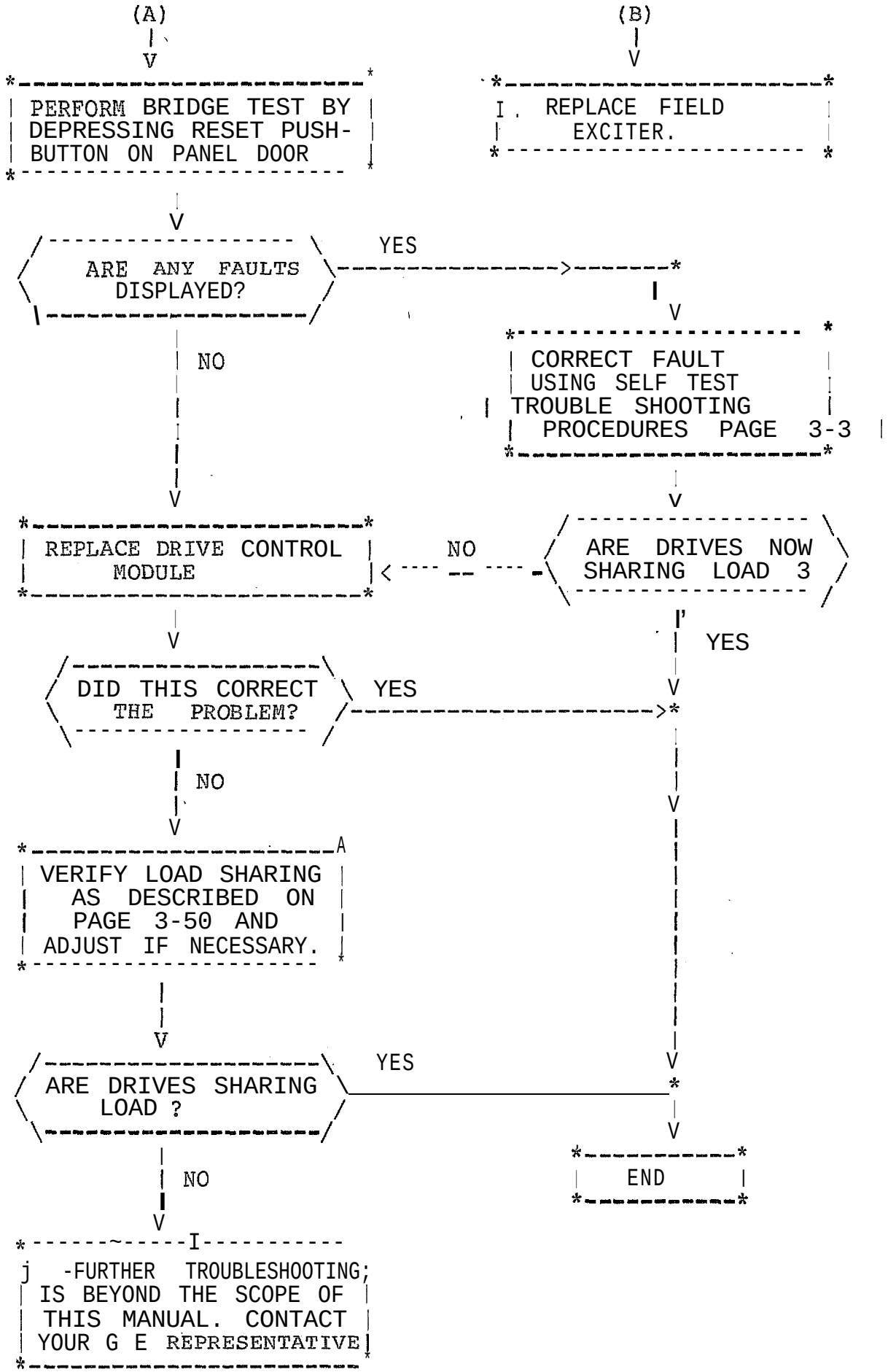
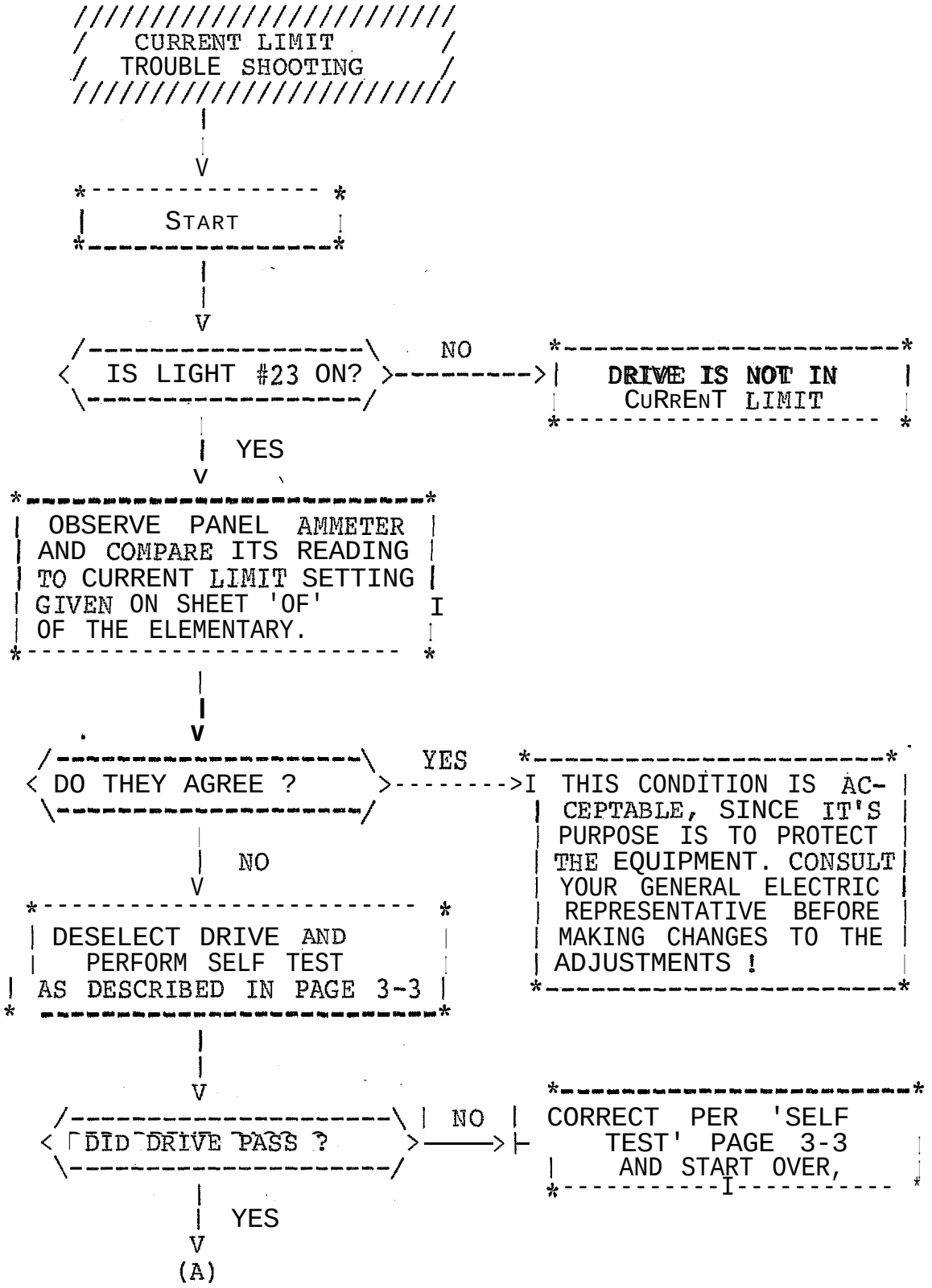


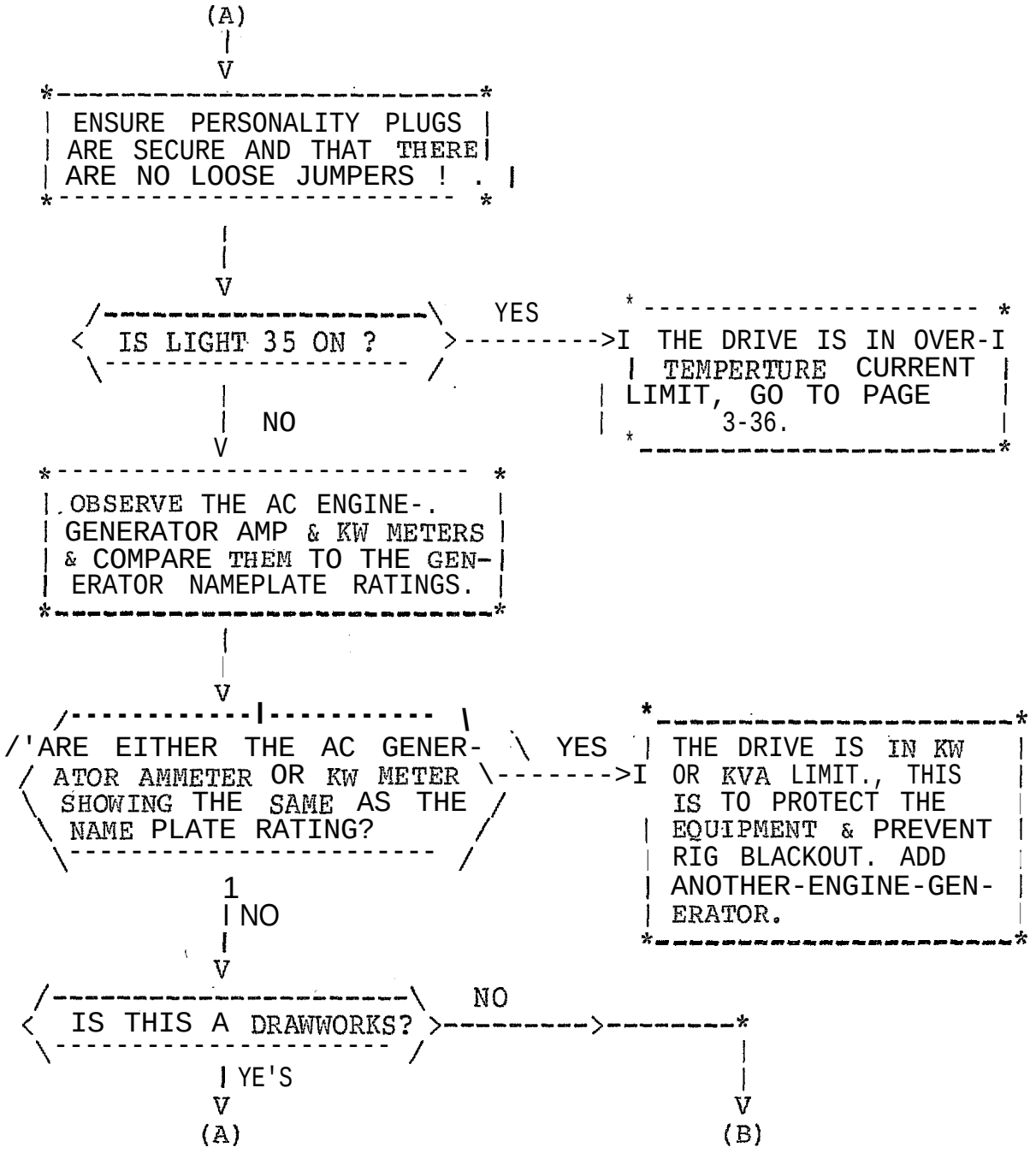
FIGURE _____

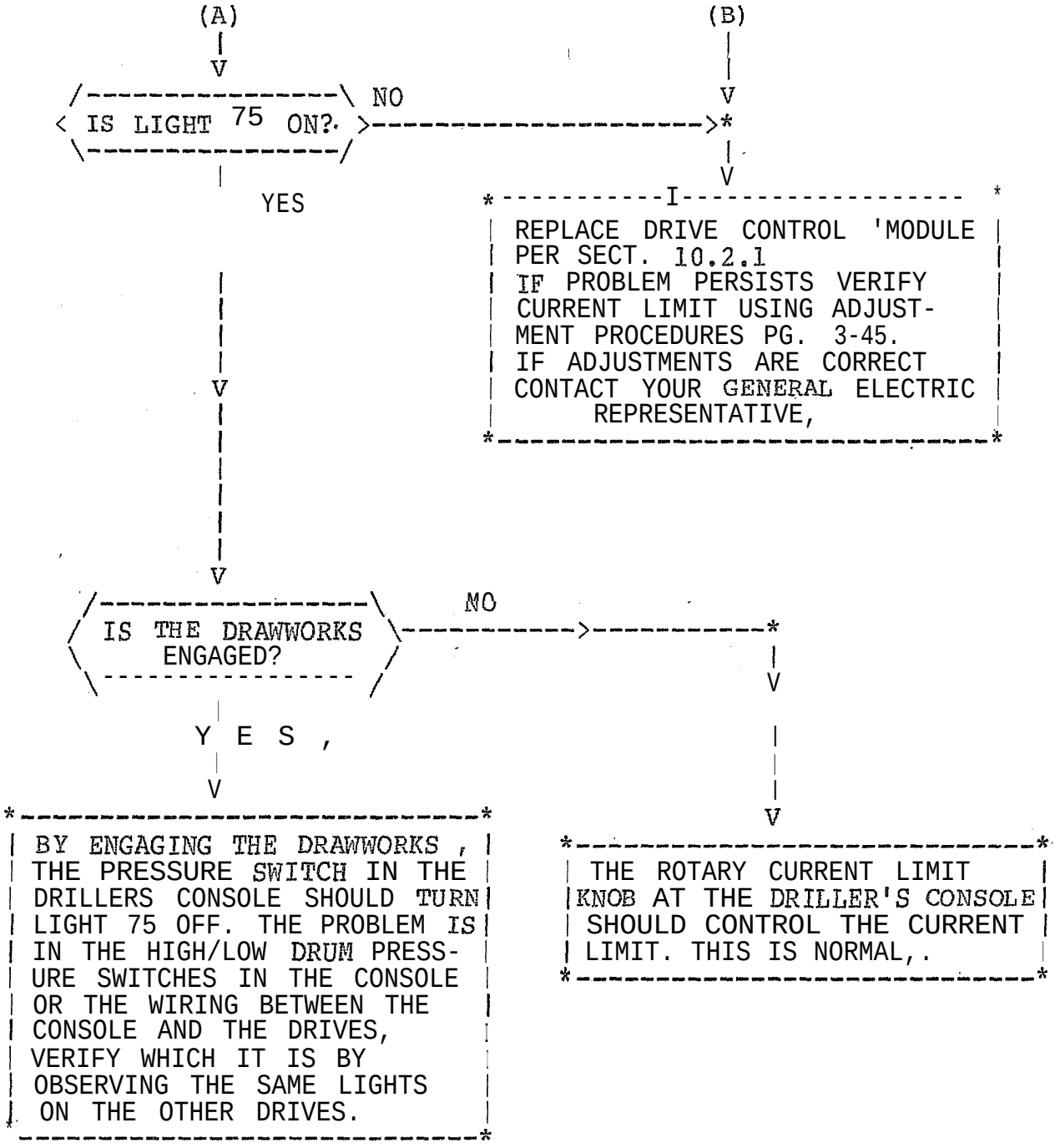
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

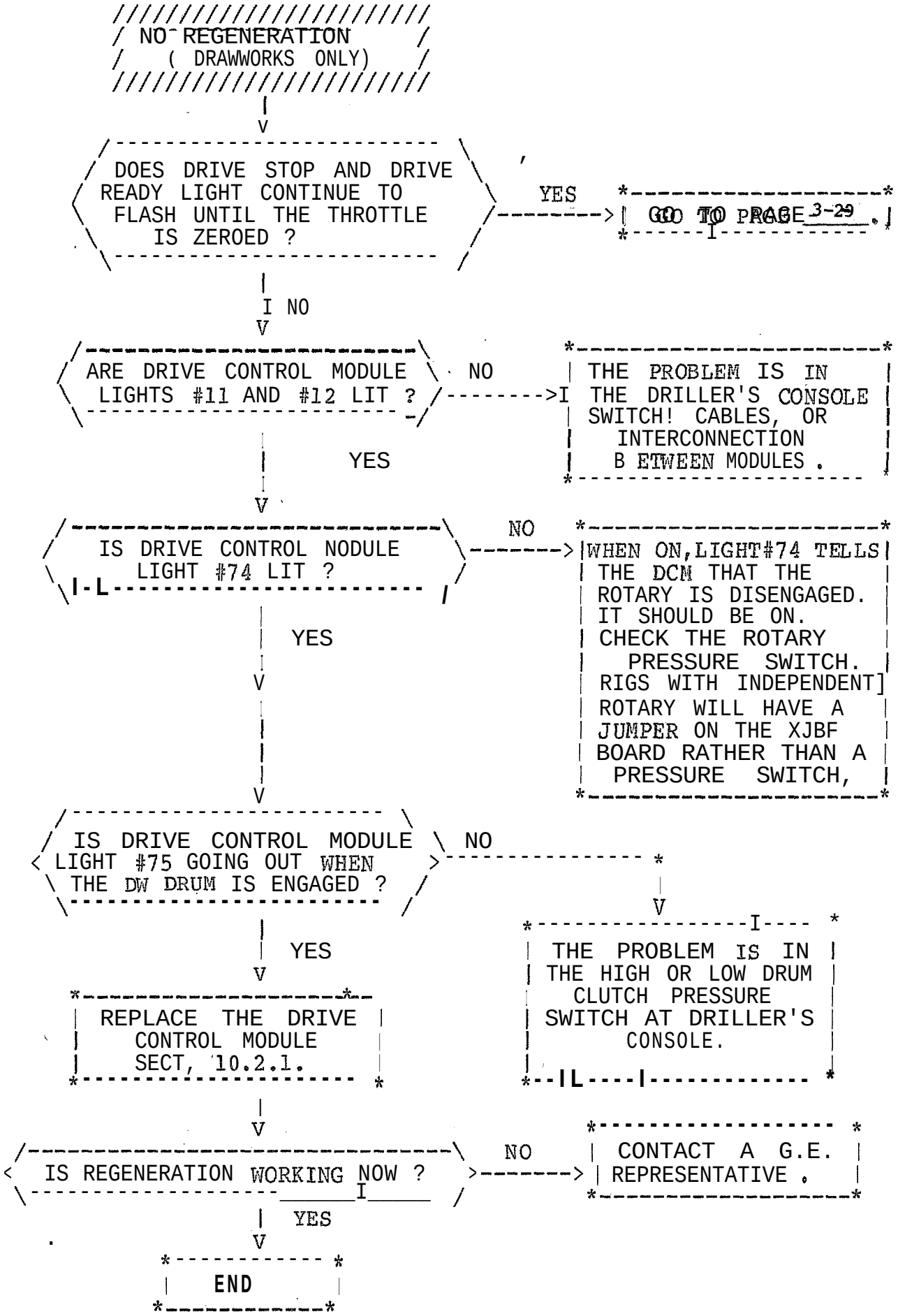
EV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
EV. 2	ISSUED			FIRST MADE FOR		
EV. 3	MADE BY			I.C. NO.		CONT. ON SH. SH. NO.

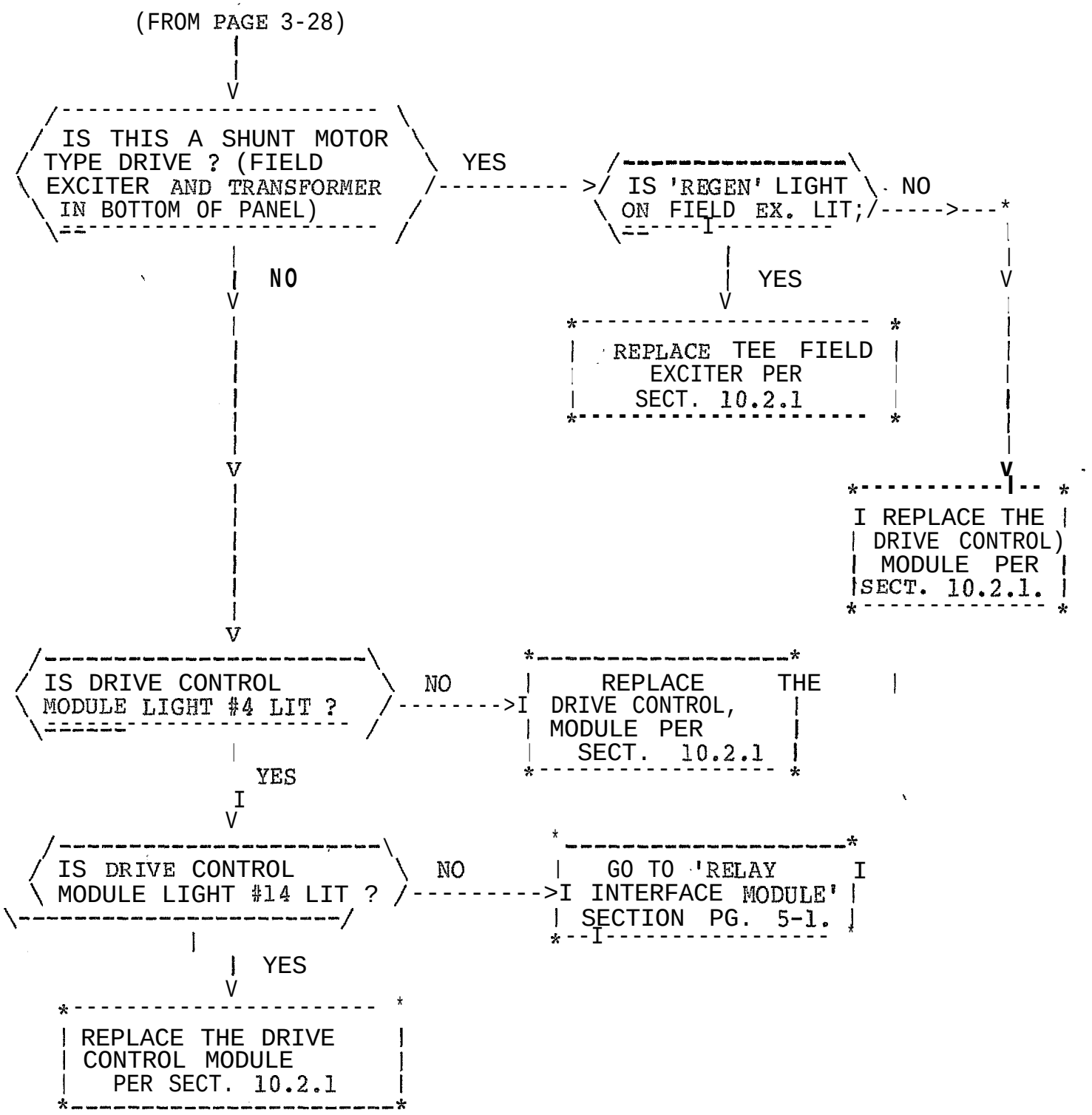

(A)
(B)
PERFORM BRIDGE TEST BY DEPRESSING RESET PUSH-BUTTON ON PANEL DOOR
REPLACE FIELD EXCITER.
ARE ANY FAULTS DISPLAYED?
YES
NO
CORRECT FAULT USING SELF TEST TROUBLE SHOOTING PROCEDURES PAGE 3-3
REPLACE DRIVE CONTROL MODULE
NO
ARE DRIVES NOW SHARING LOAD 3
YES
DID THIS CORRECT THE PROBLEM?
YES
NO
VERIFY LOAD SHARING AS DESCRIBED ON PAGE 3-50 AND ADJUST IF NECESSARY.
ARE DRIVES SHARING LOAD ?
YES
NO
END
-FURTHER TROUBLESHOOTING; IS BEYOND THE SCOPE OF THIS MANUAL. CONTACT YOUR G E REPRESENTATIVE

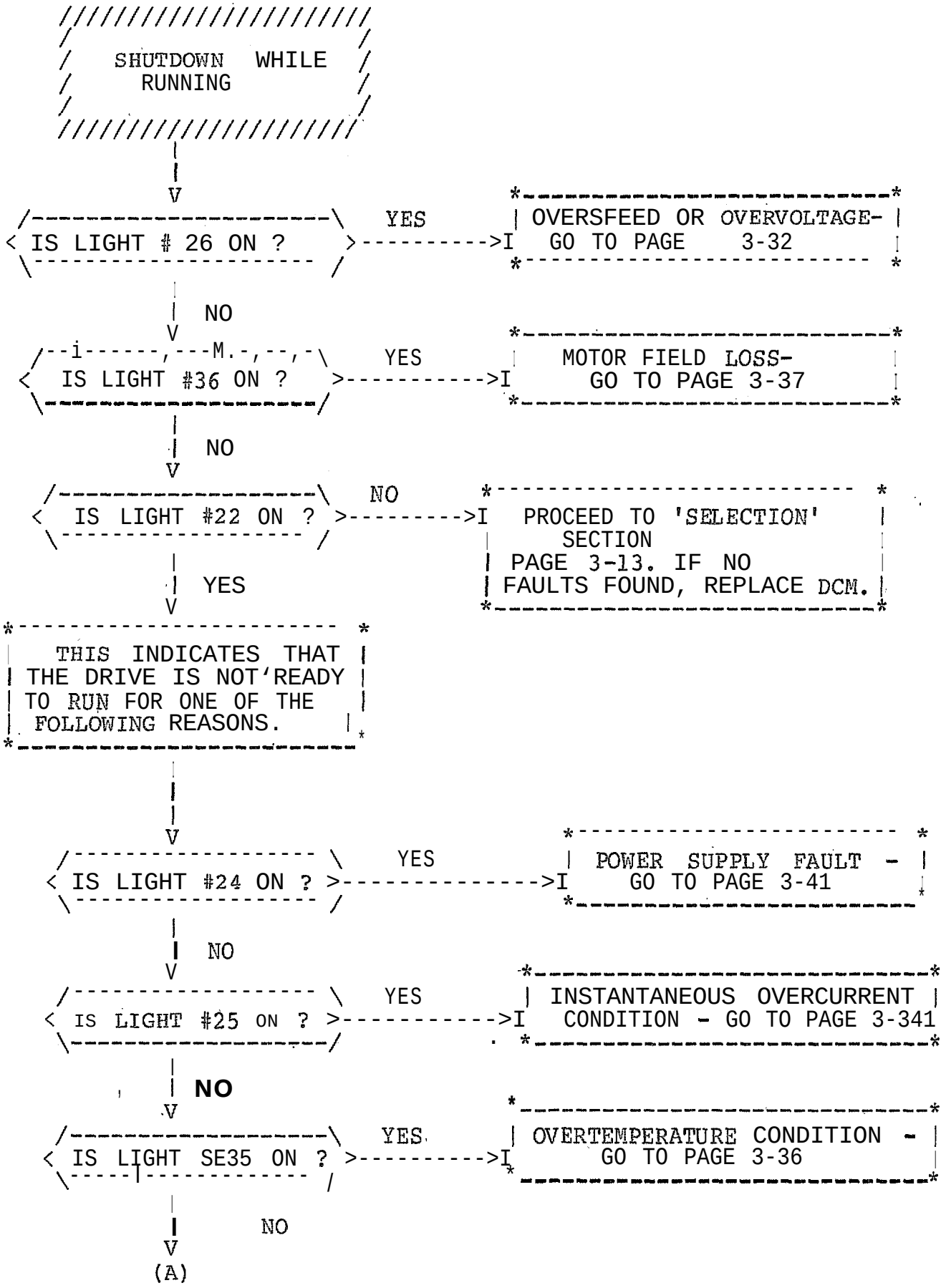

CURRENT LIMIT
TROUBLE SHOOTING
START
IS LIGHT #23 ON?
NO
DRIVE IS NOT IN CURRENT LIMIT
YES
OBSERVE PANEL AMMETER AND COMPARE ITS READING TO CURRENT LIMIT SETTING GIVEN ON SHEET 'OF' OF THE ELEMENTARY.
DO THEY AGREE ?
YES
THIS CONDITION IS ACCEPTABLE, SINCE IT'S PURPOSE IS TO PROTECT THE EQUIPMENT. CONSULT YOUR GENERAL ELECTRIC REPRESENTATIVE BEFORE MAKING CHANGES TO THE ADJUSTMENTS !
NO
DESELECT DRIVE AND PERFORM SELF TEST AS DESCRIBED IN PAGE 3-3
DID DRIVE PASS ?
NO
CORRECT PER 'SELF TEST' PAGE 3-3 AND START OVER,
YES
(A)

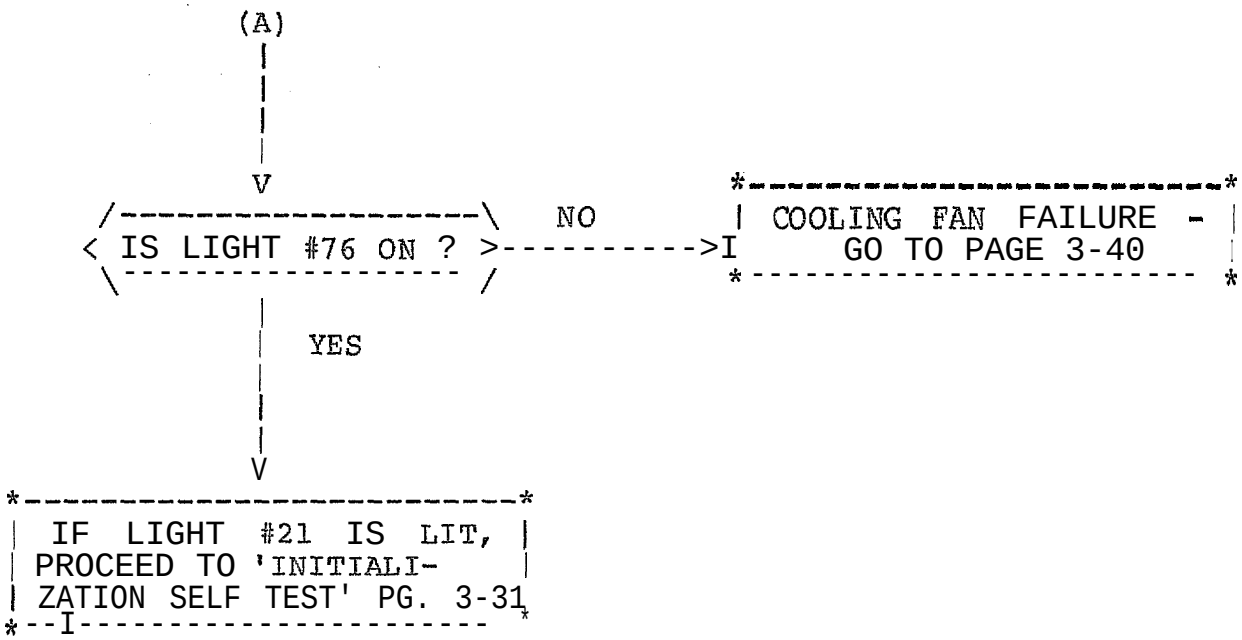

(A)
ENSURE PERSONALITY PLUGS ARE SECURE AND THAT THERE ARE NO LOOSE JUMPERS ! .
IS LIGHT 35 ON ?
YES
THE DRIVE IS IN OVER-I TEMPERTURE CURRENT LIMIT, GO TO PAGE 3-36.
NO
OBSERVE THE AC ENGINE-. GENERATOR AMP & KW METERS & COMPARE THEM TO THE GEN- ERATOR NAMEPLATE RATINGS.
'ARE EITHER THE AC GENER- ATOR AMMETER OR KW METER SHOWING THE SAME AS THE NAME PLATE RATING?
YES
THE DRIVE IS IN KW OR KVA LIMIT., THIS IS TO PROTECT THE EQUIPMENT & PREVENT RIG BLACKOUT. ADD ANOTHER-ENGINE-GEN- ERATOR.
1
NO
IS THIS A DRAWWORKS?
NO
YE'S
(A)
(B)

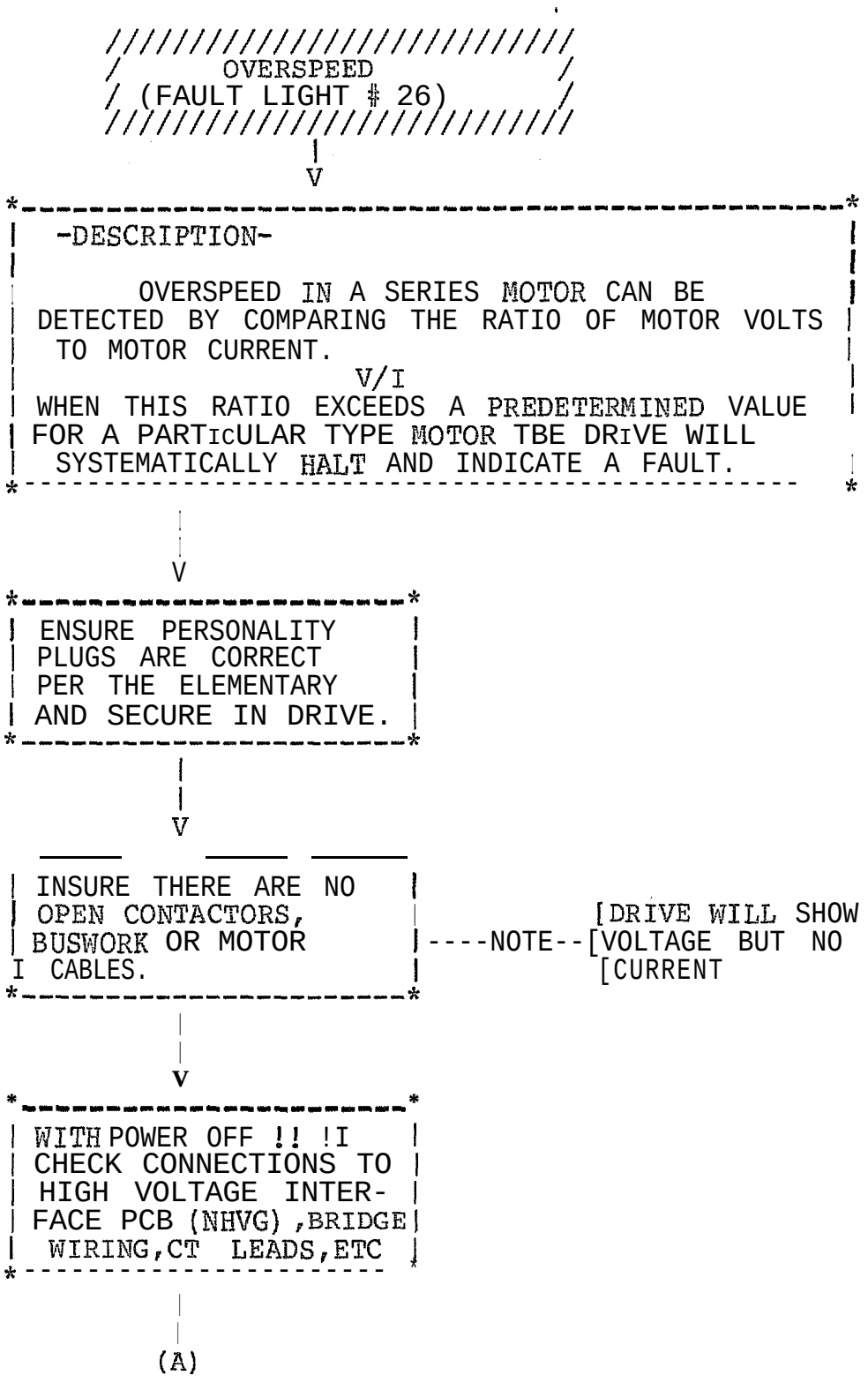

(A)
(B)
IS LIGHT 75 ON?
NO
YES
REPLACE DRIVE CONTROL 'MODULE PER SECT. 10.2.1
IF PROBLEM PERSISTS VERIFY CURRENT LIMIT USING ADJUSTMENT PROCEDURES PG. 3-45.
IF ADJUSTMENTS ARE CORRECT CONTACT YOUR GENERAL ELECTRIC REPRESENTATIVE,
IS THE DRAWWORKS ENGAGED?
NO
YES,
BY ENGAGING THE DRAWWORKS, THE PRESSURE SWITCH IN THE DRILLERS CONSOLE SHOULD TURN LIGHT 75 OFF. THE PROBLEM IS IN THE HIGH/LOW DRUM PRESSURE SWITCHES IN THE CONSOLE OR THE WIRING BETWEEN THE CONSOLE AND THE DRIVES, VERIFY WHICH IT IS BY OBSERVING THE SAME LIGHTS ON THE OTHER DRIVES.
THE ROTARY CURRENT LIMIT KNOB AT THE DRILLER'S CONSOLE SHOULD CONTROL THE CURRENT LIMIT. THIS IS NORMAL,.

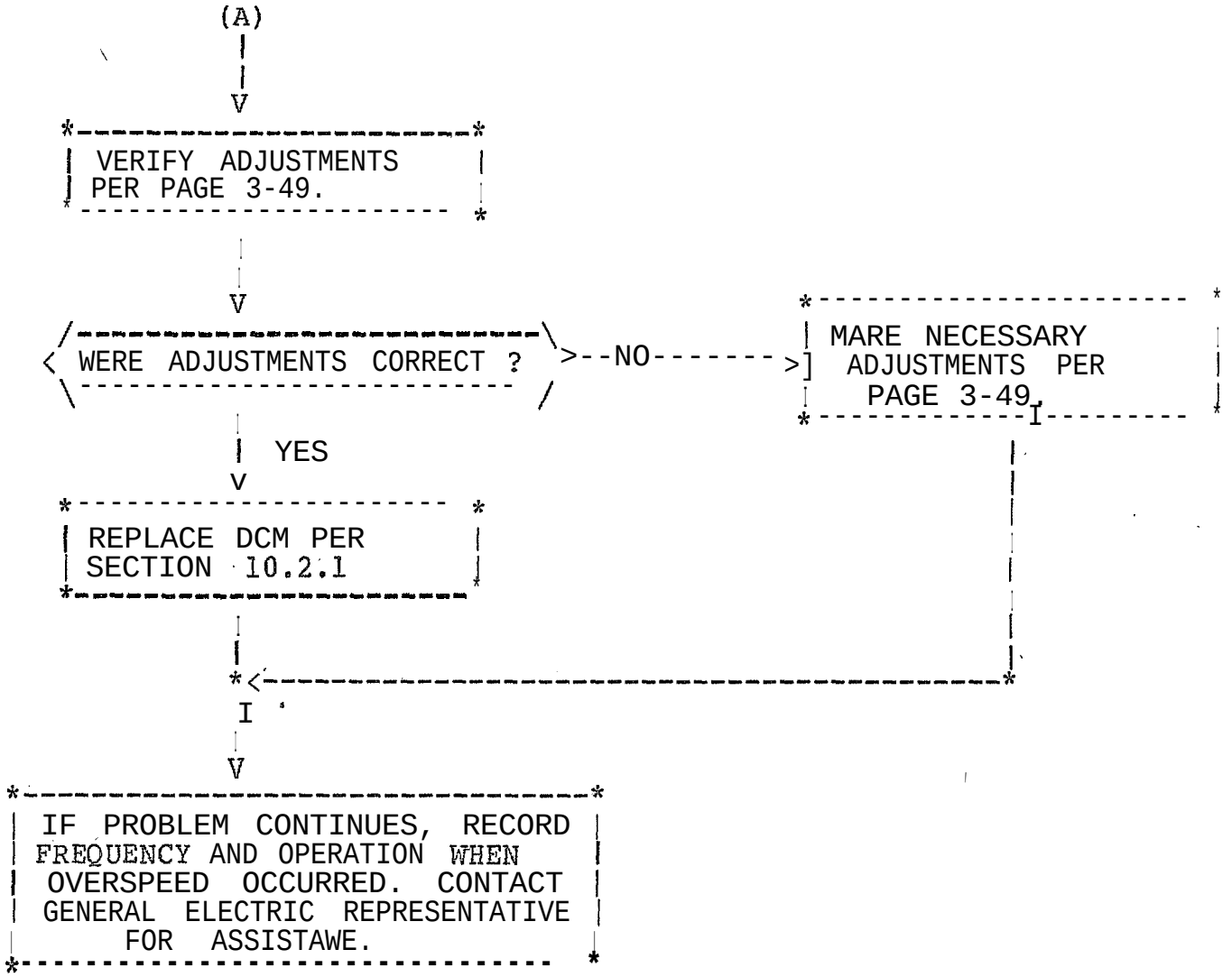

NO REGENERATION
(DRAWWORKS ONLY)
DOES DRIVE STOP AND DRIVE READY LIGHT CONTINUE TO FLASH UNTIL THE THROTTLE IS ZEROED ?
YES
GO TO PAGE 3-29
I NO
ARE DRIVE CONTROL MODULE LIGHTS #11 AND #12 LIT ?
NO
THE PROBLEM IS IN THE DRILLER'S CONSOLE SWITCH! CABLES, OR INTERCONNECTION B ETWEEN MODULES .
YES
IS DRIVE CONTROL NODULE LIGHT #74 LIT ?
NO
WHEN ON,LIGHT#74 TELLS THE DCM THAT THE ROTARY IS DISENGAGED. IT SHOULD BE ON. CHECK THE ROTARY PRESSURE SWITCH. RIGS WITH INDEPENDENT] ROTARY WILL HAVE A JUMPER ON THE XJBF BOARD RATHER THAN A PRESSURE SWITCH,
YES
IS DRIVE CONTROL MODULE LIGHT #75 GOING OUT WHEN THE DW DRUM IS ENGAGED ?
NO
THE PROBLEM IS IN THE HIGH OR LOW DRUM CLUTCH PRESSURE SWITCH AT DRILLER'S CONSOLE.
YES
REPLACE THE DRIVE CONTROL MODULE SECT, 10.2.1.
IS REGENERATION WORKING NOW ?
NO
CONTACT A G.E. REPRESENTATIVE .
YES
END

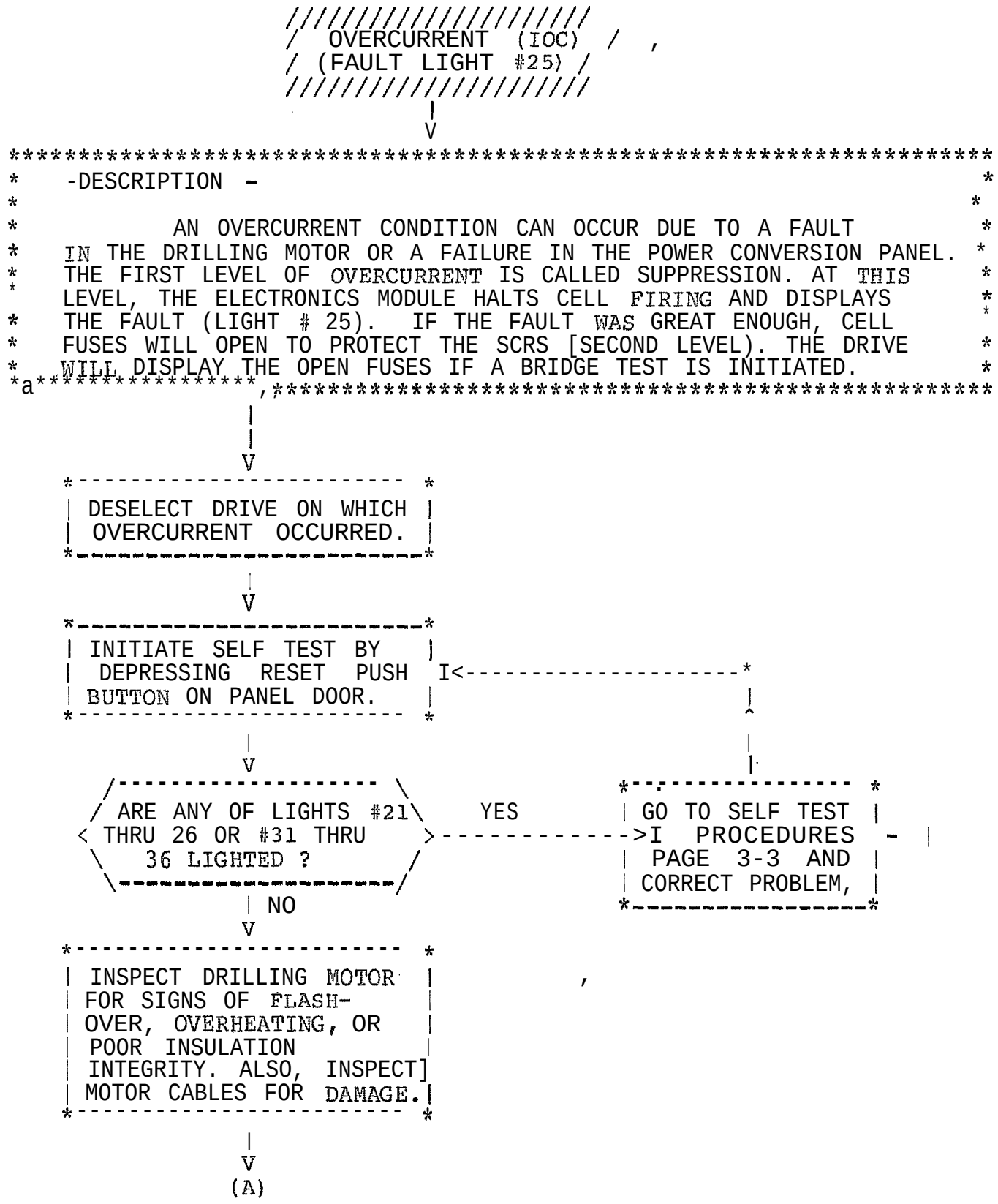

(FROM PAGE 3-28)
IS THIS A SHUNT MOTOR TYPE DRIVE ? (FIELD EXCITER AND TRANSFORMER IN BOTTOM OF PANEL)
YES
NO
IS 'REGEN' LIGHT ON FIELD EX. LIT;
NO
YES
REPLACE TEE FIELD EXCITER PER SECT. 10.2.1
I REPLACE THE DRIVE CONTROL) MODULE PER SECT. 10.2.1.
IS DRIVE CONTROL MODULE LIGHT #4 LIT ?
NO
REPLACE THE DRIVE CONTROL, MODULE PER SECT. 10.2.1
YES
IS DRIVE CONTROL MODULE LIGHT #14 LIT ?
NO
GO TO 'RELAY INTERFACE MODULE' SECTION PG. 5-1.
YES
REPLACE THE DRIVE CONTROL MODULE PER SECT. 10.2.1

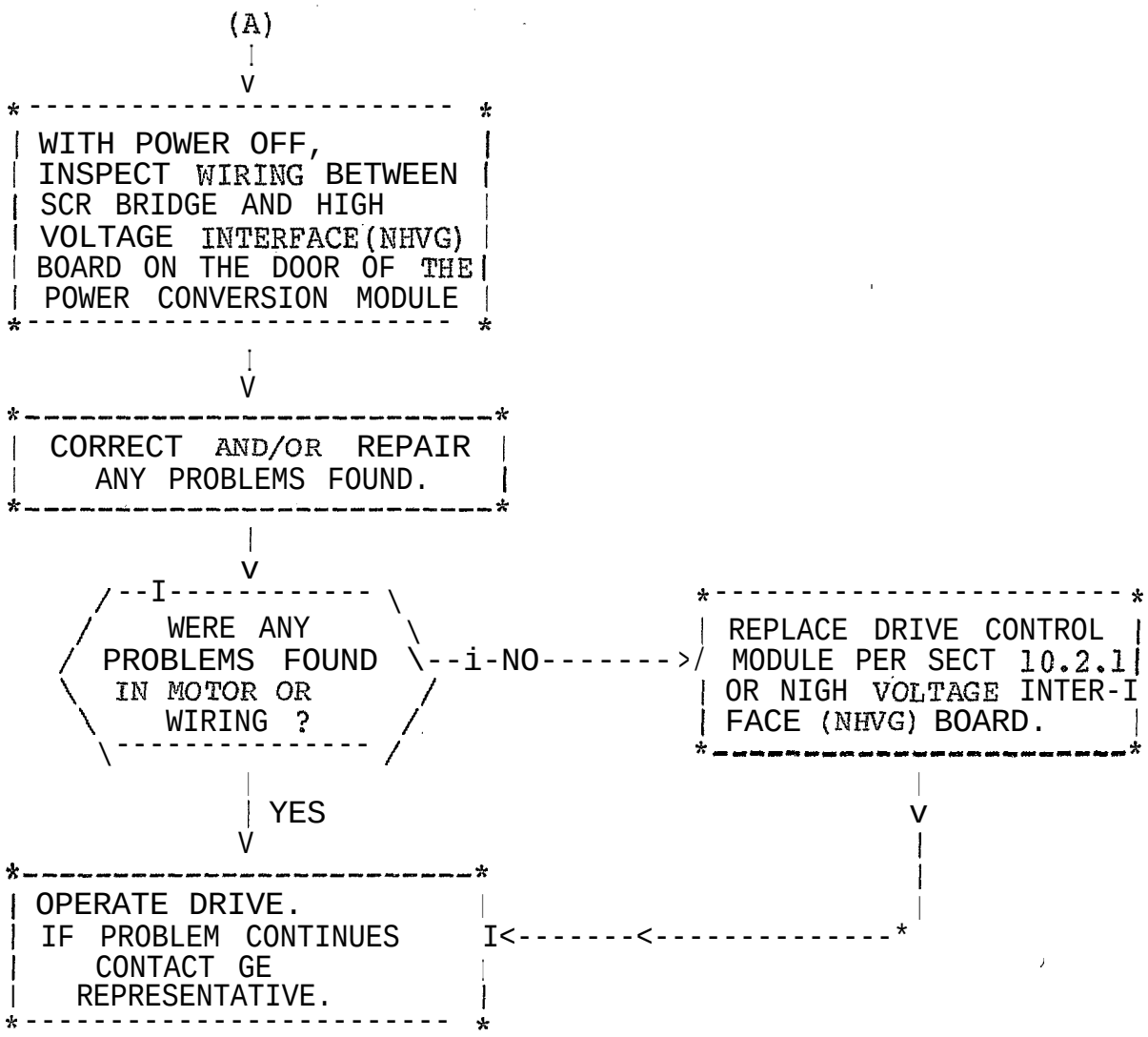

SHUTDOWN WHILE RUNNING
IS LIGHT # 26 ON ?
YES
OVERSFEED OR OVERVOLTAGE- GO TO PAGE 3-32
NO
IS LIGHT #36 ON ?
YES
MOTOR FIELD LOSS- GO TO PAGE 3-37
NO
IS LIGHT #22 ON ?
NO
PROCEED TO 'SELECTION' SECTION PAGE 3-13. IF NO FAULTS FOUND, REPLACE DCM.
YES
THIS INDICATES THAT THE DRIVE IS NOT READY TO RUN FOR ONE OF THE FOLLOWING REASONS.
IS LIGHT #24 ON ?
YES
POWER SUPPLY FAULT - GO TO PAGE 3-41
NO
IS LIGHT #25 ON ?
YES
INSTANTANEOUS OVERCURRENT CONDITION - GO TO PAGE 3-341
NO
IS LIGHT SE35 ON ?
YES
OVERTEMPERATURE CONDITION - GO TO PAGE 3-36
NO
(A)


(A)
IS LIGHT #76 ON ?
NO
COOLING FAN FAILURE - GO TO PAGE 3-40
YES
IF LIGHT #21 IS LIT, PROCEED TO 'INITIALI- ZATION SELF TEST' PG. 3-31


OVERSPEED
(FAULT LIGHT # 26)
-DESCRIPTION-
OVERSPEED IN A SERIES MOTOR CAN BE DETECTED BY COMPARING THE RATIO OF MOTOR VOLTS TO MOTOR CURRENT.
V/I
WHEN THIS RATIO EXCEEDS A PREDETERMINED VALUE FOR A PARTICULAR TYPE MOTOR TBE DRIVE WILL SYSTEMATICALLY HALT AND INDICATE A FAULT.
ENSURE PERSONALITY PLUGS ARE CORRECT PER THE ELEMENTARY AND SECURE IN DRIVE.
INSURE THERE ARE NO OPEN CONTACTORS, BUSWORK OR MOTOR CABLES.
NOTE--[DRIVE WILL SHOW [VOLTAGE BUT NO [CURRENT
WITH POWER OFF !! !I CHECK CONNECTIONS TO HIGH VOLTAGE INTER- FACE PCB (NHVG) ,BRIDGE WIRING,CT LEADS,ETC
(A)


(A)
VERIFY ADJUSTMENTS PER PAGE 3-49.
WERE ADJUSTMENTS CORRECT ?
NO
MARE NECESSARY ADJUSTMENTS PER PAGE 3-49,
YES
REPLACE DCM PER SECTION 10.2.1
IF PROBLEM CONTINUES, RECORD FREQUENCY AND OPERATION WHEN OVERSPEED OCCURRED. CONTACT GENERAL ELECTRIC REPRESENTATIVE FOR ASSISTAWE.


OVERCURRENT (IOC)
(FAULT LIGHT #25)
-DESCRIPTION -
AN OVERCURRENT CONDITION CAN OCCUR DUE TO A FAULT IN THE DRILLING MOTOR OR A FAILURE IN THE POWER CONVERSION PANEL. THE FIRST LEVEL OF OVERCURRENT IS CALLED SUPPRESSION. AT THIS LEVEL, THE ELECTRONICS MODULE HALTS CELL FIRING AND DISPLAYS THE FAULT (LIGHT # 25). IF THE FAULT WAS GREAT ENOUGH, CELL FUSES WILL OPEN TO PROTECT THE SCRS [SECOND LEVEL). THE DRIVE WILL DISPLAY THE OPEN FUSES IF A BRIDGE TEST IS INITIATED.
DESELECT DRIVE ON WHICH OVERCURRENT OCCURRED.
INITIATE SELF TEST BY DEPRESSING RESET PUSH BUTTON ON PANEL DOOR.
ARE ANY OF LIGHTS #21 THRU 26 OR #31 THRU 36 LIGHTED ?
YES
GO TO SELF TEST PROCEDURES - PAGE 3-3 AND CORRECT PROBLEM,
NO
INSPECT DRILLING MOTOR FOR SIGNS OF FLASH-OVER, OVERHEATING, OR POOR INSULATION INTEGRITY. ALSO, INSPECT] MOTOR CABLES FOR DAMAGE.
(A)

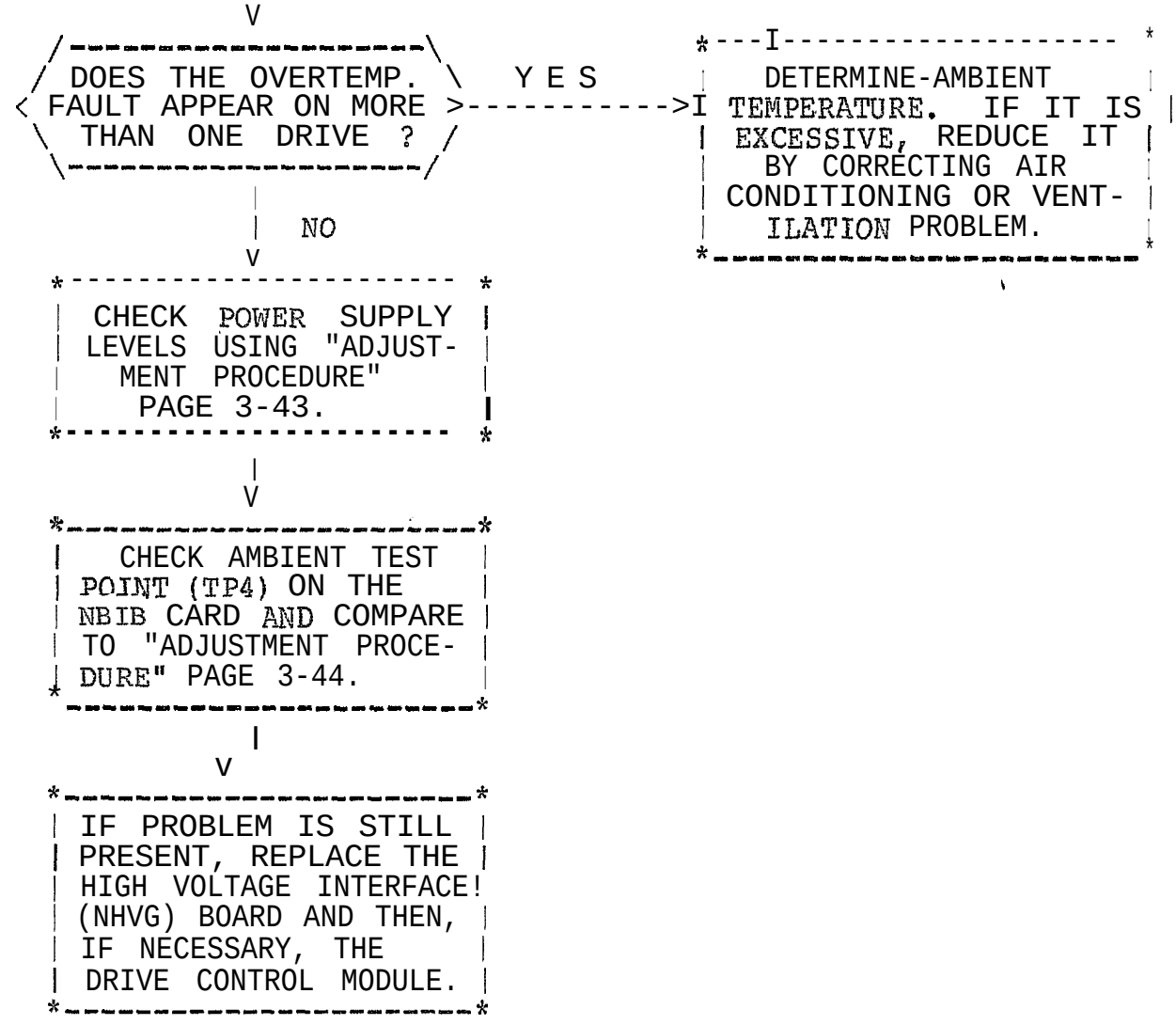

(A)
WITH POWER OFF, INSPECT WIRING BETWEEN SCR BRIDGE AND HIGH VOLTAGE INTERFACE(NHVG) BOARD ON THE DOOR OF THE POWER CONVERSION MODULE
CORRECT AND/OR REPAIR ANY PROBLEMS FOUND.
WERE ANY PROBLEMS FOUND IN MOTOR OR WIRING ?
NO
REPLACE DRIVE CONTROL MODULE PER SECT 10.2.1 OR NIGH VOLTAGE INTER-FACE (NHVG) BOARD.
YES
OPERATE DRIVE. IF PROBLEM CONTINUES CONTACT GE REPRESENTATIVE.

OVERTEMPERATURE TROUBLE SHOOTING (FAULT LIGHT #35 & #22)

DESCRIPTION:

OVERTEMPERATURE WILL ONLY SHUT A DRIVE DOWN IF THE AMBIENT TEMPERATURE OF THE POWER CONVERTER ROOM EXCEEDS 50-60 DEGREES C. (120-140 F.). SENSING IS PERFORMED BY A TEMPERATURE SENSOR ON THE HIGH VOLTAGE INTERFACE (NHVG) BOARD, 'LOCATED ON THE POWER CONVERSION MODULE DOOR, AND IS SENT TO THE DRIVE CONTROL MODULE. THE FAULT CAN BE CLEARED BY:

1. REDUCING THE TEMPERATURE
2. REPLACING A FAILED SENSOR



FIELD LOSS TROUBLE SHOOTING (FAULT LIGHT #36)

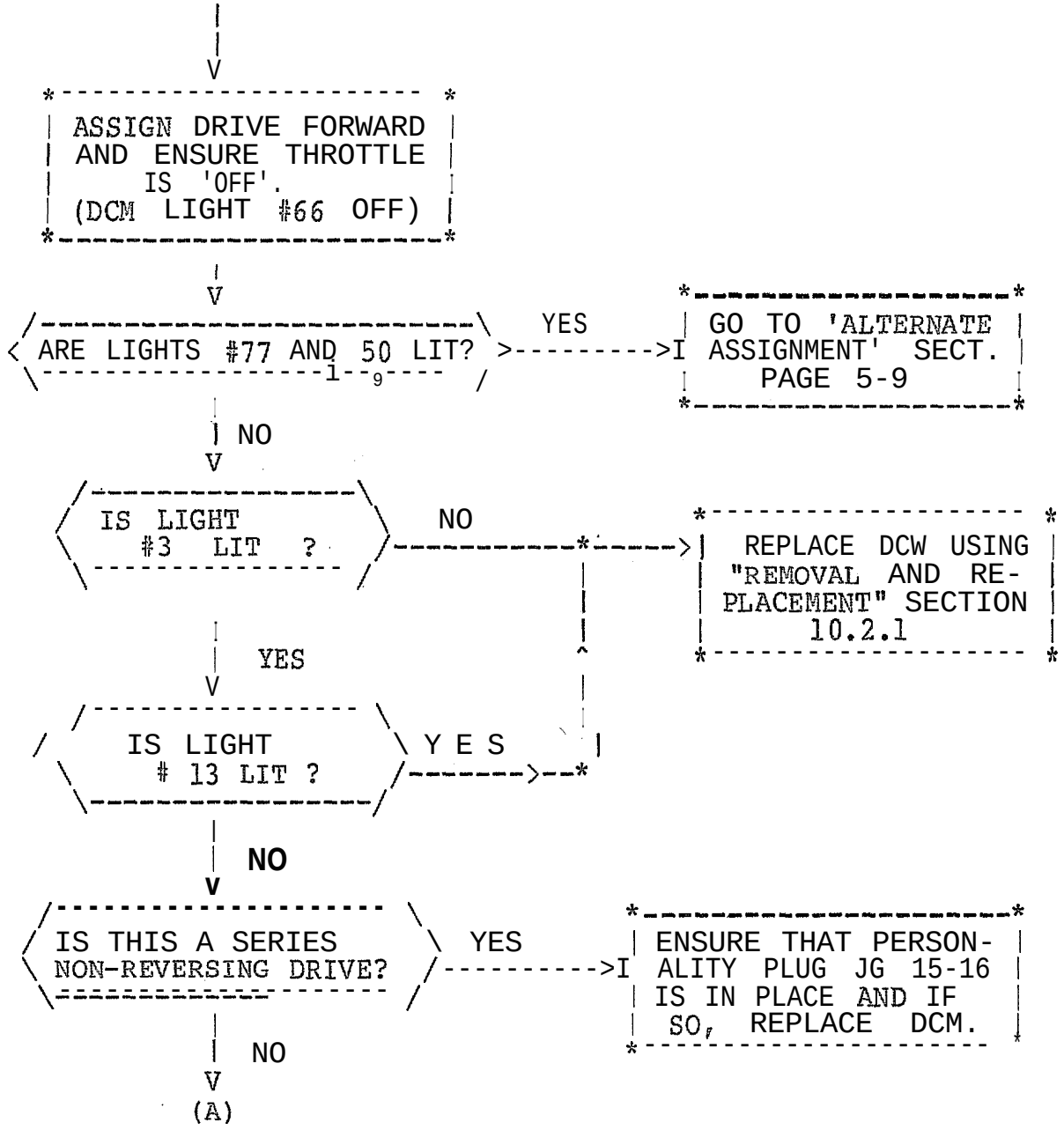
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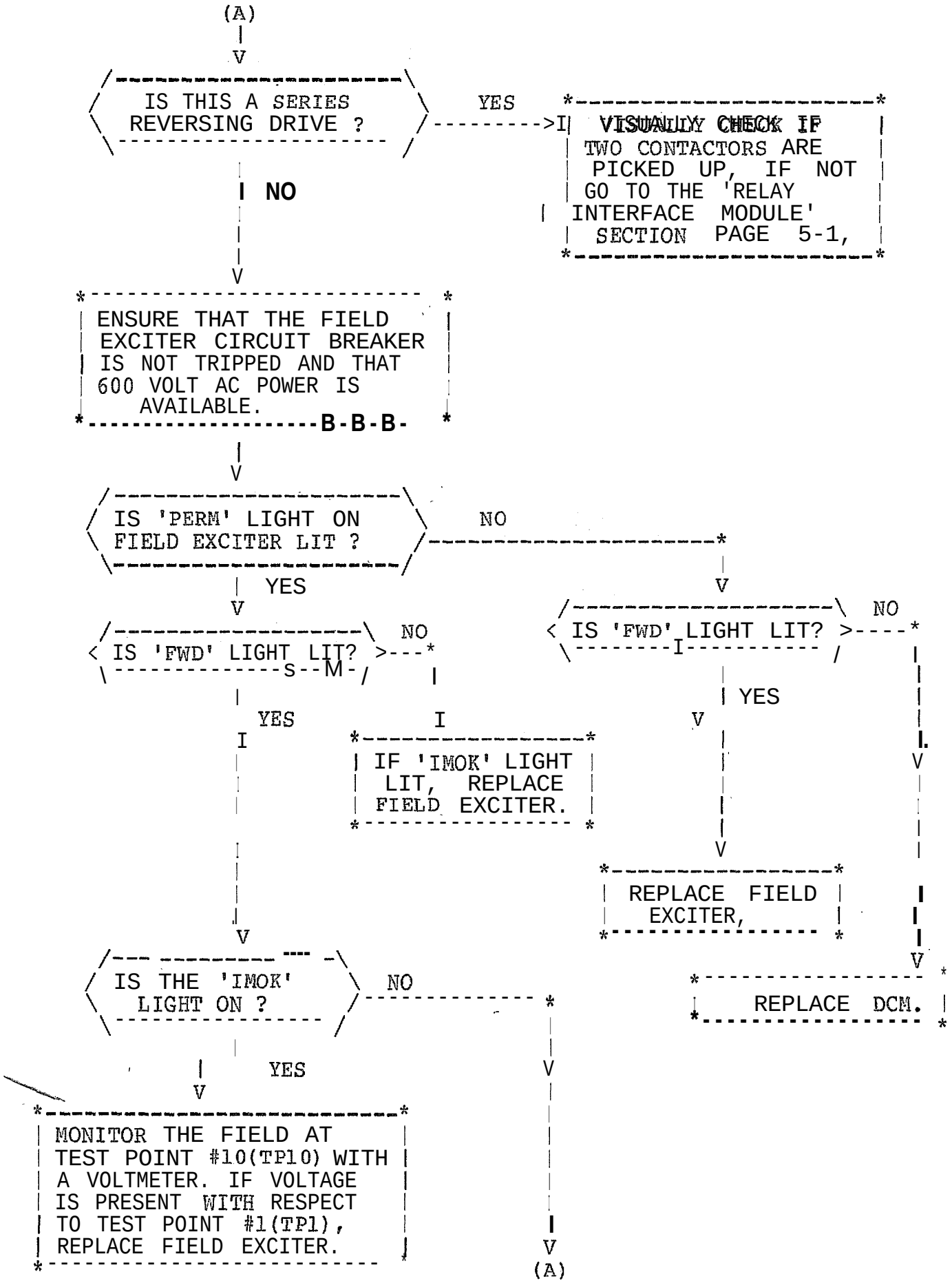
* DESCRIPTION:

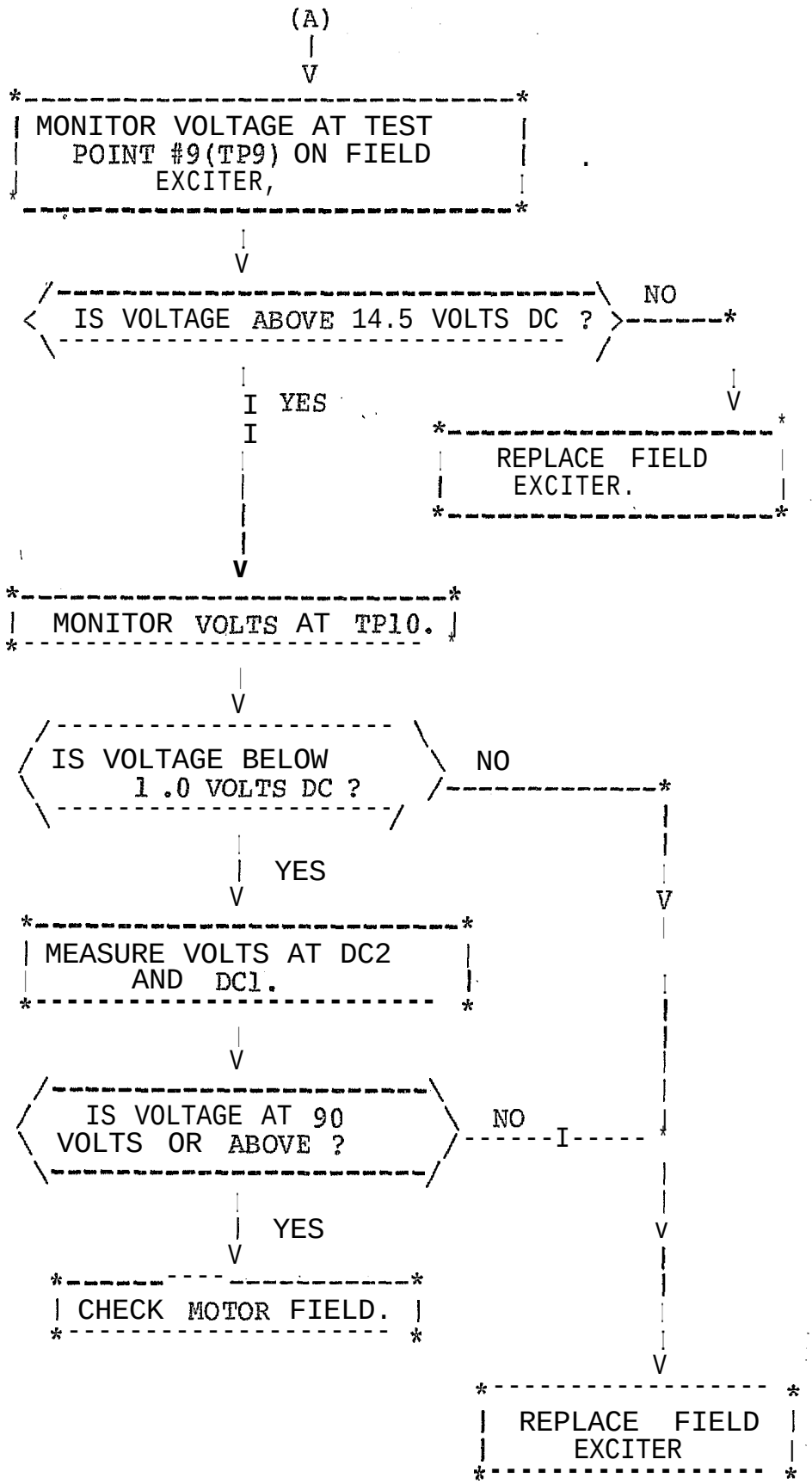
*

* A FIELD LOSS FAULT INDICATES THE LOSS OF FIELD VOLTAGE ON A SHUNT MOTOR OR THE FAILURE OF A CONTACTOR TO CLOSE ON A SERIES REVERSING DRIVE (IE. DRAWWORKS OR ROTARY). THE FOLLOWING FLOWCHART GIVES A GUIDE TO TROUBLE SHOOTING A FIELD LOSS FAULT.

****



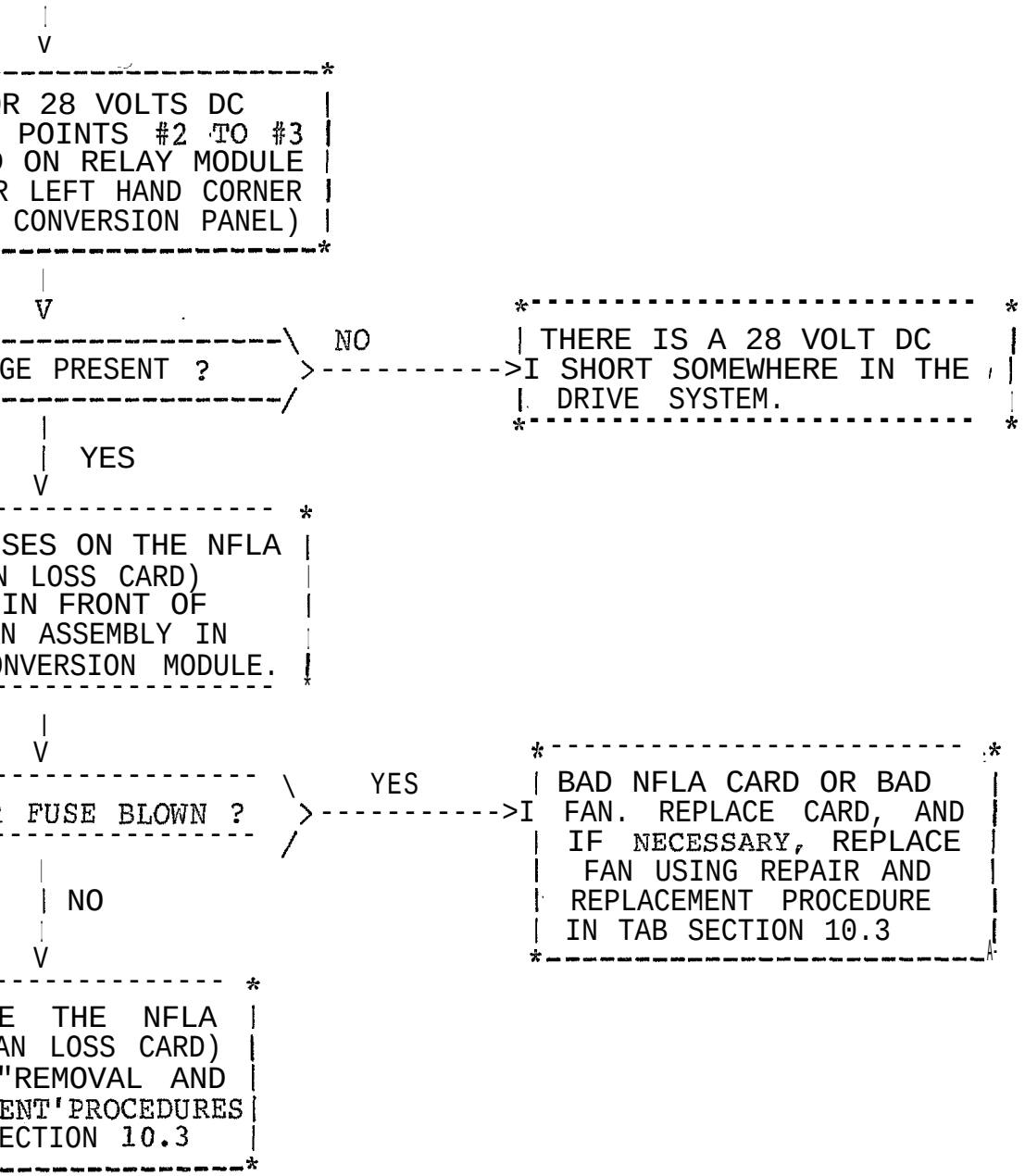

(A)
IS THIS A SERIES REVERSING DRIVE ?
YES
VISUALLY CHECK IF TWO CONTACTORS ARE PICKED UP, IF NOT GO TO THE 'RELAY INTERFACE MODULE' SECTION PAGE 5-1,
NO
ENSURE THAT THE FIELD EXCITER CIRCUIT BREAKER IS NOT TRIPPED AND THAT 600 VOLT AC POWER IS AVAILABLE.
B-B-B-
IS 'PERM' LIGHT ON FIELD EXCITER LIT ?
NO
YES
IS 'FWD' LIGHT LIT? >---*
S--M-
YES
IF 'IMOK' LIGHT LIT, REPLACE FIELD EXCITER.
IS 'FWD' LIGHT LIT? >----*
YES
REPLACE FIELD EXCITER,
REPLACE DCM.
IS THE 'IMOK' LIGHT ON ?
NO
YES
MONITOR THE FIELD AT TEST POINT #10(TP10) WITH A VOLTMETER. IF VOLTAGE IS PRESENT WITH RESPECT TO TEST POINT #1(TP1), REPLACE FIELD EXCITER.
(A)

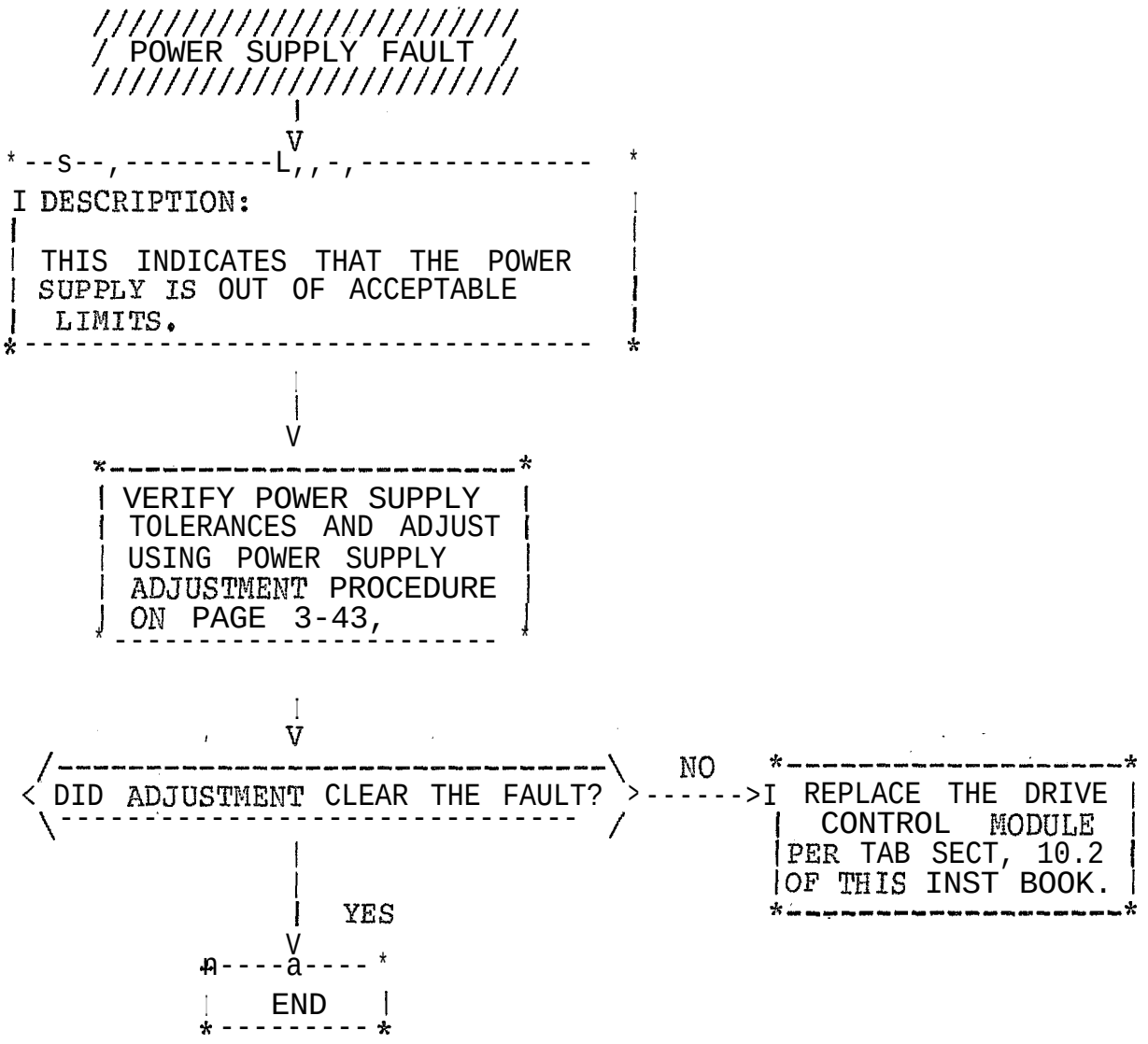

(A)
MONITOR VOLTAGE AT TEST POINT #9(TP9) ON FIELD EXCITER,
IS VOLTAGE ABOVE 14.5 VOLTS DC ?
NO
YES
REPLACE FIELD EXCITER.
MONITOR VOLTS AT TP10.
IS VOLTAGE BELOW 1 .0 VOLTS DC ?
NO
YES
MEASURE VOLTS AT DC2 AND DCl.
IS VOLTAGE AT 90 VOLTS OR ABOVE ?
NO
YES
CHECK MOTOR FIELD.
REPLACE FIELD EXCITER

FAN LOSS TROUBLESHOOTING
[FAULT LIGHT #22]

DESCRIPTION-

THE SCR HEAT SINKS ARE COOLED BY TWO(2) FANS LOCATED AT THE BOTTOM OF THE POWER CONVERSION MODULE. EACH FAN HAS ITS OWN ASSOCIATED DETECTION FOR LOSS OF AIR. IF EITHER FAN SHOULD FAIL TO OPERATE, A LOSS OF THE 'FAN OK' SIGNAL ON THE DCM WILL BE GIVEN (LIGHT #76 WILL TURN OFF).




POWER SUPPLY FAULT
DESCRIPTION:
THIS INDICATES THAT THE POWER SUPPLY IS OUT OF ACCEPTABLE LIMITS.
VERIFY POWER SUPPLY TOLERANCES AND ADJUST USING POWER SUPPLY ADJUSTMENT PROCEDURE ON PAGE 3-43,
DID ADJUSTMENT CLEAR THE FAULT?
NO
REPLACE THE DRIVE CONTROL MODULE PER TAB SECT, 10.2 OF THIS INST BOOK.
YES
END

ADJUSTMENTS

INDEX

I. INTRODUCTION

II. POWER SUPPLY (PSE)

III. POWER CONVERSION NODULE (PCM)

IV. DRIVE CONTROL MODULE (DCM)
- A. REF 1 ADJUSTMENT
- B. CURRENT LIMIT ADJUSTMENT
- C. REF 7 ADJUSTMENT

V. LOAD SHARING ADJUSTMENTS
- A. SHUNT MOTOR
- B. SERIES MOTOR

I. INTRODUCTION

THE FOLLOWING SECTIONS ARE A GUIDE TO MAKING FIELD ADJUSTMENTS ON A UDRILL 3000 RIG, EACH SECTION RELATES TO A PARTICULAR NODULE. COMSULT YOUR G.E. SERVICE REPRESENTATIVE BEFORE ATTEMPTING AN ADJUSTMENT NOT LISTED IN THIS SECTION,

II. POWER SUPPLY (PSC)

THE POWER SUPPLY(PSC) PRODUCES FIVE (5) SEPARATE VOLTAGE LEVELS. ONLY THREE (3) OF THESE LEVELS ARE ADJUSTABLE.

"ADJUSTABLE LEVELS"

VOLTAGE LEVEL	POWER SUPPLY FUSE	MONITOR VOLTAGE LEVEL AT:	ADJUST VOLTAGE LEVEL TO LISTED TOLERANCE AT RESISTOR POT:
P5 POSITIVE 5 V. DC	FU2	TEST POINT #4 (TP4) ON THE NBIB CARD	'R3' GM PSC TO +/- .1 VOLT
		TP2 ON PSC	'R3' ON PSC TO +/- .25 V.
P15 POSITIVE	FU3	TEST POINT #5 (TP5) CN THE NBIB CARD	'R9' ON PSC TO +/- .1 VOLT
		TP3 ON PSC	'R9 ON PSC TG +/- .15 V.
N15 NEGATIVE 15 v. DC-	FU4	TEST POINT #6 (TP6) ON THE NBIB CARD	'R10' ON PSC TO -i-/- .1 VOLT
		TP4 ON PSC	'R10' ON PSC TO +/- .15 V.

"NON-ADJUSTABLE LEVELS"

VOLTAGE LEVEL	POWER SUPPLY FUSE	MONITOR VOLTAGE LEVEL AT:
P28 POSITIVE +28 V. DC	FU1	POINT #2 ON TME 'PTB' OR 'IPTB' TERMINAL BOARD (LOCATED IN THE LOWER LEFT HAND SIDE OF PANEL) OR AT TP5 ON PSC
P105 POSITIVE +105 V. DC	FU5	'PTB' #5 OR 'IPTB' #24 TERMINAL BOARD POINTS (LOCATED IN PANEL GM LOWER LEFT SIDE WALL)

****
* IF ANY OF THE LEVELS ARE OUTSIDE TBE ALLOWABLE LIMITS OR *
* CANNOT BE CORRECTLY ADJUSTED, REPLACE THE POWER SUPPLY (PSC) *
* USING "REMOVAL AND REPLACEMENT' PROCEDURES IN TAB 10.4. *
****

III, POWER CONVERSION MODULE (PCM)

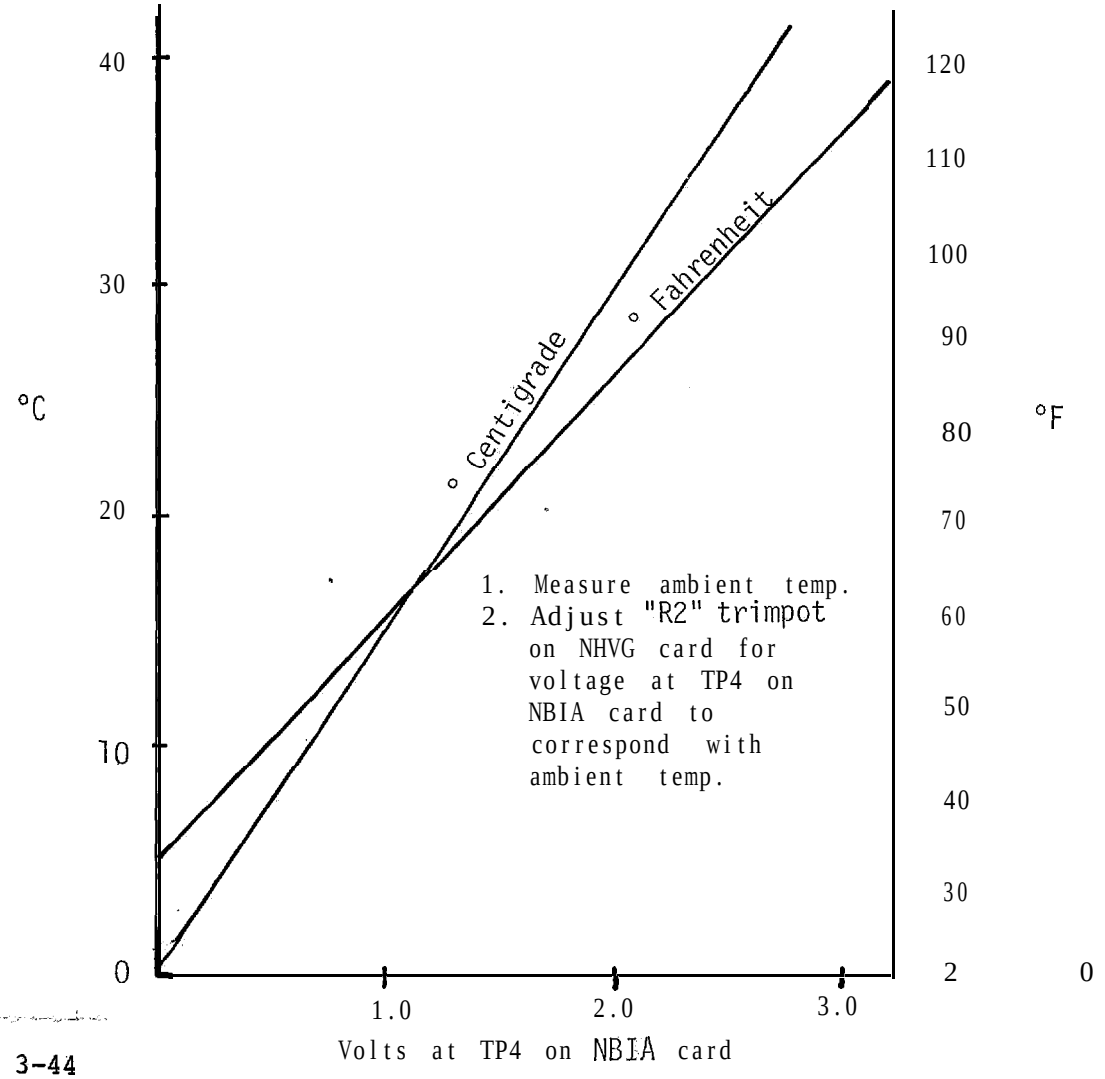
THE 'R2' RESISTOR TRIMPOT ON THE HIGH VOLTAGE INTERFACE (NHVG) BOARD, LOCATED ON THE INSIDE OF THE POWER CONVERSION MODULE DOOR, IS THE AMBIENT TEMPERATURE ADJUSTMENT POT. IF ADJUSTMENT IS EVER REQUIRED, PROCEED AS FOLLOWS:'

STEP 1- MEASURE THE AMBIENT TEMPERATURE WITH A THERMOMETER.

STEP 2- ATTACH A VOLTMETER BETWEEN TEST POINT #4 (TP4)(+) AND TEST POINT #3 (TP3)(-) OF THE NBIB CARD TO READ POSITIVE DC VOLTS (0-20 VOLT SCALE).

STEP 3- USING FIGURE 6, ADJUST THE 'R2' POT ON THE NHVG BOARD FOR A VOLTAGE READING TO CORRESPOND WITH THE AMBIENT TEMPERATURE FOUND IN STEP #1.(TURN POT CLOCKWISE TO INCREASE AND VICE VERSA)

EXAMPLE: FROM STEP #1, THE AMBIENT TEMPERATURE-WAS FOUND TO BE 85 DEGREES FAHRENHEIT, FROM FIGURE 6, IT CAN BE SEEN THAT THE VOLTAGE CORRESPONDING TO THE FARENHEIT TEMPERATURE OF 85 DEGREES IS TWO (2.0) VOLTS, NOW 'R2' SHOULD BE ADJUSTED UNTIL THE VOLTMETER READS APPROXIMATELY +2.0 VOLTS, ADJUSTMENT IS COMPLETE.



IV. DRIVE CONTROL MODULE (DCM)

THERE ARE SEVEN (7) RESISTOR TRIMPOTS LOCATED ON THE NBIA CARD'S DAUGHTER BOARD (DBIA). THESE TRPMPOTS ARE ARRANGED IN A VERTICAL ROW WITH NUMBER ONE (1) AT THE TOP DOWN TO NUMBER SEVEN (7) AT THE BOTTOM (SEE FIGURE 3-7 PG.3-40). ADJUSTMENT PROCEDURES FOR THESE TRIMPOTS ARE GIVEN IN THE FOLLOWING 'SECTIONS,

A* TRIMPOT #1 ADJUSTMENT (REF 1)

1. DRIVE MUST BE IN NORMAL MODE. SWITCH ONE (SW;) AND TWO (2) IN THE CENTER POSITION.
2. SET TMUMEWWEEL SWITCH (TWS) TO POSITION 'C'.
3. ADJUST TRIMPOT #1 (REF 1) ON THE DBIA CARD UNTIL ONLY LIGHT 'F' ON THE HFPB CARD IS LIT.(SEE FIGURE 7' PG. 3-46 FOR DETAIL OF HFPB CARD LIGHTS)

B. CURRENT LIMIT ADJUSTMENT

NOTE: CURRENT LIMITS OF ALL MOTORS THAT SHARE A COMMON SHAFT (EX. DW, MP #1 , ETC.) SHOULD BE SET THE SAME,. IF LOAD SHARING PROELEMS OCCUR, SEE "LOAD SHARING" SECTION V.

"MUD PUMP CURRENT LIMIT"

1. REMOVE THE DRIVE CONTROL MODULE COVER,
2. PLACE SWITCH ONE (SW1) AND TWO (SW2) IN THE 'UP' POSITION,
3. PLACE THUMBWHEEL SWITCH (TWS) IN POSITION '6'.
4. AT DRILLER'S CONSOLE, ASSIGN BOTH MUD PUMP MOTORS AND SELECT THE DESIRED MUD PUMP,
   NOTE: IF ONLY A SINGLE MOTOR MUD PUMP; GO TO STEP 8.
5. CHECK DCM TO SEE THAT LIGHT # 18 IS ON. THIS INDICATES THAT TWO MOTORS ARE ASSIGNED.

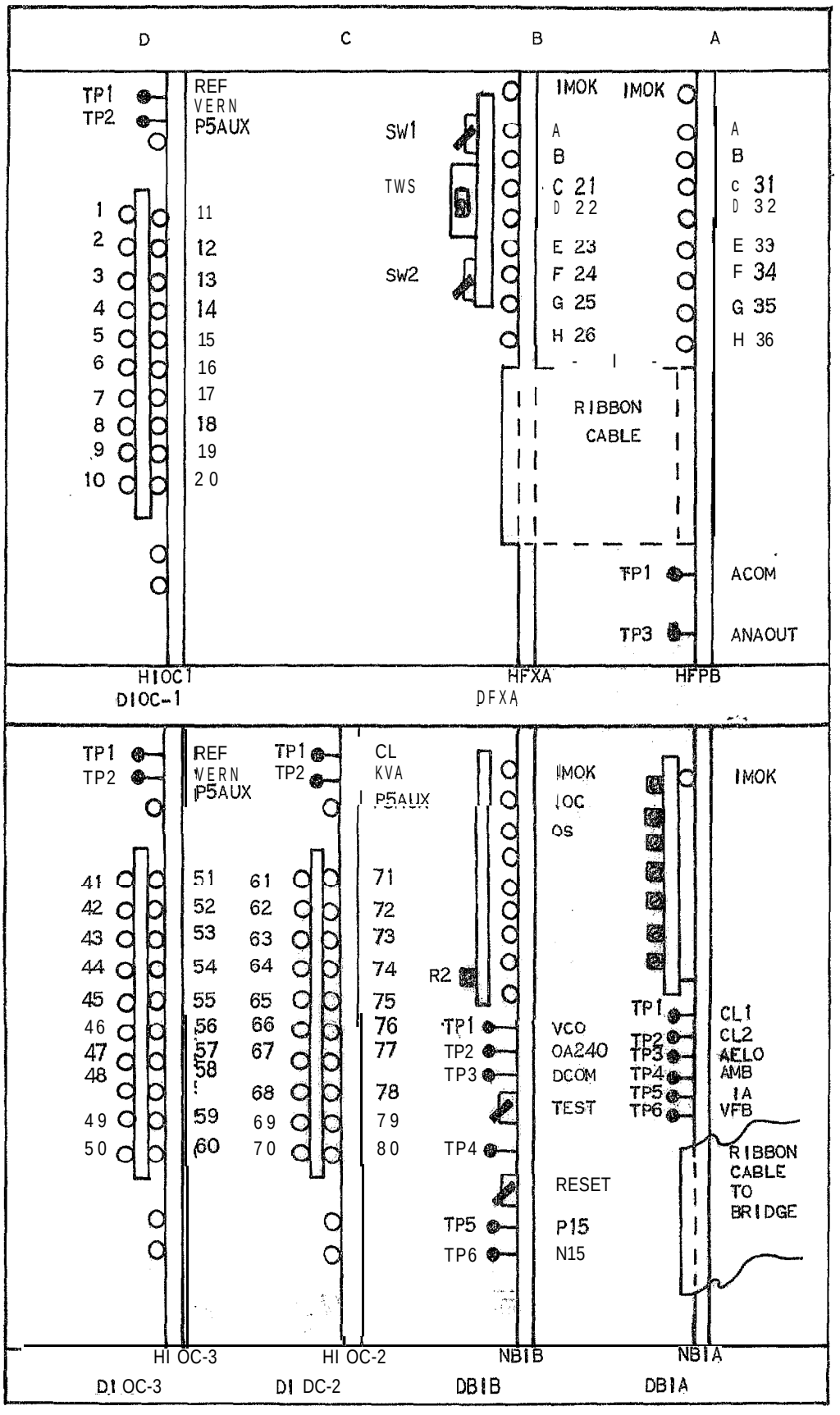


FIGURE _____

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

V. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
V. 2	ISSUED			FIRST MADE FOR		
V. 3	MADE BY			I.C. NO.		CONT. ON SH. SH. NO.

6. USING TABLE 2, ADJUST 'REF 3' ON DBIA 'CARD TO OBTAIN DESIRED CURRENT LIMIT.

FOR A CURRENT LIMIT OF: (PER MOTOR)	THESE HFXA CARD LIGHTS SHOULD BE LIT:	THESE HFPB CARD LIGHTS SHOULD BE LIT:
900 AMPS	NONE	G,F,C,AND B
1000 AMPS	A	NONE
1050 AMPS	A	D,C, AND A
1100 AMPS	A	E,D, AND B
1200 AMPS	A	F,E,B, AND A
1300 AMPS	A	G,D, AND C
1400 AMPS	A	G,F,C, AND B

"TABLE 2"

7. DE-ASSIGN ONE MUD PUMP MOTOR AT DRILLER'S CONSOLE, ADJUST CURRENT LIMIT ON THE ASSIGNED MOTOR.

8. CHECK THAT DCM LIGHT # 18 IS OFF. THIS INDICATES THAT ONLY ONE MOTOR IS ASSIGNED.

9. USING TABLE 2, ADJUST 'REF 5' TRIMPOT ON THE DBIB CARD TO OBTAIN DESIRED SINGLE MOTOR CURRENT LIMIT. (SEE FIGURE 3-1 FOR CALCULATION OF CURRENT LIMIT SETTINGS NOT LISTED IN TABLE 1).

10. FOR TWO MOTOR MUDPUMPS, REPEAT STEPS 7 THRU 9 FOR OTHER MOTOR.

11. PLACE SWITCH ONE (SW1) AND TWO (SW2) IN CENTER POSITION. PLACE THUMBWHEEL SWITCH (TWS) IN POSITION '0'. REPLACE DCM COVER.

"DRAWWORKS CURRENT LIMIT"

1. REMOVE THE DRIVE CONTROL MODULE COVER.
2. PLACE SWITCH ONE (SW1) AND TWO (SW2) IN THE 'UP' POSITION.
3. PLACE THUMBWHEEL SWITCH (TWS) IN POSITION '6'.
4. AT DRILLER'S CONSOLE, SELECT DRAWWORKS AND ASSIGN BOTH MOTORS, NOTE: IF ONLY A SINGLE' MOTOR DRAWWORKS, GO TO STEP 8.
5. IF DCM LIGHT # 20 IS OFF, GO TO STEP 8.
6. USING TABLE 3, ADJUST 'REF 2' TRIMPOT ON EACH DRAWWORKS DRIVE TO OBTAIN DESIRED TWO MOTOR CURRENT LIMIT.

FOR A CURRENT LIMIT OF: (PER MOTOR)	THESE HFXA CARD LIGHTS SHOULD BE LIT:	THESE HFPB CARD LIGHTS SHOULD'BE LIT:
900 AMPS	NONE	G,F,C,AND B
1000 AMPS	A	NONE
1050 AMPS	A	D,C, AND A
1100 AMPS	A	E,D, AND B
1200 A M P S	A	F,E,B, AND A
1300 AMPS	A	G,D, AND C
1400 AMPS	A	G,F,C, AND B

"TABLE 3"

7. AT DRILLER'S CONSOLE, DE- ASSIGN ONE DRAWWORKS MOTOR. ADJUST CURRENT LIMIT ON THE'ASSIGNED MOTOR,
   NOTE: DCM LIGHT # 20 SHOULD NOW HE OFF.
8. USING TABLE 3, ADJUST 'REF 4' TRIMPOT ON EACH ASSIGNED DRIVE TO OBTAIN THE DESIRED CURRENT LIMIT.
9. IF STEP #6 WAS PERFORMED, REPEAT STEPS #7 AND #8 FOR THE.OTHER DRAWWORKS MOTOR(S) .
10. PLACE SWITCH ONE (SW1) AND TWO (SW2) IN CENTER POSITION, PLACE THUMBWHEEL SNITCH (TWS) IN POSITION '0'. REPLACE DCM COVER,

C. TRIMPOT #7 (REF 7) ADJUSTAENT

1. REMOVE DRIVE CONTROL MODULE COVER,, PLACE SWITCH ONE (SW1) IN THE CENTER POSITION AND SWITCH TWO (SW2) IN THE 'UP' POSITION,
NOTE: DRIVE WILL NOT SELECT IN THIS MODE.

2. SET THUMBWHEEL SWITCH (TWS) TO POSITION '8' .

3. TOGGLE SWITCH ONE (SW1) DOWN TWICE (A CLICK SHOULD BE HEARD AFTER EACH TOGGLE).

4. TURN THROTTLE AT DRILLER'S CONSOLE TO FULL ON (MAXIMUM) POSITION.

5. ADJUST 'REF 7' TRIMPOT TO THE DESIRED VALUE USING THE LIGHTS ON THE HFPB CARD (FIGURE 3-7 PG.3-46). SEE EXAMPLES BELOW.

EXAMPLE : FOR SHUNT MOTORS

FOR A MAXIMUM VOLTAGE OF 750 VOLTS, ADJUST 'REF 7' TRIMPOT UNTIL ONLY HFPB CARD LIGHTS 'G' , 'F' 'E' , AND 'D' ARE LIT.(SEE FIGURE 3-7 PG. 3-46 FOE; DETAILS OF HFPB LIGHTS;)

EXAMPL E: FOR SERIES MOTORS

FOR A MAXIMUM SPEED OF: (PER MOTOR)	HFPB CARD LIGHTS SHOULD BE LIT :
800 RPM	F,E,B, AND A
900 RPM	F,E,D, AND B
1000 RPM	G
1100 RPM	G,C, AND B

6. RETURN SWITCH ONE (SW1) AND TWO (SW2) TO THE CENTER POSITION. RETURN THE THUMBWHEEL SWITCH (TWS) TO POSITION '0' AND ZERO THE THROTTLE AT THE DRILLERS CONSOLE.

7. DEPRESS 'RESET' PUSHBUTTON ON PANEL DOOR.

V. LOAD SHARING

A, SHUNT MOTORS

1. REMOVE DRIVE CONTROL MODULE COVER.

2. PLACE SWITCH ONE (SW1) IN THE CENTER POSITION AND SWITCH TWO (SW2) IN THE 'UP' POSITION.

3. PLACE THUMBWHEEL SWITCH (TWS) IN POSITION '8'.

```
**************************************************
*    CAUTION :  OUTPUTS WILL BE ENABLED IN THIS  *
*               MODE (EX. BLOWER MOTOR, INERTIA BRAKE, *
*               MOTOR FIELD, ETC.)  USE SAFETY  *
*               PRECAUTIONS TO AVOID DAMAGE OR INJURY  *
*             . TO EQUIPMENT AND PERSONNEL.     *
**************************************************
```

4. TOGGLE SWITCH ONE (SW1) DOWN FIVE(5) TIMES.(A CLICKING SOUND SHOULD BE HEARD AFTER EACH TOGGLE)

5. PLACE A VOLTMETER BETWEEN FIELD EXCITER TEST POINTS TP1(+) AND TP10(-) TO READ DC VOLTS (0-20 V. SCALE).

6. ADJUST THE FIELD EXCITER 'REF' TRIMPOT TO THE VALUE GIVEN IN FIGURE 1 (CORRESPONDING TO THE MOTOR TYPE BEING USED) .

7. WITH DRIVE DESELECTED, TURN THE THROTTLE TO FULL. CHECK THAT THE HFXA AND HFPB CARD LIGHTS, ON THE DRIVES THAT MUST SHARE LOAD, ARE LIT IN THE SAME PATTERN.. IF NOT,. GO TO 'ADJUSTMENTS' SECTION IV(C).

8. PLACE SWITCH ONE (SW1) AND TWO (SW2) IN THE CENTER POSITION. PLACE THE THUMBWHEEL SWITCH (TWS) IN POSITION '0'.

9. SELECT THE DRIVE AT THE DRILLER'S CONSOLE AND RUN AT 1/4 T H R O T T L E .

10. CONNECT A VOLTMETER (DIGITAL PREFERRED) BETWEEN NBIA CARD TEST POINTS #5 (TP5) OF THE DRIVES THAT MUST SHARE LQAD (SEE FIGURE 3-8 PG.4-51 FOR VOLTMETER CONNECTION). CONNECT THE DRIVES TOGETHER TWO AT A TIME FOR DRIVES WITH MORE THAN TWO MOTORS (EX. THREE(3) MOTOR DRAWWORKS).

11. ADJUST 'REF 7' TRIMPOT ON THE NBIB DAUGHTER BOARD(DBIB) FOR THE LOWEST VOLTMETER READING.

1%. IF A STEADY, CONSTANT LOAD IS AVAILABLE, REPEAT STEP #11 TO MAXIMIZE: LOAD SHARING AT RATED LOAD.

NOTE: DO NOT ADJUST MOTOR FIELD EXCITERS TO OBTAIN LOAD SHARING.

13. TURN THE THROTTLE OFF, PLACE THUMBWHEEL SWITCH (TWS) IN ZERO(0) POSITION, AND SWITCH #2 (SW2) IN CENTER. REPLACE THE DRIVE CONTROL MODULE COVER.

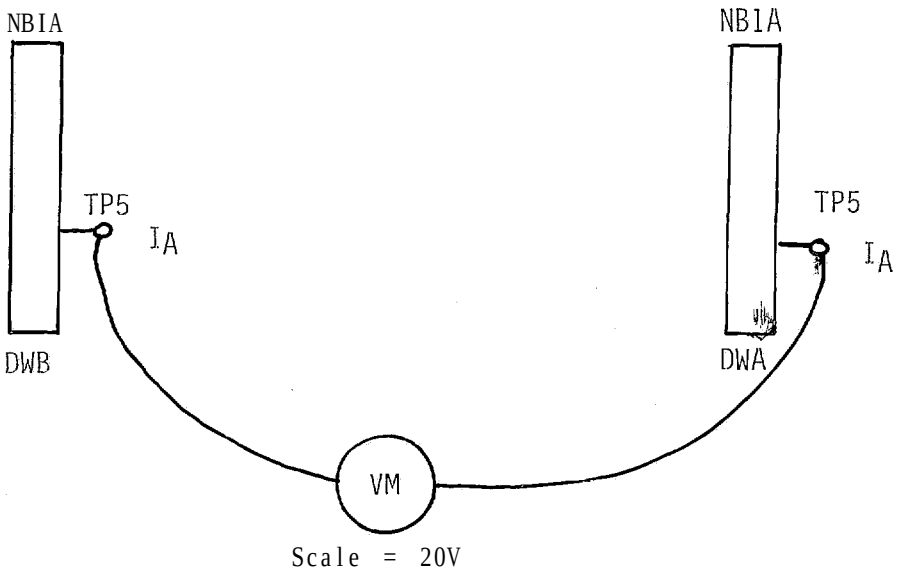


FIGURE 8

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

B. SERIES MOTOR LOAD SHARING

1. REMOVE DCM COVER.

2. REMOVE THE NBIB CARD. DISCONNECT THE DBIB (DAUGHTER BOARD) FROM THE NEIB CARD.

3. MONITOR RESISTANCE BETWEEN DBIB CARD POINTS #3 AND #4 WITH AN OHMMETER.

4. ADJUST THE 'R2' TRIMPOT ON THE DBIB CARD FOR A METER' READING OF 5 K.OHMS. (TURN POT COUNTER CLOCKWISE TO INCREASE).

5. REPLACE THE DBIB CARD ON THE NBIB CARD AND PUT THE NBIB CARD BACK INTO THE DCM.

6. PLACE SWITCH ONE (SW1) IN CENTER POSITION AND SWITCH TWO (SW2) IN THE 'UP' POSITION. PLACE THE THUWBWHEEL SWITCH (TWS) IN POSITION '8'. .

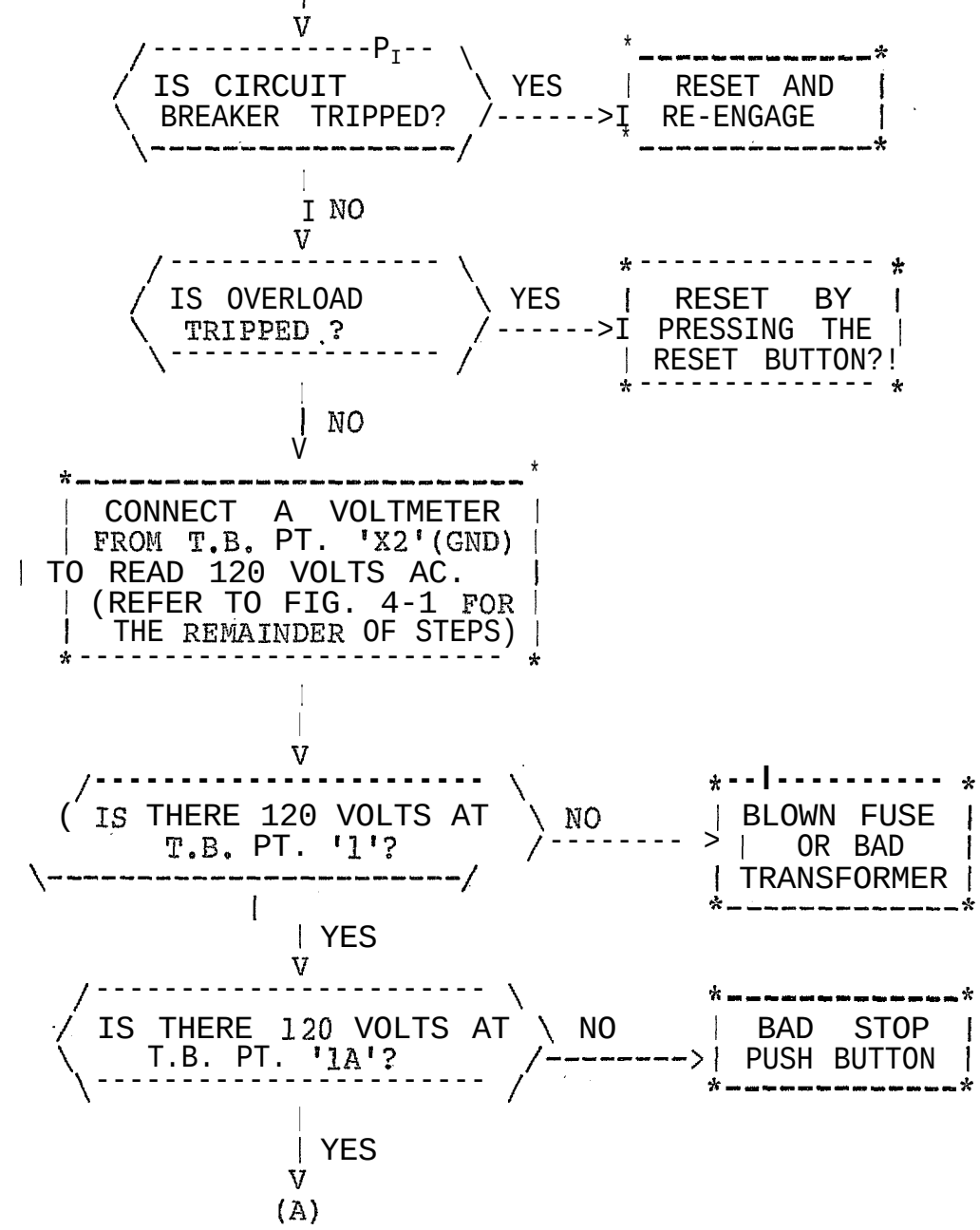
7. WITH DRIVE DESELECTED, TURN THROTTLE FULL ON. CHECK TEAT THE HFXA AND HFPB CARD LIGHTS, ON DRIVES THAT MUST SHARE LOAD (EX. DRAWWORKS, MUDPUMP #1, ETC.), ARE LIT IN THE SAME PATTERN. IF NOT, GO TO 'ADJUSTMENTS' SECTION IV(C).

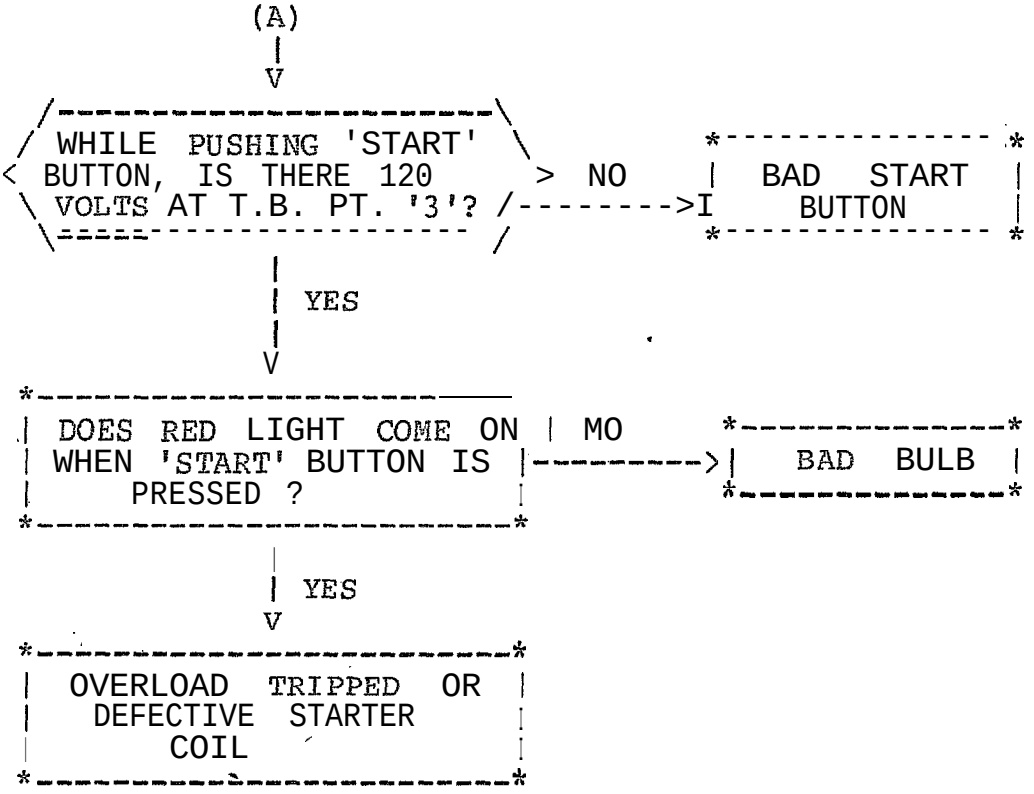
8. TURN THE THROTTLE OFF, PLACE SWITCH (SW1) AND TWO (SW2) IN THE CENTER POSITION* PLACE THUMBWHEEL SWITCH (TWS) IN THE POSITION ZERO (0).

9. REPLACE DCM COVER,

MANUAL STARTER TROUBLESHOOTING FLOWCHART

THE STEPS IN THE CHART BELOW SHOULD BE USED IF A MOTOR FAILS TO START WHEN THE START PUSH BUTTON IS PRESSED. ALSO, USE "FIGURE 4-1 PG.473 TO UNDERSTAND STARTER COMPONENTS AND OPERATION.




(A)
WHILE PUSHING 'START' BUTTON, IS THERE 120 VOLTS AT T.B. PT. '3'?
NO
BAD START BUTTON
YES
DOES RED LIGHT COME ON WHEN 'START' BUTTON IS PRESSED ?
MO
BAD BULB
YES
OVERLOAD TRIPPED OR DEFECTIVE STARTER COIL

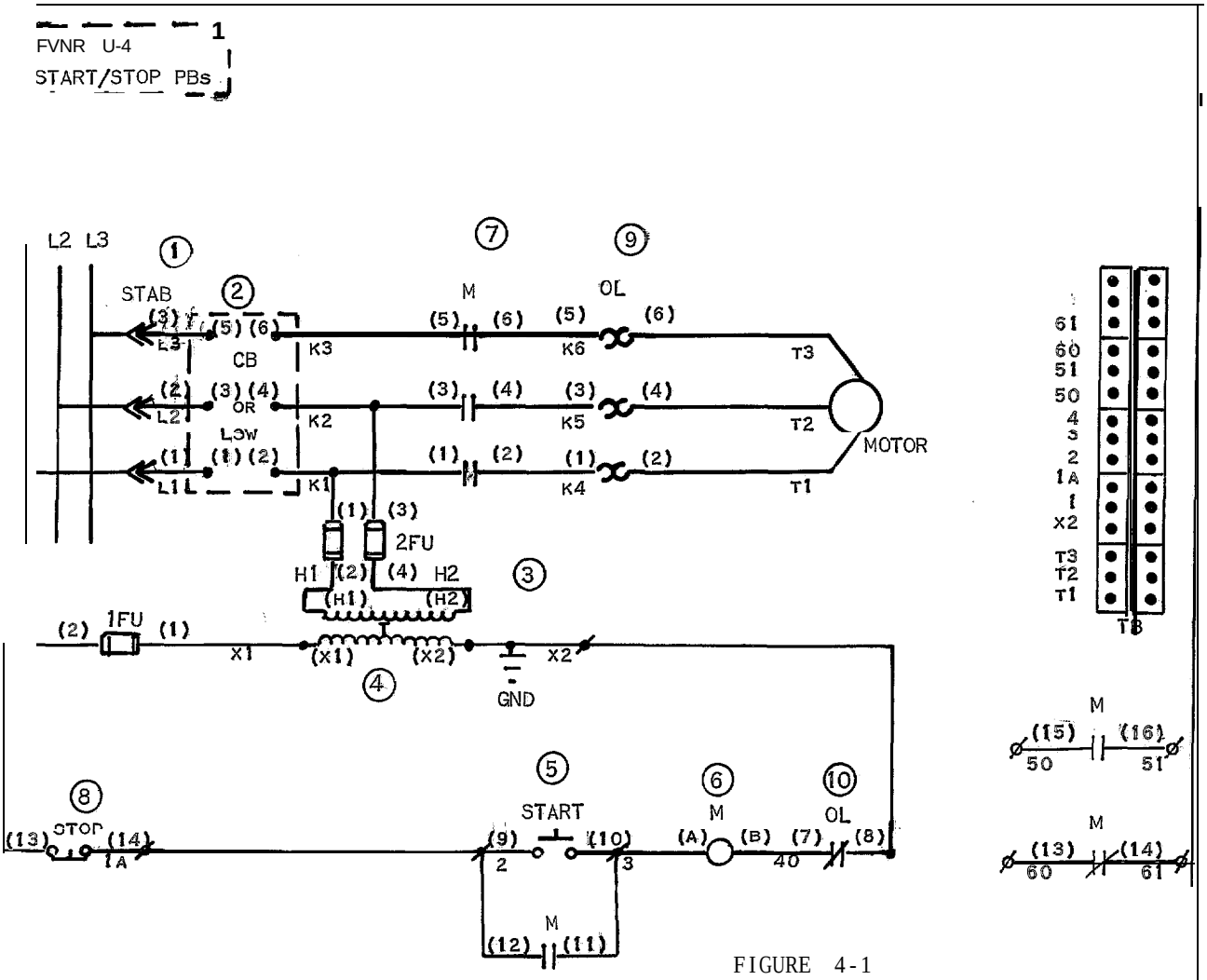
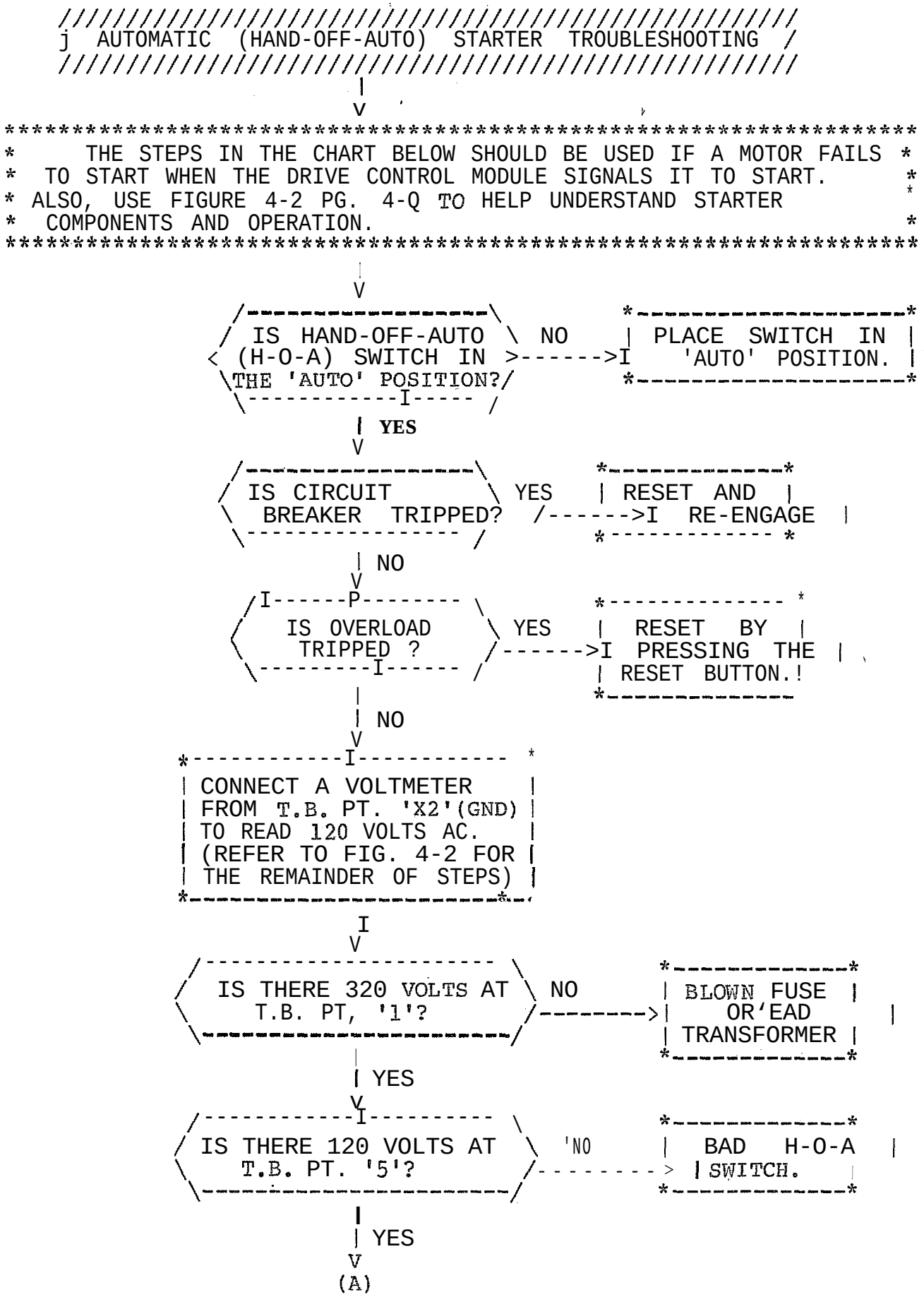


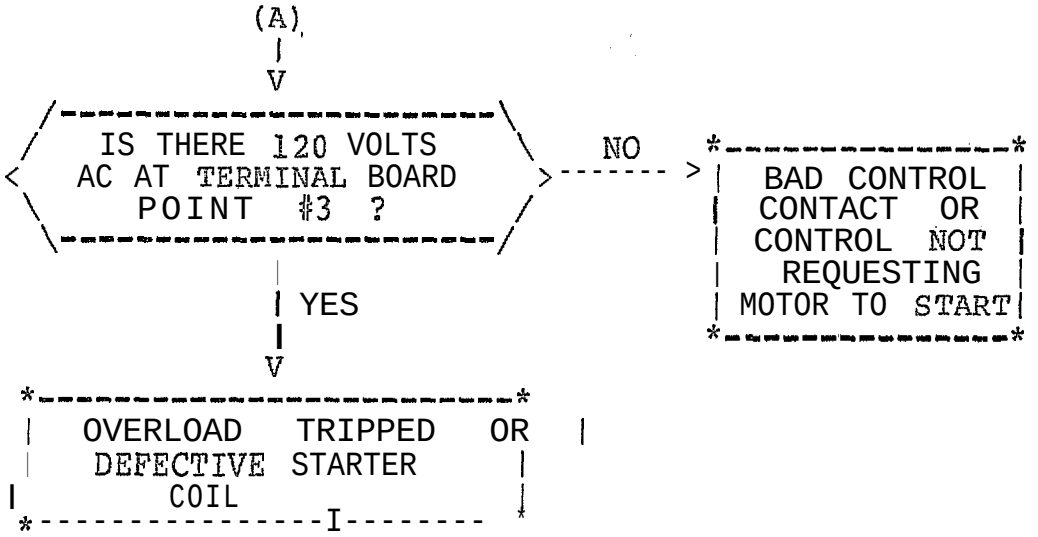
FIGURE 4-1

1) Three phase power enters starter here.
2) When circuit breaker is closed, power is applied to high side of transformer through fuses.
3) Transformer 480/120 single phase.
4) This low side of transformer applies 120 VAC to control circuit.
5) When this start pushbutton is pressed, the control circuit is completed.
6) The motor starter coil "M" is energized,
7) The motor starter contacts "M" are closed, sealing the circuit across the start button 5 and applying power to the motor.
8) When the stop pushbutton is pressed, the circuit is broken. The motor starter coil"M" (6) is de-energized. Motor starter contacts (7) open.
9) The overload thermal elements "OL" protect the motor from overcurrents.
10) If an overcurrent condition occurs, the overload contacts "OL" will open, accomplishing the same function as pressing the stop pushbutton 9.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

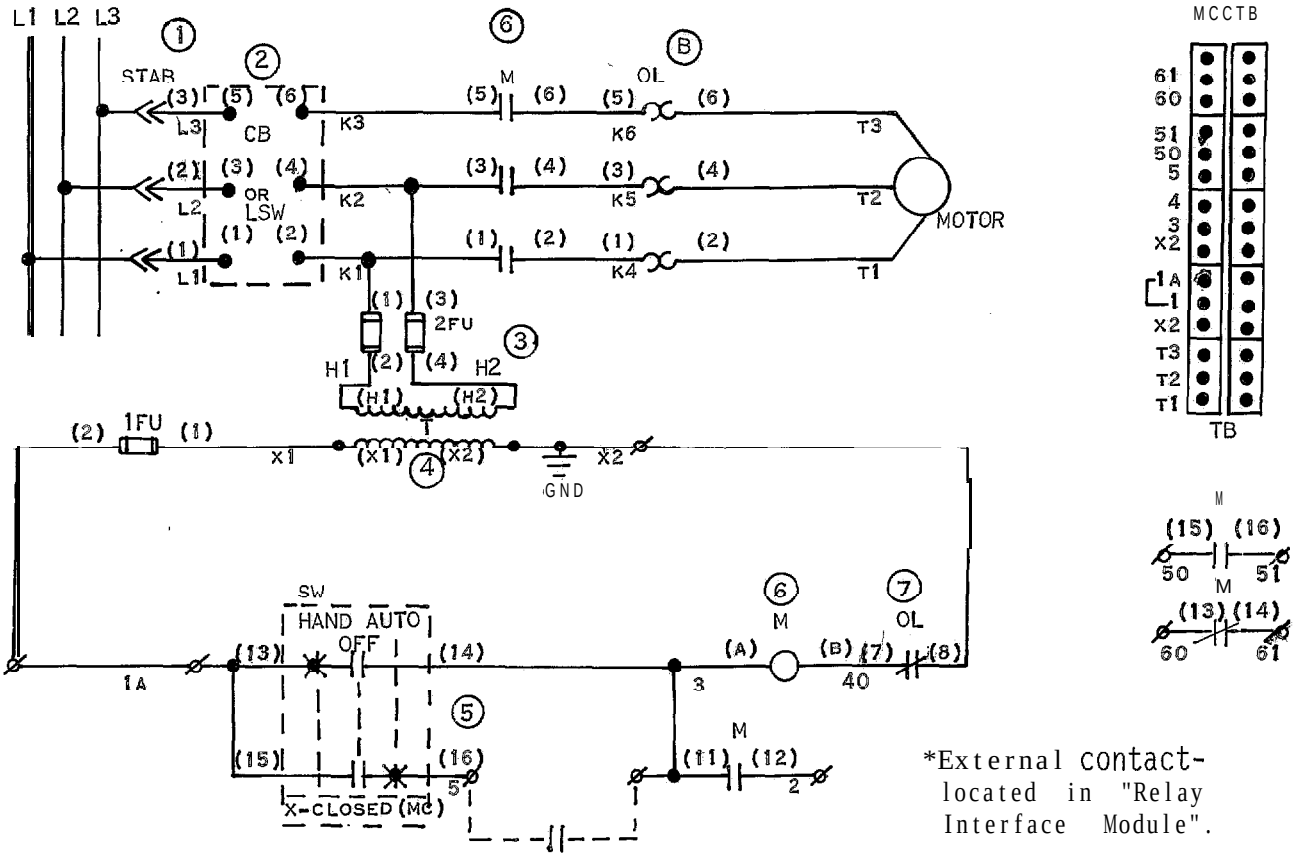
V. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	ELEMENTARY DIAGRAM
V. 2	ISSUED			FIRST MADE FOR	GENERAL ELECTRIC
V. 3	MADE BY			I.C. NO.	CONT. ON SH. / SH. NO.


AUTOMATIC (HAND-OFF-AUTO) STARTER TROUBLESHOOTING
THE STEPS IN THE CHART BELOW SHOULD BE USED IF A MOTOR FAILS TO START WHEN THE DRIVE CONTROL MODULE SIGNALS IT TO START.
ALSO, USE FIGURE 4-2 PG. 4-Q TO HELP UNDERSTAND STARTER COMPONENTS AND OPERATION.
IS HAND-OFF-AUTO (H-O-A) SWITCH IN THE 'AUTO' POSITION?
NO
PLACE SWITCH IN 'AUTO' POSITION.
YES
IS CIRCUIT BREAKER TRIPPED?
YES
RESET AND RE-ENGAGE
NO
IS OVERLOAD TRIPPED ?
YES
RESET BY PRESSING THE RESET BUTTON.!
NO
CONNECT A VOLTMETER FROM T.B. PT. 'X2'(GND) TO READ 120 VOLTS AC. (REFER TO FIG. 4-2 FOR THE REMAINDER OF STEPS)
IS THERE 320 VOLTS AT T.B. PT, '1'?
NO
BLOWN FUSE OR'EAD TRANSFORMER
YES
IS THERE 120 VOLTS AT T.B. PT. '5'?
'NO
BAD H-O-A SWITCH.
YES
(A)


(A)
IS THERE 120 VOLTS AC AT TERMINAL BOARD POINT #3 ?
NO
BAD CONTROL CONTACT OR CONTROL NOT REQUESTING MOTOR TO START
YES
OVERLOAD TRIPPED OR DEFECTIVE STARTER COIL

FVNR SIZE 1-4
HAND-OFF-AUTO SW

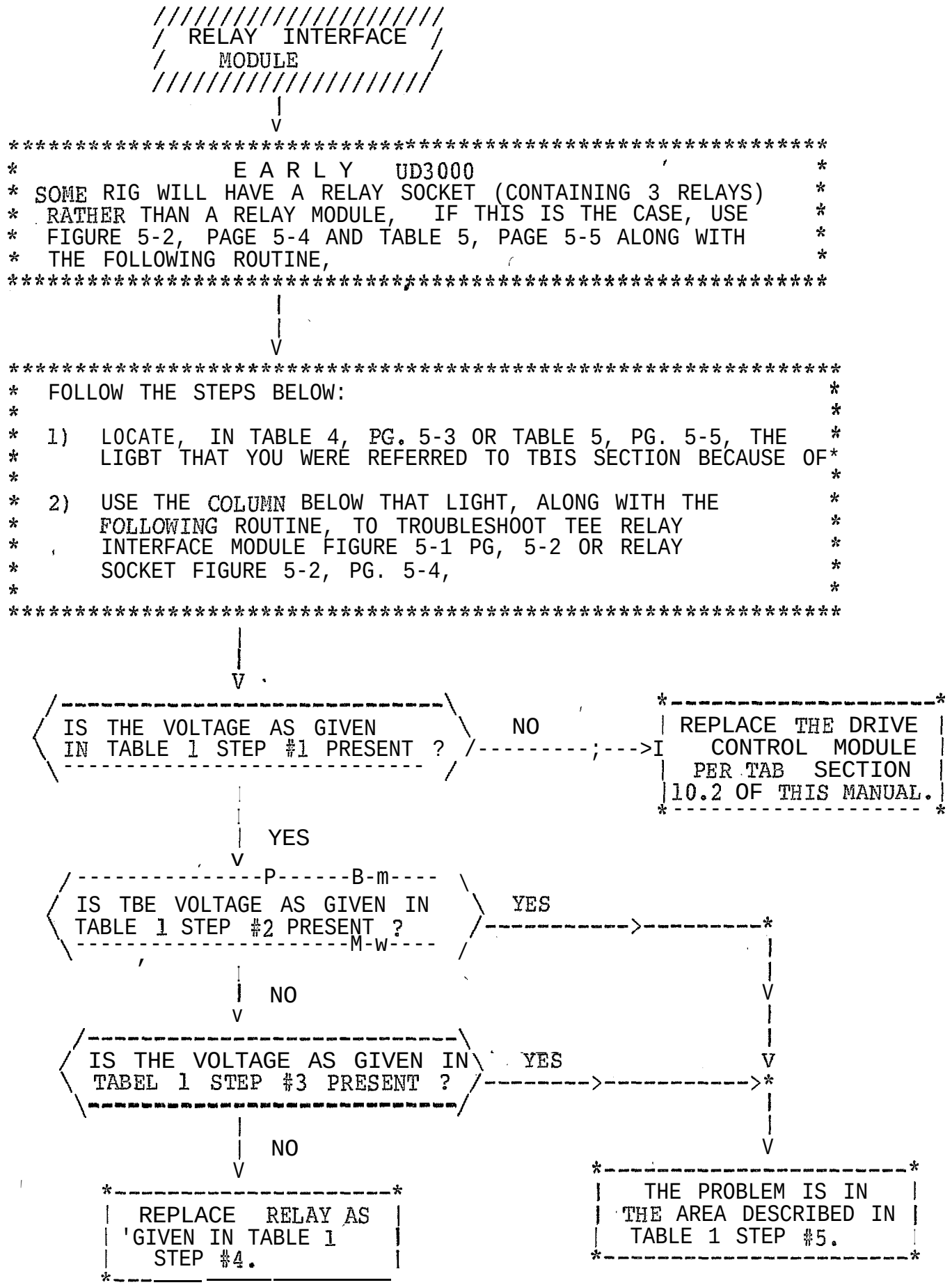
FIGURE 4-2



1) Three phase power enters starter here.
2) When circuit breaker is closed, power is applied to high side of transformer through fuses.
3) Transformer 480/120 single phase.
4) This low side of transformer applies 120 VAC to control circuit.
5) When the H-O-A switch is placed in "Hand" or when the switch is in "Auto" and the external contact picks up, the control circuit is completed.
6) The motor starter coil "M" is energized and the "M" contacts (7) close, applying power to the motor.
7) When H-O-A is put in "Off" or when the H-O-A is in "Auto and the external contact opens, the circuit is broken. The motor starter coil "M" 6 is de-energized. Motor starter contacts 8 open.
8) The overload thermal elements "OL" protect the motor from overcurrents.
9) If an overcurrent condition occurs, the overload contacts "OL" will open, accomplishing the same function as Step #7.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

EV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
EV. 2	ISSUED			FIRST MADE FOR		
EV. 3	MADE BY			I.C. NO.		CONT. ON SH. SH. NO.


RELAY INTERFACE MODULE
E A R L Y UD3000
SOME RIG WILL HAVE A RELAY SOCKET (CONTAINING 3 RELAYS) RATHER THAN A RELAY MODULE, IF THIS IS THE CASE, USE FIGURE 5-2, PAGE 5-4 AND TABLE 5, PAGE 5-5 ALONG WITH THE FOLLOWING ROUTINE,
FOLLOW THE STEPS BELOW:
1) LOCATE, IN TABLE 4, PG. 5-3 OR TABLE 5, PG. 5-5, THE LIGBT THAT YOU WERE REFERRED TO TBIS SECTION BECAUSE OF
2) USE THE COLUMN BELOW THAT LIGHT, ALONG WITH THE FOLLOWING ROUTINE, TO TROUBLESHOOT TEE RELAY INTERFACE MODULE FIGURE 5-1 PG, 5-2 OR RELAY SOCKET FIGURE 5-2, PG. 5-4,
IS THE VOLTAGE AS GIVEN IN TABLE 1 STEP #1 PRESENT ?
NO
REPLACE THE DRIVE CONTROL MODULE PER TAB SECTION 10.2 OF THIS MANUAL.
YES
IS TBE VOLTAGE AS GIVEN IN TABLE 1 STEP #2 PRESENT ?
YES
NO
IS THE VOLTAGE AS GIVEN IN TABEL 1 STEP #3 PRESENT ?
YES
NO
REPLACE RELAY AS 'GIVEN IN TABLE 1 STEP #4.
THE PROBLEM IS IN THE AREA DESCRIBED IN TABLE 1 STEP #5.

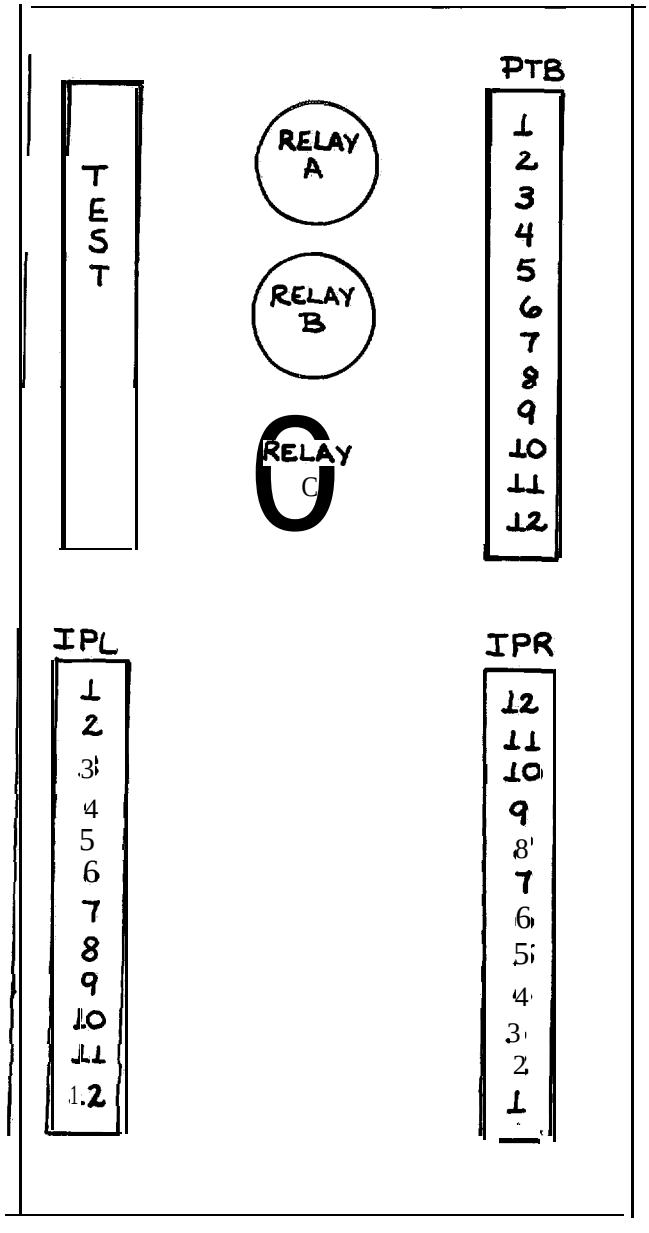


FIGURE 11

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

STEPS	LIGHT #15 (BLOWER)	LIGHT #72 (AUXILIARY)	LIGHT #73 (S.CHARGER)	LIGHT #13 (FWD. CONT.)	LIGHT #14 (REV. CONT.)
#1	TEST POINT PTB-4 TO PTB-3 28 VOLT DC	TEST POINT IPL-9 TO PTB-3 28 VOLT DC	TEST POINT IPL-6 TO PTB-3 28 VOLT DC	TEST POINT IPL-9 TO PTB-3 28 VOLT DC	TEST POINT IPL-6 TO PTB-3 28 VOLT DC
#2	TEST POINT IPR-6 TO GND. 120 VOLT AC	TEST POINT IPR-4 TO GND. 120 VOLT AC	TEST POINT IPL-3 TO GND. 120 VOLT AC	TEST POINT IPR-4 TO GND. 120 VOLT AC	TEST POINT IPL-4 GND. 120 VOLT AC
#3	TEST POINT IPR-5 TO GND. 120 VOLT AC	TEST POINT IPR-3 TO GND. 120 VOLT AC	TEST POINT IPL-4 TO GND. 120 VOLT AC	TEST POINT IPR-3 TO GND. 120 VOLT AC	TEST POINT IPL-3 TO GND. 120 VOLT AC
#4	RELAY 'A'	RELAY 'B'	RELAY 'C'	RELAY 'B'	RELAY 'C'
#5	MCC BLOWER	MCC AUXILIARY	MCC SUPERCHARGER	FORWARD CONTACTOR	REVERSE CONTACTOR

" TABLE 4 "

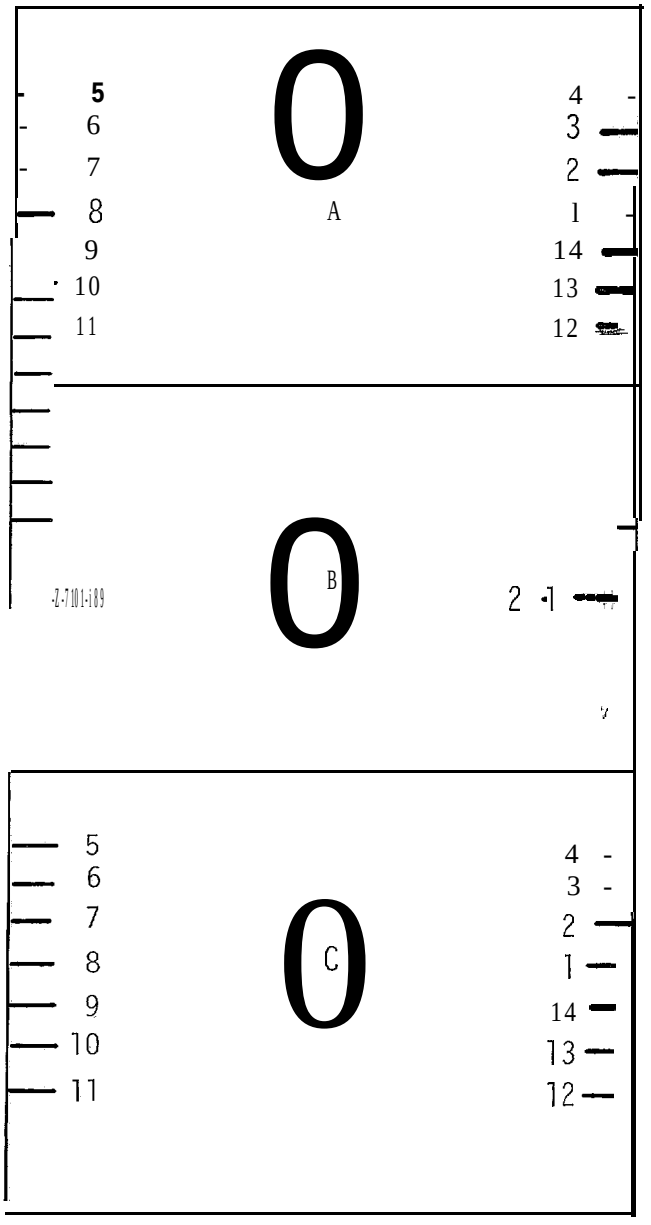


FIGURE 12

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

V. 1	REV. 4 REV. 5 PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
V. 2	ISSUED	FIRST MADE FOR		
V. 3	MADE BY	I.C. N d		CONT. CN SH. SH. NO.

	LIGHT #15	LIGHT #72	LIGHT #73	LIGHT #13	LIGHT #14
1	TEST POINT A10-11 28V DC	TEST POINT B10-11 P28	TEST POINT C10-11 P28	TEST POINT B10-11 p28	TEST POINT C10-11 P28
2	TEST POINT A2-GND 120V AC	TEST POINT B2-GND	TEST POINT C2-GND	TEST POINT B2-GND	TEST POINT C2-GND
3	TEST POINT A-3	TEST POINT B-3	TEST POINT C-3	TEST POINT B-3	TEST POINT C-3
4	RELAY a	RELAY B	RELAY C	RELAY B	RELAY C
5	MCC BLOWER	MCC AUX.	MCC SUPER-CHARGER	MCC CONT.	MCC CONT.

TABLE 5
-W--S--

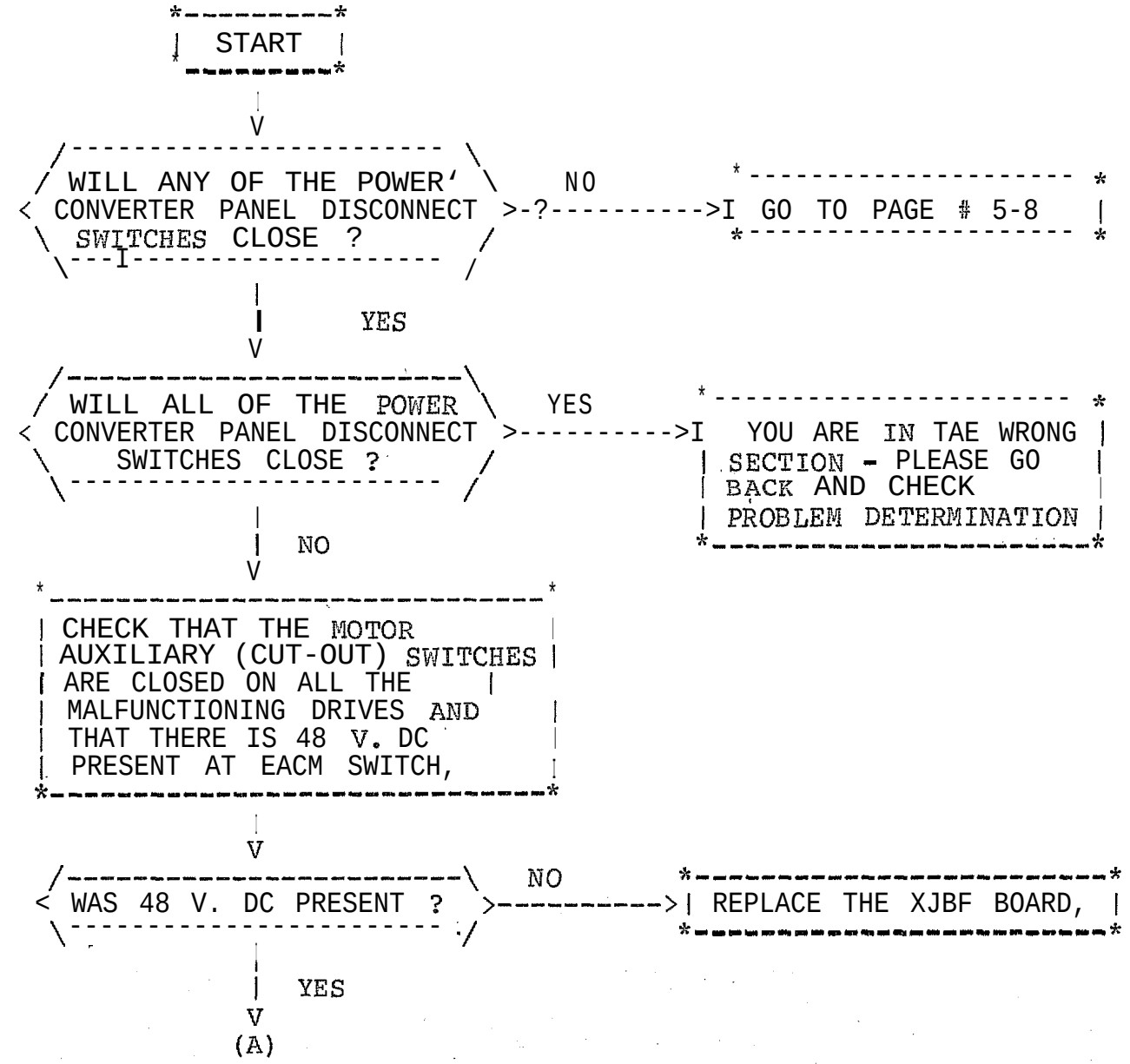
48 VOLT D.C. SUPPLY TROUBLESBOOTING

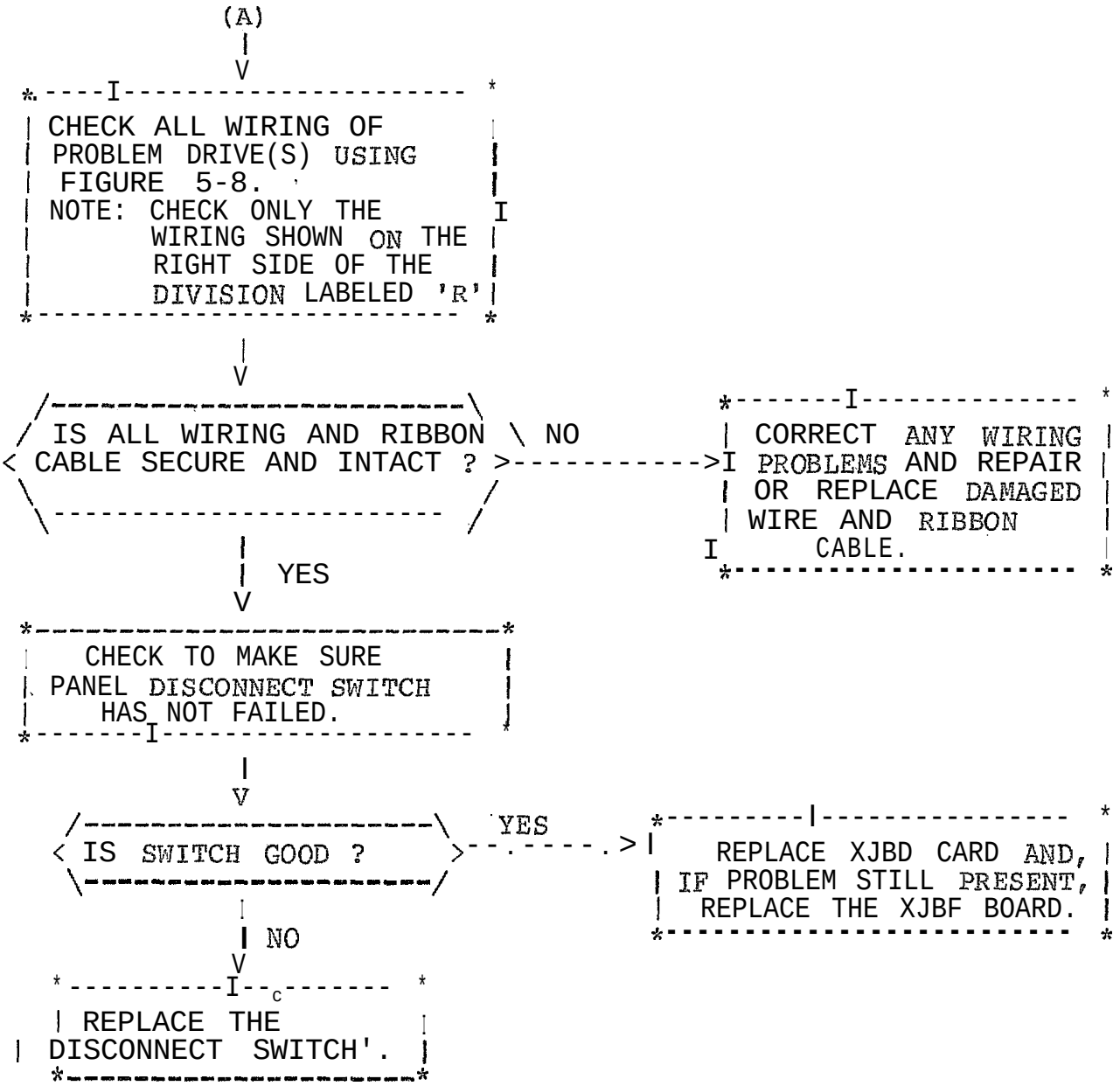
TNIS SECTION COVERS THE FOLLOWING TROUBLE SYMPTOMS:

1. ALL POWER CONVERTER PANEL DISCONNECT SWITCHES WILL NOT CLOSE.

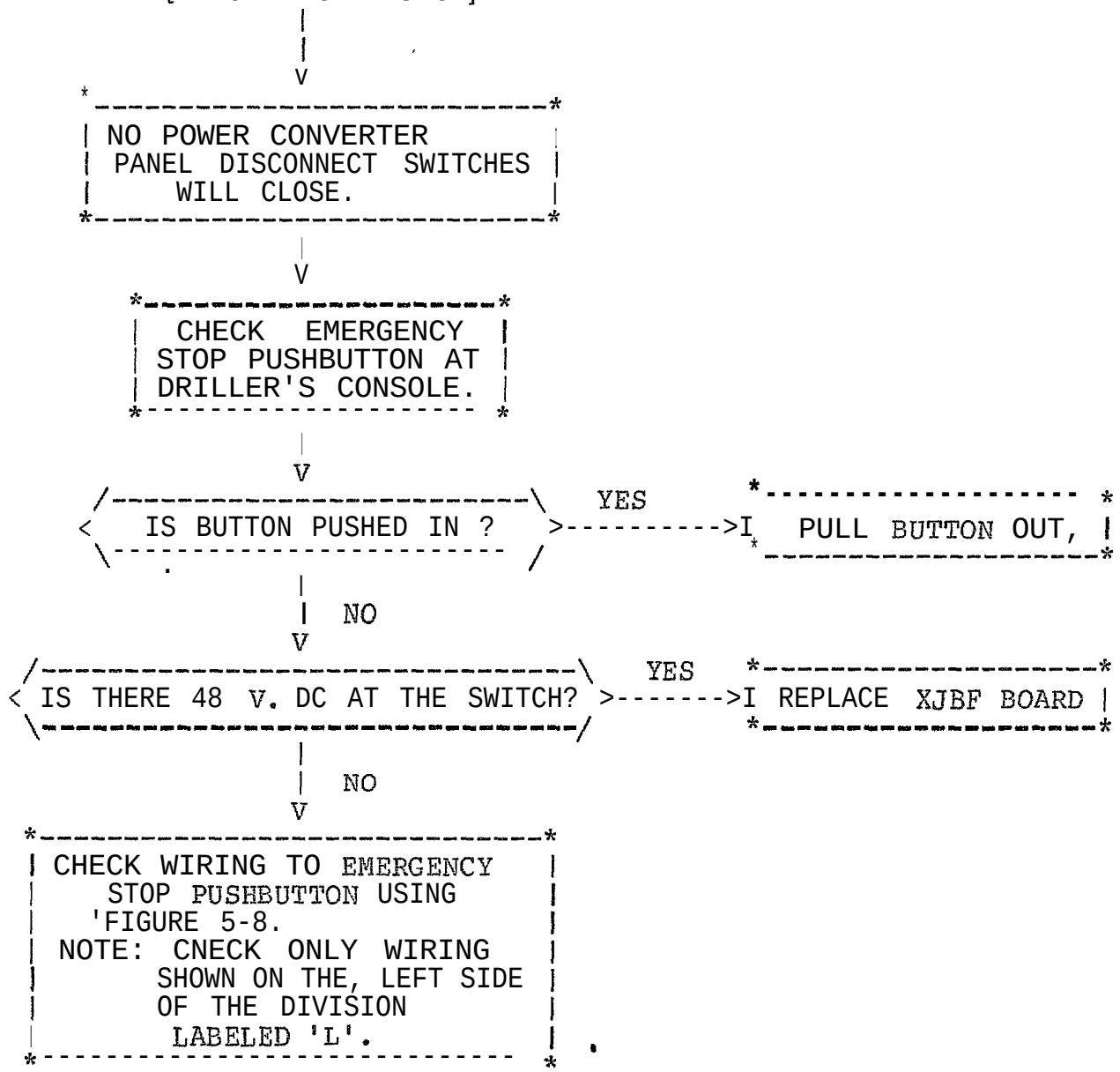
2. A FEW, BUT NO ALL, DISCONNECT SWITCHES WILL NOT CLOSE.

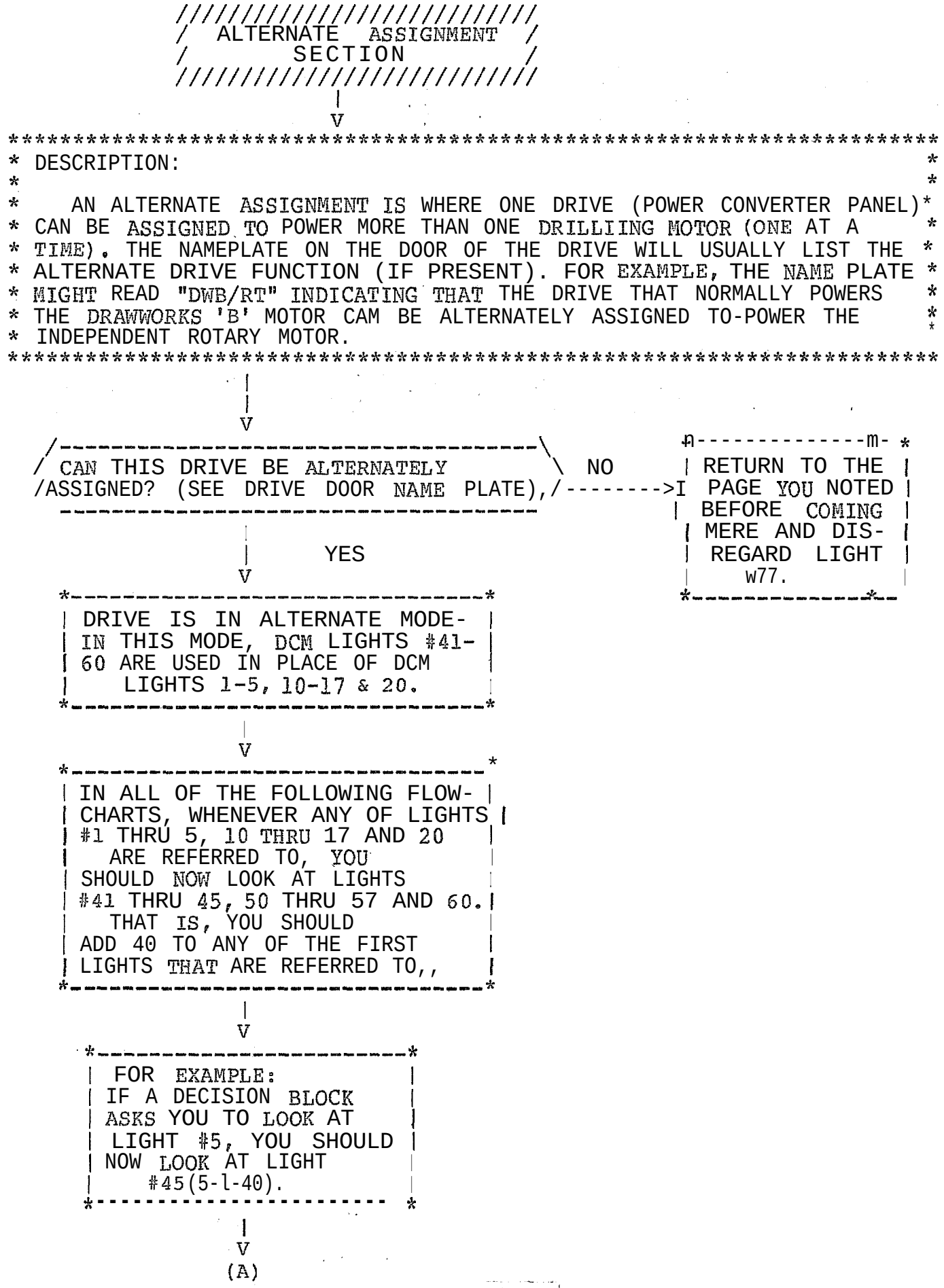
IF YOUR PROBLEM DOES NOT FALL INTO ONE OF THE ABOVE CATAGORIES, GO TO ANOTHER SECTION,

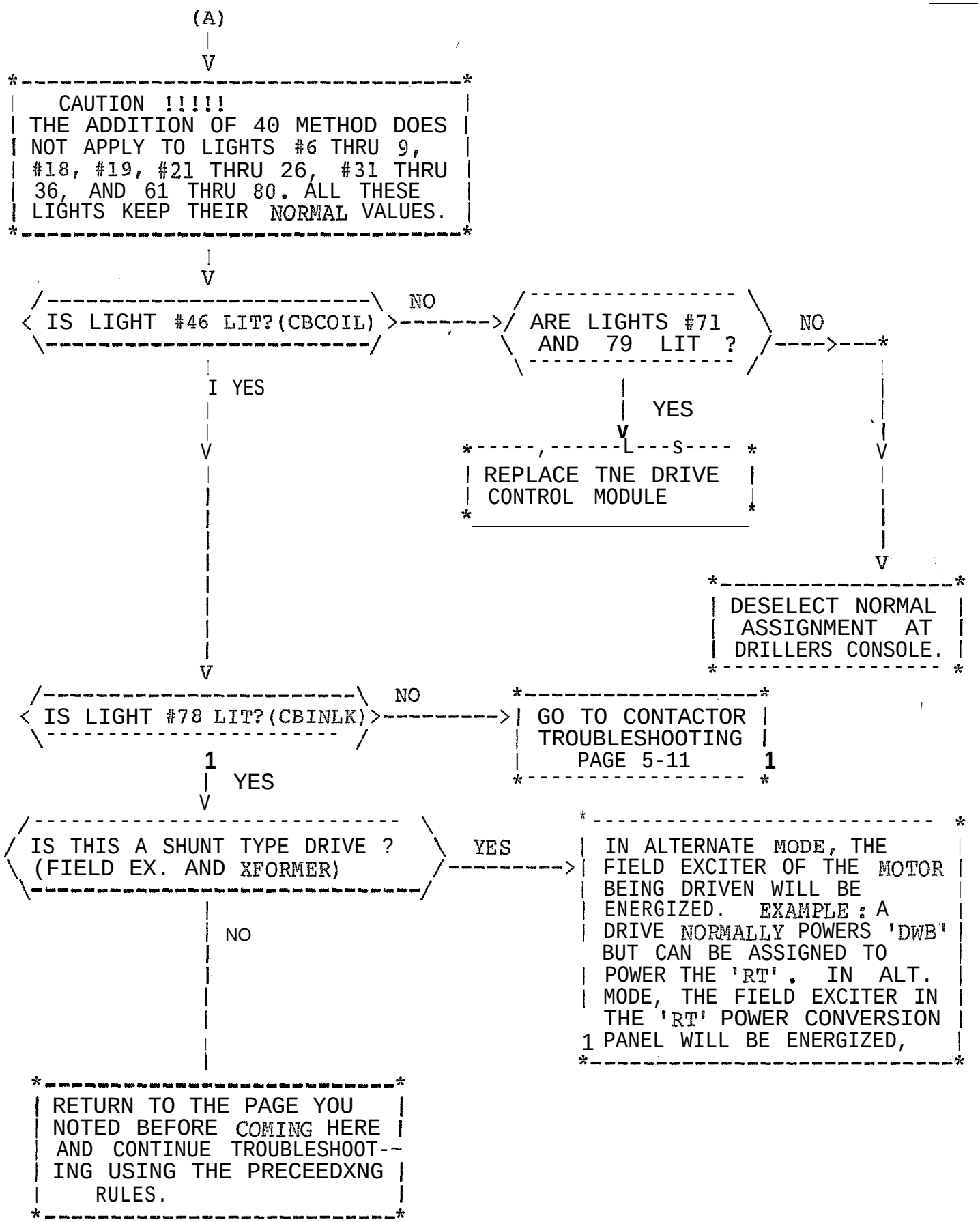



(A)
CHECK ALL WIRING OF PROBLEM DRIVE(S) USING FIGURE 5-8.
NOTE: CHECK ONLY THE WIRING SHOWN ON THE RIGHT SIDE OF THE DIVISION LABELED 'R'
IS ALL WIRING AND RIBBON CABLE SECURE AND INTACT ?
NO
CORRECT ANY WIRING PROBLEMS AND REPAIR OR REPLACE DAMAGED WIRE AND RIBBON CABLE.
YES
CHECK TO MAKE SURE PANEL DISCONNECT SWITCH HAS NOT FAILED.
IS SWITCH GOOD ?
YES
REPLACE XJBD CARD AND, IF PROBLEM STILL PRESENT, REPLACE THE XJBF BOARD.
NO
REPLACE THE DISCONNECT SWITCH'.

[FROM PAGE # 5-6]

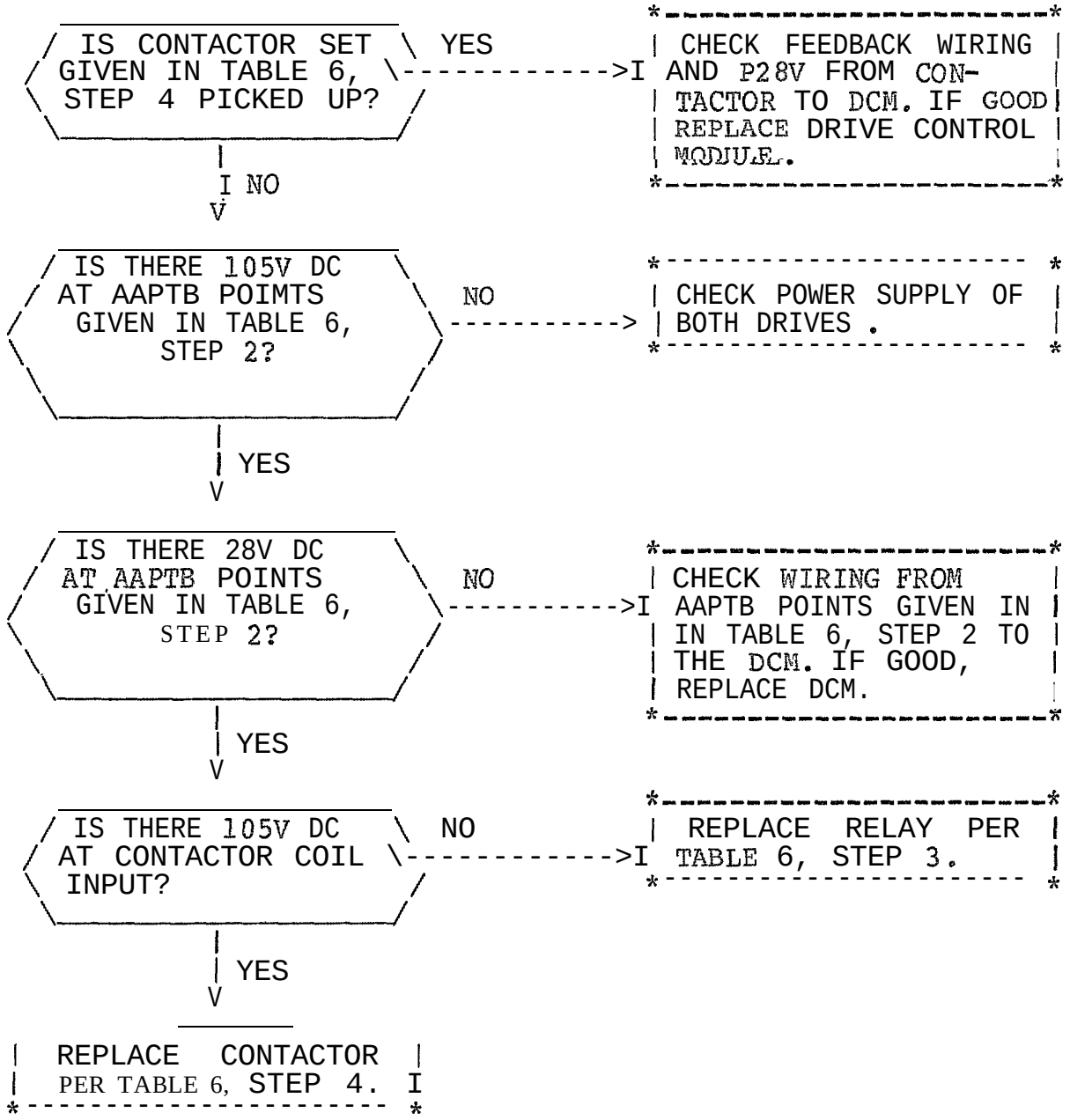

NO POWER CONVERTER PANEL DISCONNECT SWITCHES WILL CLOSE.
CHECK EMERGENCY STOP PUSHBUTTON AT DRILLER'S CONSOLE.
IS BUTTON PUSHED IN ?
YES
PULL BUTTON OUT,
NO
IS THERE 48 V. DC AT THE SWITCH?
YES
REPLACE XJBF BOARD
NO
CHECK WIRING TO EMERGENCY STOP PUSHBUTTON USING 'FIGURE 5-8.
NOTE: CNECK ONLY WIRING SHOWN ON THE, LEFT SIDE OF THE DIVISION LABELED 'L'.


ALTERNATE ASSIGNMENT SECTION
DESCRIPTION:
AN ALTERNATE ASSIGNMENT IS WHERE ONE DRIVE (POWER CONVERTER PANEL) CAN BE ASSIGNED TO POWER MORE THAN ONE DRILLIING MOTOR (ONE AT A TIME). THE NAMEPLATE ON THE DOOR OF THE DRIVE WILL USUALLY LIST THE ALTERNATE DRIVE FUNCTION (IF PRESENT). FOR EXAMPLE, THE NAME PLATE MIGHT READ "DWB/RT" INDICATING THAT THE DRIVE THAT NORMALLY POWERS THE DRAWWORKS 'B' MOTOR CAM BE ALTERNATELY ASSIGNED TO-POWER THE INDEPENDENT ROTARY MOTOR.
CAN THIS DRIVE BE ALTERNATELY ASSIGNED? (SEE DRIVE DOOR NAME PLATE),
NO
RETURN TO THE PAGE YOU NOTED BEFORE COMING MERE AND DIS- REGARD LIGHT w77.
YES
DRIVE IS IN ALTERNATE MODE- IN THIS MODE, DCM LIGHTS #41- 60 ARE USED IN PLACE OF DCM LIGHTS 1-5, 10-17 & 20.
IN ALL OF THE FOLLOWING FLOW- CHARTS, WHENEVER ANY OF LIGHTS #1 THRU 5, 10 THRU 17 AND 20 ARE REFERRED TO, YOU SHOULD NOW LOOK AT LIGHTS #41 THRU 45, 50 THRU 57 AND 60. THAT IS, YOU SHOULD ADD 40 TO ANY OF THE FIRST LIGHTS THAT ARE REFERRED TO,,
FOR EXAMPLE: IF A DECISION BLOCK ASKS YOU TO LOOK AT LIGHT #5, YOU SHOULD NOW LOOK AT LIGHT #45(5-l-40).
(A)


(A)
CAUTION !!!!!
THE ADDITION OF 40 METHOD DOES NOT APPLY TO LIGHTS #6 THRU 9, #18, #19, #21 THRU 26, #31 THRU 36, AND 61 THRU 80. ALL THESE LIGHTS KEEP THEIR NORMAL VALUES.
IS LIGHT #46 LIT?(CBCOIL)
NO
ARE LIGHTS #71 AND 79 LIT ?
NO
YES
REPLACE TNE DRIVE CONTROL MODULE
DESELECT NORMAL ASSIGNMENT AT DRILLERS CONSOLE.
I YES
IS LIGHT #78 LIT?(CBINLK)
NO
GO TO CONTACTOR TROUBLESHOOTING PAGE 5-11
1
1
YES
IS THIS A SHUNT TYPE DRIVE ? (FIELD EX. AND XFORMER)
YES
IN ALTERNATE MODE, THE FIELD EXCITER OF THE MOTOR BEING DRIVEN WILL BE ENERGIZED. EXAMPLE: A DRIVE NORMALLY POWERS 'DWB' BUT CAN BE ASSIGNED TO POWER THE 'RT'. IN ALT. MODE, THE FIELD EXCITER IN THE 'RT' POWER CONVERSION 1 PANEL WILL BE ENERGIZED,
NO
RETURN TO THE PAGE YOU NOTED BEFORE COMING HERE AND CONTINUE TROUBLESHOOTING USING THE PRECEEDXNG RULES.

CONTACTOR TROUBLE SHOOTING

(SEE ELEMENTARY, SECTION 11 OF THIS INSTRUCTION BOOK.)



	DRIVE THAT CAN POWER 2 MOTORS AND LT.377 LIT. (SEE DOOR NAMEPLATE) ,	DRIVE THAT CAN POWER 2 MOTORS AND #77 NOT LIT,	DRIVE WHOSE MOTOR CAN BE ALTERNATELY ASSIGNED TO ANOTHER DRIVE,
1	AAPTB 5-11	AAPTB 5-11	AAPTB 6-12
2	AAPTB 2-4	AAPTB 1-4	AAPTB 3-4
3	AA	ALT	NORM
4	CONT.B	CONT.A	CONT.C

TABLE 6



GE Drive Systems

General Electric Company
1501 Roanoke Boulevard
Salem, *Virginia 24 153*
(703) 387- 7000